decksh reference

Version

2026-07-13-1.1.0

Introduction

decksh is a Domain-Specific Language (DSL) for making presentations, visualizations, and information displays.

This reference describes the keywords and elements of the language, how to structure decksh code, along with how to use variables, assignments, and binary operations.

Also included is a color reference and a detailed description, with examples, for all decksh elements.

Keywords

Structure

deck/edeck
slide/eslide
canvas
def/edef
for/efor
func
grid
import
include
if/else/eif

Text

text
btext
ctext
etext
para/epara
rtext
arctext
textblock
textblockfile
textfile
textcode

Graphics

acircle
arc
circle
curve
ellipse
dline
hline
line
pill
polygon
polyline
rect
rrect
ruler
square
star
vline

Lists

list
blist
nlist
clist
li
elist

Braces

lbrace
rbrace
ubrace
dbrace
lbracket
rbracket
dbracket
ubracket

Arrows

arrow
rcarrow
lcarrow
ucarrow
dcarrow

Maps

geoarc
geoborder
geobound
geocanvas
geoimage
geolabel
geoloc
geomark
geopath
geopathfile
georegion
georuler

Images

image
cimage

Charts

chartbbox
dchart
legend
area
barchart
hbar
wbar
linechart
scatter
slopechart
donut
pie
waffle
pgrid
pmap
bowtie
fanchart

Assign

polar
polarx
polary
random
area
format
substr
vmap
dump

Math

geocoord
cosine
sine
sqrt
tangent

Data

data
edata
content

Keywords and arguments

keyword	arguments		
	mandatory	optional	
`text`	`"..string...." x y n`	`"font" "color" op`	
`text`	`"hello, world" 80 50 2`		hello, world
`text`	`"hello, world" 80 40 2`	`"serif"`	hello, world
`text`	`"hello, world" 80 30 2`	`"serif" "red"`	hello, world
`text`	`"hello, world" 80 20 2`	`"serif" "red" 50`	hello, world

Structure

```
// This is a comment
deck
    canvas 1920 1080
    x=20      // define x
    y=80
    slide
        text "first" x y 2
    eslide
    slide "black" "white"
        include "file.dsh"
    eslide
edeck
```

comment

canvas size hint (width height)

inline comment

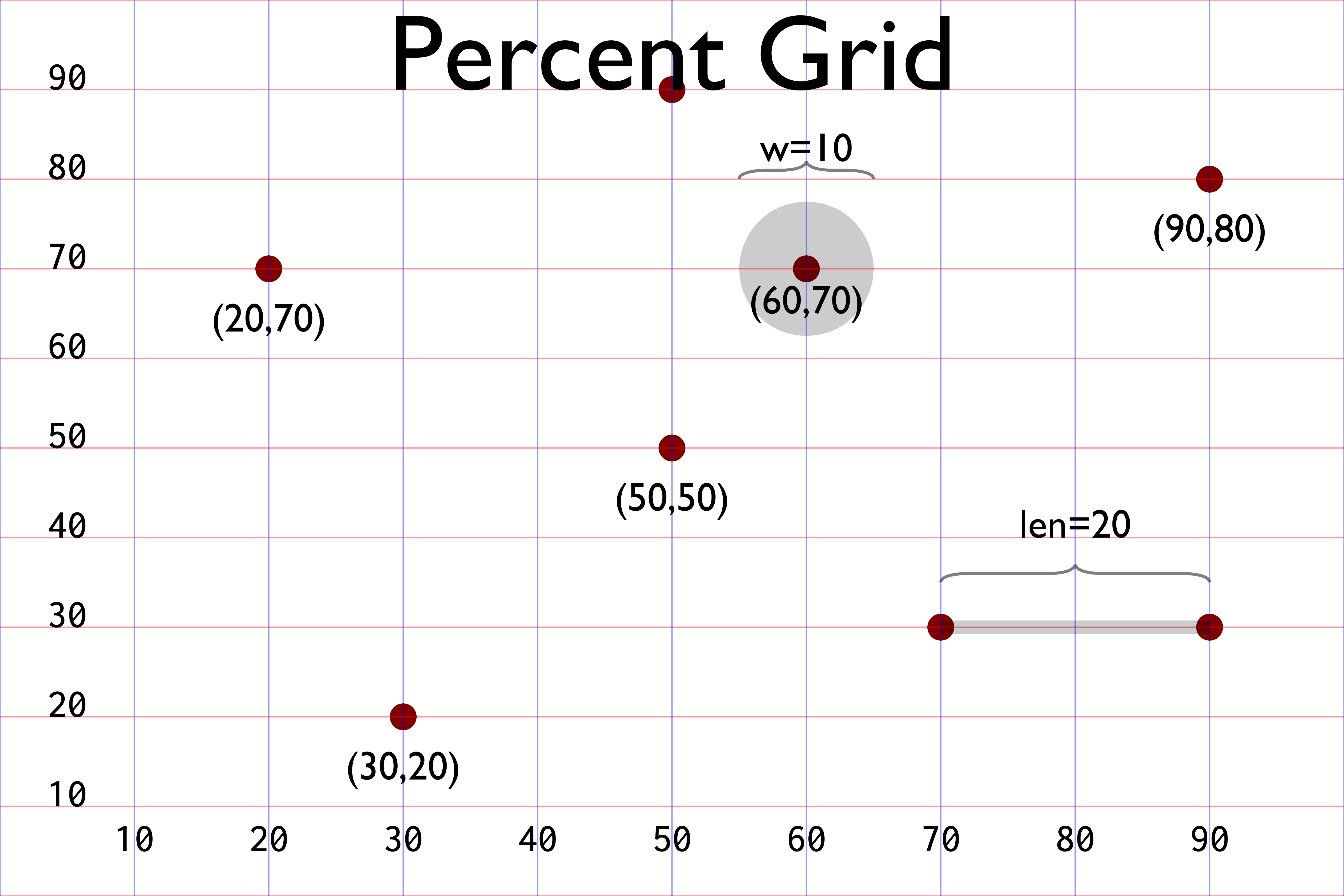
deck

variables

slide 1

slide 2

```
ctext "hello, world" 50 25 10
circle 50 0 100 "blue"
for x=20 80 10
    circle x 75 2
efor
```


Percent Grid
90
80
70
60
50
40
30
20
10
w=10
(90,80)
(20,70)
(60,70)
(50,50)
len=20
(30,20)
10
20
30
40
50
60
70
80
90

Defining and using variables

name is thing	
`x = 3.14159265`	number
`s = "hello"`	string
`y = x`	another variable
`x = a + b`	binary operation
`x *= 10`	assignment operator
`text s x y 2`	variable use
`dump [var list]`	dump the value of variables

Note: the reserved variable deckshVersion contains the version of decksh `2026-07-13-1.1.0`

Binary operators

`x = a + b`	addition
`x = a - b`	subtraction
`x = a * b`	multiplication
`x = a / b`	division
`x = a % b`	modulo

Assignment operators

`x += 10`	increase x by 10
`x -= 10`	decrease x by 10
`x *= 10`	multiply x by 10
`x /= 10`	divide x by 10

Special Assignments

`p=(expr, expr)`	coordinates (p_x, p_y)
`p=geocoord "loc" or geocoord lat long`	geo mapped coordinates (p_x, p_y)
`p=polar cx cy r theta`	polar coordinate (p_x, p_y)
`x=polarx cx cy r theta`	polar coordinate (x)
`y=polary cx cy r theta`	polar coordinate (y)
`v=format string expr`	number formatting
`v=substr string begin end`	substring
`v=random v1 v2`	random number
`v=vmap data v1 v2 v3 v4`	range map
`v=area expr`	area
`v=cosine expr`	cosine
`v=sine expr`	sine
`v=sqrt expr`	square root
`v=tangent expr`	tangent

Colors, fonts, opacity, gradient

Colors	Fonts		Opacity (0-100)
"steelblue"	"sans"	Sans Serif	100
"#4682b4"	"serif"	Serif	50
"rgb(70,130,180)"	"mono"	Monospace	25
"hsv(207,61,71)"	"symbol"	❁✷❃❄❅	10

"red/blue/90"

(applies to rect and square only)

Color Index

name	hex	RGB	name	hex	RGB
aliceblue	#f0f8ff	rgb(240,248,255)	coral	#ff7f50	rgb(255,127,80)
antiquewhite	#faebd7	rgb(250,235,215)	cornflowerblue	#6495ed	rgb(100,149,237)
aqua	#00ffff	rgb(0,255,255)	cornsilk	#fff8dc	rgb(255,248,220)
aquamarine	#7fffd4	rgb(127,255,212)	crimson	#dc143c	rgb(220,20,60)
azure	#f0ffff	rgb(240,255,255)	cyan	#00ffff	rgb(0,255,255)
beige	#f5f5dc	rgb(245,245,220)	darkblue	#00008b	rgb(0,0,139)
bisque	#ffe4c4	rgb(255,228,196)	darkcyan	#008b8b	rgb(0,139,139)
black	#000000	rgb(0,0,0)	darkgoldenrod	#b8860b	rgb(184,134,11)
blanchedalmond	#ffebcd	rgb(255,235,205)	darkgray	#a9a9a9	rgb(169,169,169)
blue	#0000ff	rgb(0,0,255)	darkgreen	#006400	rgb(0,100,0)
blueviolet	#8a2be2	rgb(138,43,226)	darkgrey	#a9a9a9	rgb(169,169,169)
brown	#a52a2a	rgb(165,42,42)	darkkhaki	#bdb76b	rgb(189,183,107)
burlywood	#deb887	rgb(222,184,135)	darkmagenta	#8b008b	rgb(139,0,139)
cadetblue	#5f9ea0	rgb(95,158,160)	darkolivegreen	#556b2f	rgb(85,107,47)
chartreuse	#7fff00	rgb(127,255,0)	darkorange	#ff8c00	rgb(255,140,0)
chocolate	#d2691e	rgb(210,105,30)	darkorchid	#9932cc	rgb(153,50,204)

Color Index (2)

name	hex	RGB	name	hex	RGB
darkred	#8b0000	rgb(139,0,0)	fuchsia	#ff00ff	rgb(255,0,255)
darksalmon	#e9967a	rgb(233,150,122)	gainsboro	#dcdcdc	rgb(220,220,220)
darkseagreen	#8fbc8f	rgb(143,188,143)	ghostwhite	#f8f8ff	rgb(248,248,255)
darkslateblue	#483d8b	rgb(72,61,139)	gold	#ffd700	rgb(255,215,0)
darkslategray	#2f4f4f	rgb(47,79,79)	goldenrod	#daa520	rgb(218,165,32)
darkslategrey	#2f4f4f	rgb(47,79,79)	gray	#808080	rgb(128,128,128)
darkturquoise	#00ced1	rgb(0,206,209)	green	#008000	rgb(0,128,0)
darkviolet	#9400d3	rgb(148,0,211)	greenyellow	#adff2f	rgb(173,255,47)
deeppink	#ff1493	rgb(255,20,147)	grey	#808080	rgb(128,128,128)
deepskyblue	#00bfff	rgb(0,191,255)	honeydew	#f0fff0	rgb(240,255,240)
dimgray	#696969	rgb(105,105,105)	hotpink	#ff69b4	rgb(255,105,180)
dimgrey	#696969	rgb(105,105,105)	indianred	#cd5c5c	rgb(205,92,92)
dodgerblue	#1e90ff	rgb(30,144,255)	indigo	#4b0082	rgb(75,0,130)
firebrick	#b22222	rgb(178,34,34)	ivory	#fffff0	rgb(255,255,240)
floralwhite	#fffaf0	rgb(255,250,240)	khaki	#f0e68c	rgb(240,230,140)
forestgreen	#228b22	rgb(34,139,34)	lavender	#e6e6fa	rgb(230,230,250)

Color Index (3)

name	hex	RGB	name	hex	RGB
lavenderblush	#fff0f5	rgb(255,240,245)	lightsteelblue	#b0c4de	rgb(176,196,222)
lawngreen	#7cfc00	rgb(124,252,0)	lightyellow	#ffffe0	rgb(255,255,224)
lemonchiffon	#fffacd	rgb(255,250,205)	lime	#00ff00	rgb(0,255,0)
lightblue	#add8e6	rgb(173,216,230)	limegreen	#32cd32	rgb(50,205,50)
lightcoral	#f08080	rgb(240,128,128)	linen	#faf0e6	rgb(250,240,230)
lightcyan	#e0ffff	rgb(224,255,255)	magenta	#ff00ff	rgb(255,0,255)
lightgoldenrodyellow	#fafad2	rgb(250,250,210)	maroon	#800000	rgb(128,0,0)
lightgray	#d3d3d3	rgb(211,211,211)	mediumaquamarine	#66cdaa	rgb(102,205,170)
lightgreen	#90ee90	rgb(144,238,144)	mediumblue	#0000cd	rgb(0,0,205)
lightgrey	#d3d3d3	rgb(211,211,211)	mediumorchid	#ba55d3	rgb(186,85,211)
lightpink	#ffb6c1	rgb(255,182,193)	mediumpurple	#9370db	rgb(147,112,219)
lightsalmon	#ffa07a	rgb(255,160,122)	mediumseagreen	#3cb371	rgb(60,179,113)
lightseagreen	#20b2aa	rgb(32,178,170)	mediumslateblue	#7b68ee	rgb(123,104,238)
lightskyblue	#87cefa	rgb(135,206,250)	mediumspringgreen	#00fa9a	rgb(0,250,154)
lightslategray	#778899	rgb(119,136,153)	mediumturquoise	#48d1cc	rgb(72,209,204)
lightslategrey	#778899	rgb(119,136,153)	mediumvioletred	#c71585	rgb(199,21,133)

Color Index (4)

name	hex	RGB	name	hex	RGB
midnightblue	#191970	rgb(25,25,112)	papayawhip	#ffefd5	rgb(255,239,213)
mintcream	#f5fffa	rgb(245,255,250)	peachpuff	#ffdab9	rgb(255,218,185)
mistyrose	#ffe4e1	rgb(255,228,225)	peru	#cd853f	rgb(205,133,63)
moccasin	#ffe4b5	rgb(255,228,181)	pink	#ffc0cb	rgb(255,192,203)
navajowhite	#ffdead	rgb(255,222,173)	plum	#dda0dd	rgb(221,160,221)
navy	#000080	rgb(0,0,128)	powderblue	#b0e0e6	rgb(176,224,230)
oldlace	#fdf5e6	rgb(253,245,230)	purple	#800080	rgb(128,0,128)
olive	#808000	rgb(128,128,0)	red	#ff0000	rgb(255,0,0)
olivedrab	#6b8e23	rgb(107,142,35)	rosybrown	#bc8f8f	rgb(188,143,143)
orange	#ffa500	rgb(255,165,0)	royalblue	#4169e1	rgb(65,105,225)
orangered	#ff4500	rgb(255,69,0)	saddlebrown	#8b4513	rgb(139,69,19)
orchid	#da70d6	rgb(218,112,214)	salmon	#fa8072	rgb(250,128,114)
palegoldenrod	#eee8aa	rgb(238,232,170)	sandybrown	#f4a460	rgb(244,164,96)
palegreen	#98fb98	rgb(152,251,152)	seagreen	#2e8b57	rgb(46,139,87)
paleturquoise	#afeeee	rgb(175,238,238)	seashell	#fff5ee	rgb(255,245,238)
palevioletred	#db7093	rgb(219,112,147)	sienna	#a0522d	rgb(160,82,45)

Color Index (5)

name	hex	RGB
silver	#c0c0c0	rgb(192,192,192)
skyblue	#87ceeb	rgb(135,206,235)
slateblue	#6a5acd	rgb(106,90,205)
slategray	#708090	rgb(112,128,144)
slategrey	#708090	rgb(112,128,144)
snow	#fffafa	rgb(255,250,250)
springgreen	#00ff7f	rgb(0,255,127)
steelblue	#4682b4	rgb(70,130,180)
tan	#d2b48c	rgb(210,180,140)
teal	#008080	rgb(0,128,128)
thistle	#d8bfd8	rgb(216,191,216)
tomato	#ff6347	rgb(255,99,71)
turquoise	#40e0d0	rgb(64,224,208)
violet	#ee82ee	rgb(238,130,238)
wheat	#f5deb3	rgb(245,222,179)
white	#ffffff	rgb(255,255,255)

name	hex	RGB
whitesmoke	#f5f5f5	rgb(245,245,245)
yellow	#ffff00	rgb(255,255,0)
yellowgreen	#9acd32	rgb(154,205,50)

Neutrals

name	hex	RGB	name	hex	RGB
aliceblue	#f0f8ff	rgb(240,248,255)	dimgray	#696969	rgb(105,105,105)
antiquewhite	#faebd7	rgb(250,235,215)	dimgrey	#696969	rgb(105,105,105)
azure	#f0ffff	rgb(240,255,255)	floralwhite	#fffaf0	rgb(255,250,240)
beige	#f5f5dc	rgb(245,245,220)	gainsboro	#dcdcdc	rgb(220,220,220)
bisque	#ffe4c4	rgb(255,228,196)	ghostwhite	#f8f8ff	rgb(248,248,255)
black	#000000	rgb(0,0,0)	gray	#808080	rgb(128,128,128)
blanchedalmond	#ffebcd	rgb(255,235,205)	grey	#808080	rgb(128,128,128)
brown	#a52a2a	rgb(165,42,42)	honeydew	#f0fff0	rgb(240,255,240)
burlywood	#deb887	rgb(222,184,135)	ivory	#fffff0	rgb(255,255,240)
chocolate	#d2691e	rgb(210,105,30)	lavender	#e6e6fa	rgb(230,230,250)
cornsilk	#fff8dc	rgb(255,248,220)	lavenderblush	#fff0f5	rgb(255,240,245)
darkgray	#a9a9a9	rgb(169,169,169)	lightgray	#d3d3d3	rgb(211,211,211)
darkgrey	#a9a9a9	rgb(169,169,169)	lightgrey	#d3d3d3	rgb(211,211,211)
darksalmon	#e9967a	rgb(233,150,122)	lightslategray	#778899	rgb(119,136,153)
darkslategray	#2f4f4f	rgb(47,79,79)	lightslategrey	#778899	rgb(119,136,153)
darkslategrey	#2f4f4f	rgb(47,79,79)	linen	#faf0e6	rgb(250,240,230)

Neutrals (2)

name	hex	RGB
mintcream	#f5fffa	rgb(245,255,250)
mistyrose	#ffe4e1	rgb(255,228,225)
moccasin	#ffe4b5	rgb(255,228,181)
navajowhite	#ffdead	rgb(255,222,173)
oldlace	#fdf5e6	rgb(253,245,230)
papayawhip	#ffefd5	rgb(255,239,213)
peachpuff	#ffdab9	rgb(255,218,185)
peru	#cd853f	rgb(205,133,63)
rosybrown	#bc8f8f	rgb(188,143,143)
saddlebrown	#8b4513	rgb(139,69,19)
salmon	#fa8072	rgb(250,128,114)
sandybrown	#f4a460	rgb(244,164,96)
seashell	#fff5ee	rgb(255,245,238)
sienna	#a0522d	rgb(160,82,45)
silver	#c0c0c0	rgb(192,192,192)
slategray	#708090	rgb(112,128,144)

name	hex	RGB
slategrey	#708090	rgb(112,128,144)
snow	#fffafa	rgb(255,250,250)
tan	#d2b48c	rgb(210,180,140)
wheat	#f5deb3	rgb(245,222,179)
white	#ffffff	rgb(255,255,255)
whitesmoke	#f5f5f5	rgb(245,245,245)

Reds

name	hex	RGB
coral	#ff7f50	rgb(255,127,80)
crimson	#dc143c	rgb(220,20,60)
darkmagenta	#8b008b	rgb(139,0,139)
darkred	#8b0000	rgb(139,0,0)
deeppink	#ff1493	rgb(255,20,147)
firebrick	#b22222	rgb(178,34,34)
fuchsia	#ff00ff	rgb(255,0,255)
hotpink	#ff69b4	rgb(255,105,180)
indianred	#cd5c5c	rgb(205,92,92)
lightcoral	#f08080	rgb(240,128,128)
lightpink	#ffb6c1	rgb(255,182,193)
lightsalmon	#ffa07a	rgb(255,160,122)
magenta	#ff00ff	rgb(255,0,255)
maroon	#800000	rgb(128,0,0)
orangered	#ff4500	rgb(255,69,0)
orchid	#da70d6	rgb(218,112,214)
palevioletred	#db7093	rgb(219,112,147)
pink	#ffc0cb	rgb(255,192,203)
plum	#dda0dd	rgb(221,160,221)
red	#ff0000	rgb(255,0,0)
thistle	#d8bfd8	rgb(216,191,216)
tomato	#ff6347	rgb(255,99,71)

Greens

name	hex	RGB
aquamarine	#7fffd4	rgb(127,255,212)
chartreuse	#7fff00	rgb(127,255,0)
darkgreen	#006400	rgb(0,100,0)
darkkhaki	#bdb76b	rgb(189,183,107)
darkolivegreen	#556b2f	rgb(85,107,47)
darkseagreen	#8fbc8f	rgb(143,188,143)
forestgreen	#228b22	rgb(34,139,34)
green	#008000	rgb(0,128,0)
greenyellow	#adff2f	rgb(173,255,47)
lawngreen	#7cfc00	rgb(124,252,0)
lightgreen	#90ee90	rgb(144,238,144)
lightseagreen	#20b2aa	rgb(32,178,170)
lime	#00ff00	rgb(0,255,0)
limegreen	#32cd32	rgb(50,205,50)
mediumseagreen	#3cb371	rgb(60,179,113)
mediumspringgreen	#00fa9a	rgb(0,250,154)
olive	#808000	rgb(128,128,0)
olivedrab	#6b8e23	rgb(107,142,35)
palegreen	#98fb98	rgb(152,251,152)
seagreen	#2e8b57	rgb(46,139,87)
springgreen	#00ff7f	rgb(0,255,127)
teal	#008080	rgb(0,128,128)
yellowgreen	#9acd32	rgb(154,205,50)

Blues

name	hex	RGB
aqua	#00ffff	rgb(0,255,255)
blue	#0000ff	rgb(0,0,255)
cadetblue	#5f9ea0	rgb(95,158,160)
cornflowerblue	#6495ed	rgb(100,149,237)
cyan	#00ffff	rgb(0,255,255)
darkblue	#00008b	rgb(0,0,139)
darkcyan	#008b8b	rgb(0,139,139)
darkslateblue	#483d8b	rgb(72,61,139)
darkturquoise	#00ced1	rgb(0,206,209)
deepskyblue	#00bfff	rgb(0,191,255)
dodgerblue	#1e90ff	rgb(30,144,255)
lightblue	#add8e6	rgb(173,216,230)
lightcyan	#e0ffff	rgb(224,255,255)
lightskyblue	#87cefa	rgb(135,206,250)
lightsteelblue	#b0c4de	rgb(176,196,222)
mediumaquamarine	#66cdaa	rgb(102,205,170)

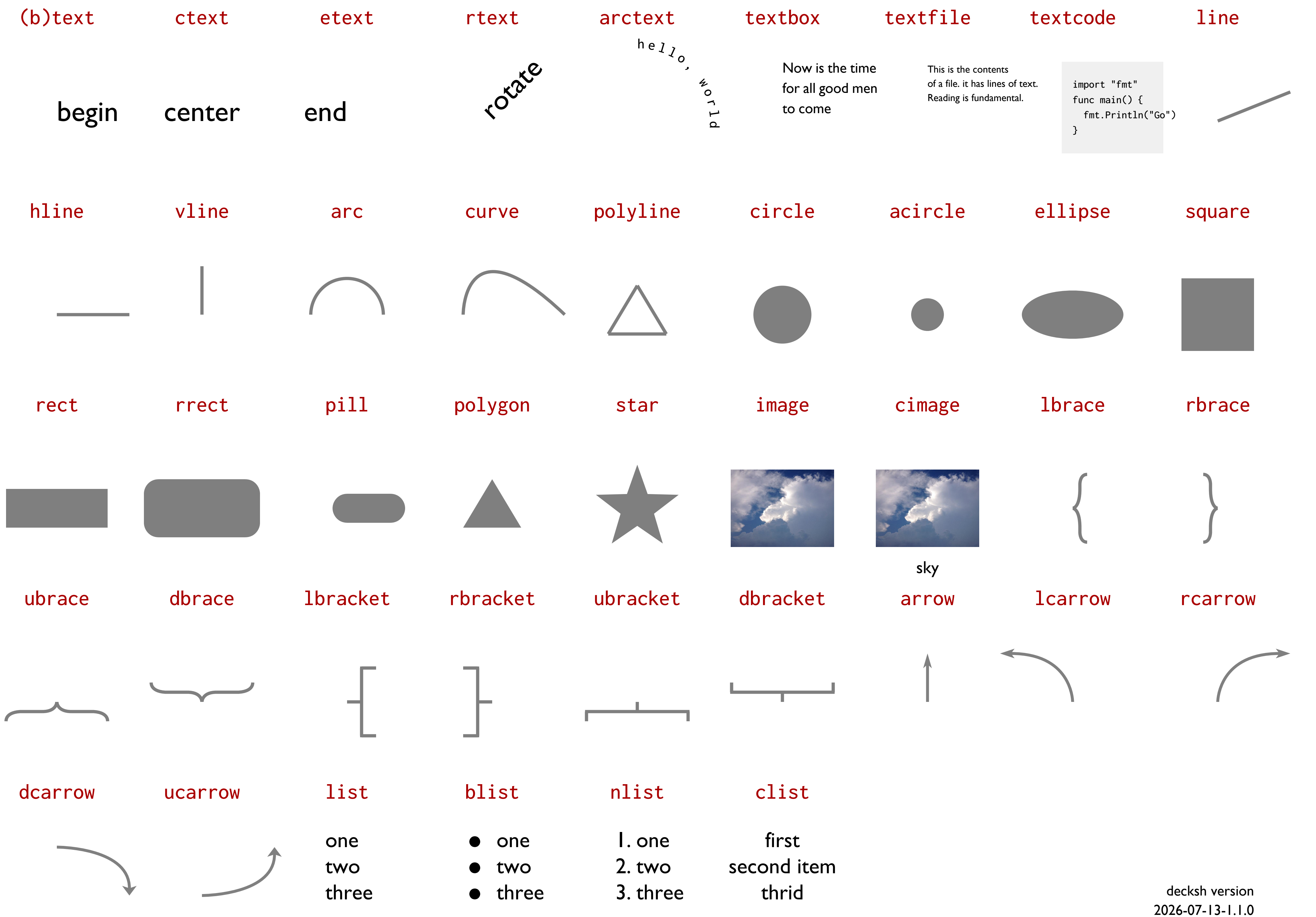
name	hex	RGB
mediumblue	#0000cd	rgb(0,0,205)
mediumslateblue	#7b68ee	rgb(123,104,238)
mediumturquoise	#48d1cc	rgb(72,209,204)
midnightblue	#191970	rgb(25,25,112)
navy	#000080	rgb(0,0,128)
paleturquoise	#afeeee	rgb(175,238,238)
powderblue	#b0e0e6	rgb(176,224,230)
royalblue	#4169e1	rgb(65,105,225)
skyblue	#87ceeb	rgb(135,206,235)
slateblue	#6a5acd	rgb(106,90,205)
steelblue	#4682b4	rgb(70,130,180)
turquoise	#40e0d0	rgb(64,224,208)

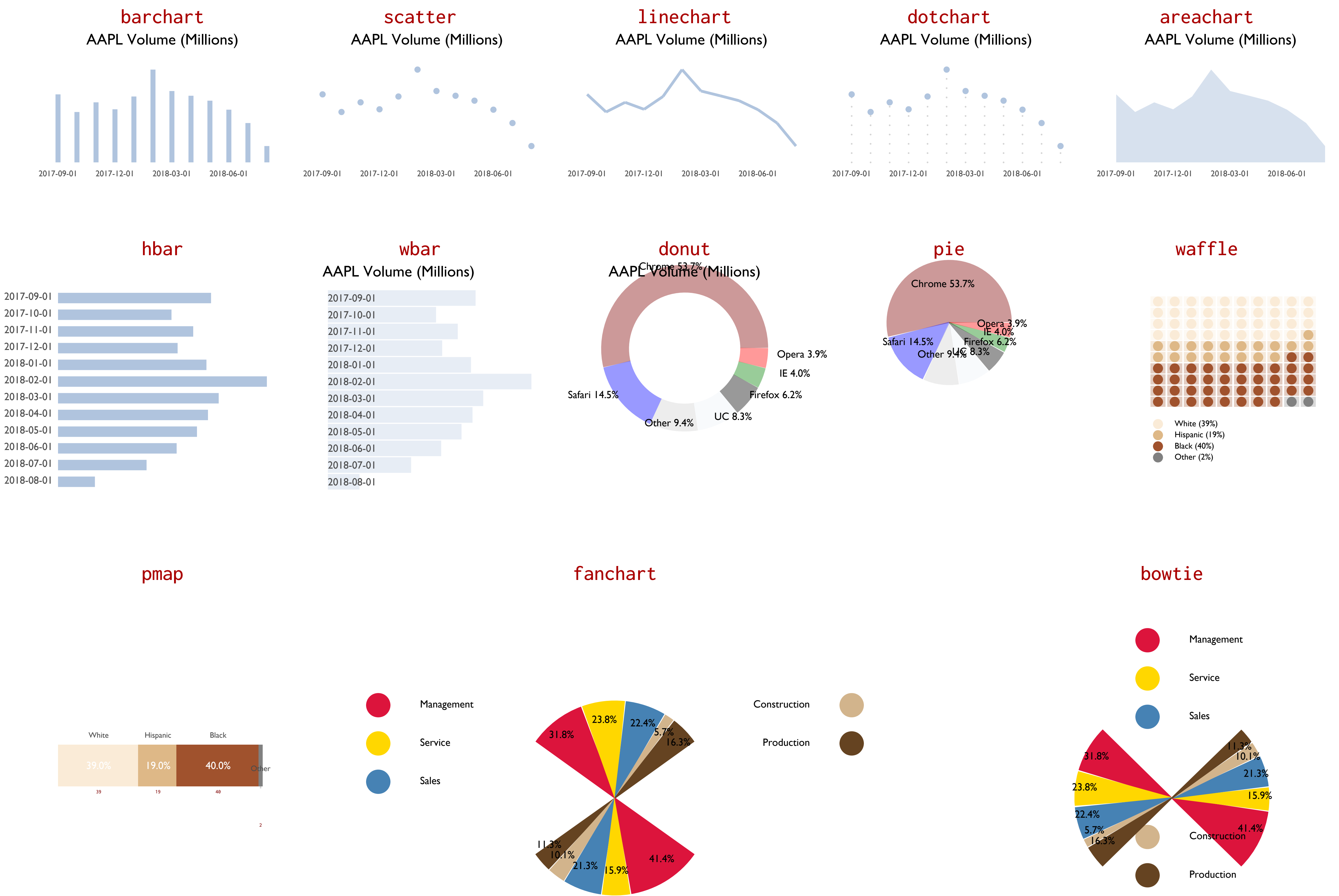
Violets

	name	hex	RGB
	blueviolet	#8a2be2	rgb(138,43,226)
	darkorchid	#9932cc	rgb(153,50,204)
	darkviolet	#9400d3	rgb(148,0,211)
	indigo	#4b0082	rgb(75,0,130)
	mediumorchid	#ba55d3	rgb(186,85,211)
	mediumpurple	#9370db	rgb(147,112,219)
	mediumvioletred	#c71585	rgb(199,21,133)
	purple	#800080	rgb(128,0,128)
	violet	#ee82ee	rgb(238,130,238)

Yellows

	name	hex	RGB
	darkgoldenrod	#b8860b	rgb(184,134,11)
	darkorange	#ff8c00	rgb(255,140,0)
	gold	#ffd700	rgb(255,215,0)
	goldenrod	#daa520	rgb(218,165,32)
	khaki	#f0e68c	rgb(240,230,140)
	lemonchiffon	#fffacd	rgb(255,250,205)
	lightgoldenrodyellow	#fafad2	rgb(250,250,210)
	lightyellow	#ffffe0	rgb(255,255,224)
	orange	#ffa500	rgb(255,165,0)
	palegoldenrod	#eee8aa	rgb(238,232,170)
	yellow	#ffff00	rgb(255,255,0)


(b)text
ctext
etext
rtext
arctext
textbox
textfile
textcode
line
begin
center
end
rotate
hello, world
Now is the time
for all good men
to come
This is the contents
of a file. it has lines of text.
Reading is fundamental.
import "fmt"
func main() {
fmt.Println("Go")
}
hline
vline
arc
curve
polyline
circle
acircle
ellipse
square
rect
rrect
pill
polygon
star
image
cimage
lbrace
rbrace
sky
ubrace
dbrace
lbracket
rbracket
ubracket
dbracket
arrow
lcarrow
rcarrow
dcarrow
ucarrow
list
blist
nlist
clist
one
two
three
● one
● two
● three
1. one
2. two
3. three
first
second item
thrid
decksh version
2026-07-13-1.1.0

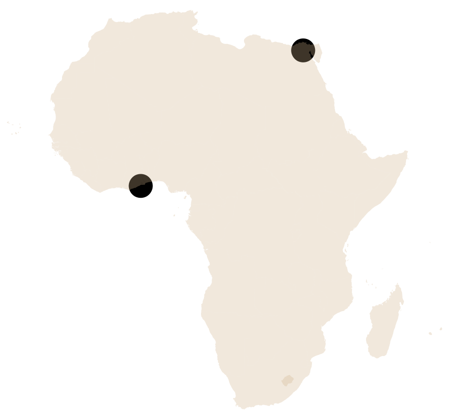

barchart
AAPL Volume (Millions)
2017-09-01
2017-12-01
2018-03-01
2018-06-01
scatter
AAPL Volume (Millions)
linechart
AAPL Volume (Millions)
dotchart
AAPL Volume (Millions)
areachart
AAPL Volume (Millions)
hbar
2017-09-01
2017-10-01
2017-11-01
2017-12-01
2018-01-01
2018-02-01
2018-03-01
2018-04-01
2018-05-01
2018-06-01
2018-07-01
2018-08-01
wbar
AAPL Volume (Millions)
donut
AAPL Volume (Millions)
Chrome 53.7%
Opera 3.9%
IE 4.0%
Firefox 6.2%
UC 8.3%
Other 9.4%
Safari 14.5%
pie
waffle
White (39%)
Hispanic (19%)
Black (40%)
Other (2%)
pmap
White
Hispanic
Black
Other
39.0%
19.0%
40.0%
39
19
40
2
fanchart
Management
Service
Sales
Construction
Production
31.8%
23.8%
22.4%
5.7%
16.3%
11.3%
10.1%
21.3%
15.9%
41.4%
bowtie

georegion

geoborder

geolabel

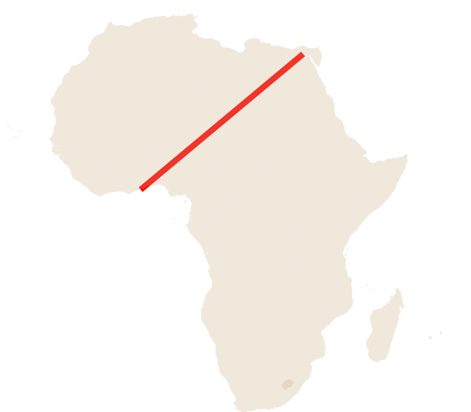
geomark



geoloc

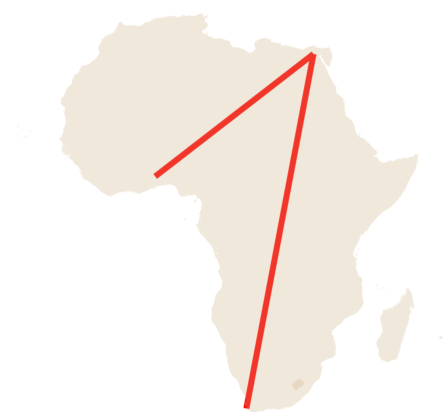
geoimage

geopath



geoarc

geopathfile



Textual Elements

description	keyword	mandatory	optional
Left-aligned	text	"..." x y fontsize	font color op link
Centered	ctext	"..." x y fontsize	font color op link
End-aligned	etext	"..." x y fontsize	font color op link
Rotated	rtext	"..." x y angle fontsize	font color op link
Text on an arc	arctext	"..." x y rad a1 a2 fontsize	font color op link
Block text	textblock	"..." x y w fontsize	font color op link
Paragraphs	para...epara	x y w fontsize	font color op link
Block text from file	textblockfile	"file" x y w fontsize	font color op link
File contents	textfile	"file" x y fontsize	font color op spacing
Code listing	textcode	"file" x y w fontsize	font color

hello, world

(x,y)

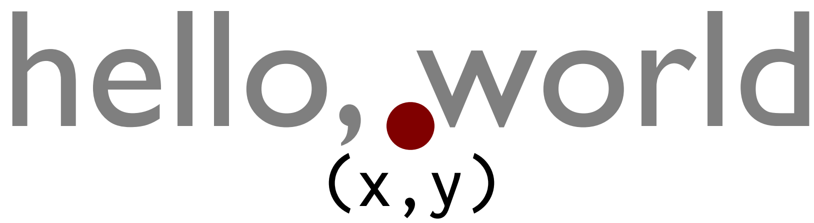
```
text "..." x y fontsize font color op link
```

abc

```
text "abc" 20 20 4
```

abc

```
text "abc" 75 20 7 "mono" "maroon"
```




```
ctext "..." x y fontsize font color op link
```


```
ctext "abc" 20 20 4
```


```
ctext "abc" 80 20 7 "mono" "maroon"
```

hello, world

(x,y)

```
etext "..." x y fontsize font color op link
```

abc

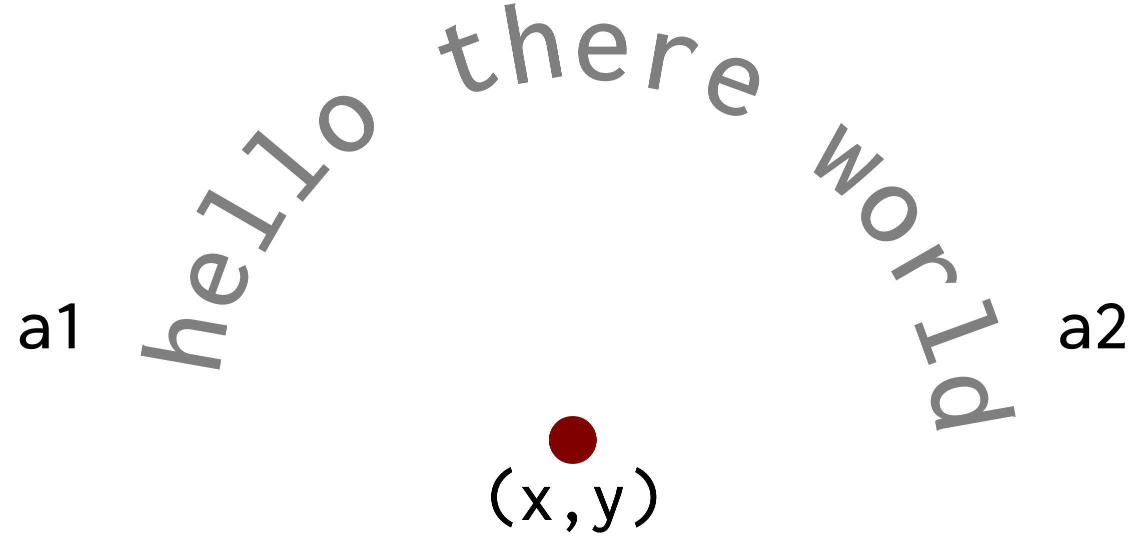
```
etext "abc" 20 20 4
```

abc

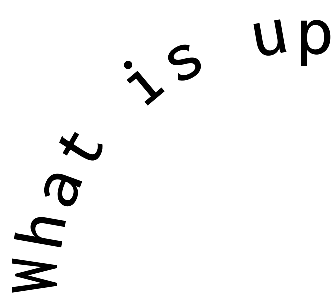
```
etext "abc" 80 20 7 "mono" "maroon"
```


```
rtext "..." x y angle fontsize font color op link
```

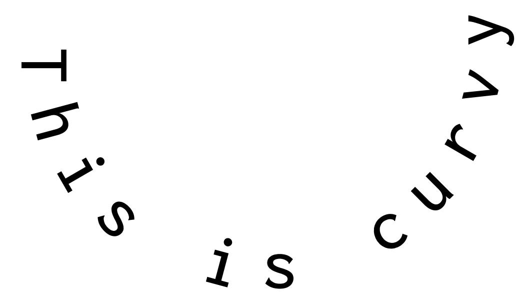

```
rtext 20 20 30 3        rtext 50 20 90 5        rtext 80 20 270 4 "sans" "maroon"
```



arctext "..." x y radius a1 a2 fontsize font color op link



arctext "What is up" 25 20 10 180 90 3 "mono"



arctext "This is curvy" 75 30 10 180 360 3 "mono" w

(x,y) “Where justice is denied, where poverty is enforced,
where ignorance prevails, and where any one class
is made to feel that society is an organized conspiracy
to oppress, rob and degrade them, neither persons
nor property will be safe.”

```
textblock "..." x y w fontsize font color op link
```

“Where justice is denied, where poverty is enforced,
where ignorance prevails, and where any one class
is made to feel that society is an organized conspiracy
to oppress, rob and degrade them, neither persons
nor property will be safe.”

```
textblock "..." 10 35 30 2
```

“Where justice is denied,
where poverty is enforced,
where ignorance prevails,
and where any one class is
made to feel that society
is an organized conspiracy
to oppress, rob and degrade
them, neither persons nor
property will be safe.”

```
textblock "..." 50 35 10 1 "sans" "maroon"
```

w

(x,y) We must rapidly begin...we must rapidly begin the shift from a thing-oriented society to a person-oriented society.

When machines and computers, profit motives and property rights, are considered more important than people, the giant triplets of racism, extreme materialism, and militarism are incapable of being conquered.

A true revolution of values will soon cause us to question the fairness and justice of many of our past and present policies. On the one hand, we are called to play the Good Samaritan on life's roadside, but that will be only an initial act.

```
para x y w fontsize font color op link ... epara
```

For, lo, the winter is passed, the rain is over and gone.

The flowers appear on the earth, the time for the singing of birds is come, And the voice of the turtle is heard on the land.

```
para 5 35 20 1.8 2.5 "serif" "gray"
    For, lo, the winter is passed, the rain is over and gone.

    The flowers appear on the earth, the time
    for the singing of birds is come,
    And the voice of the turtle is heard on the land.

epara
```

(x,y) This is the contents
of a file. it has lines of text.
Reading is fundamental.

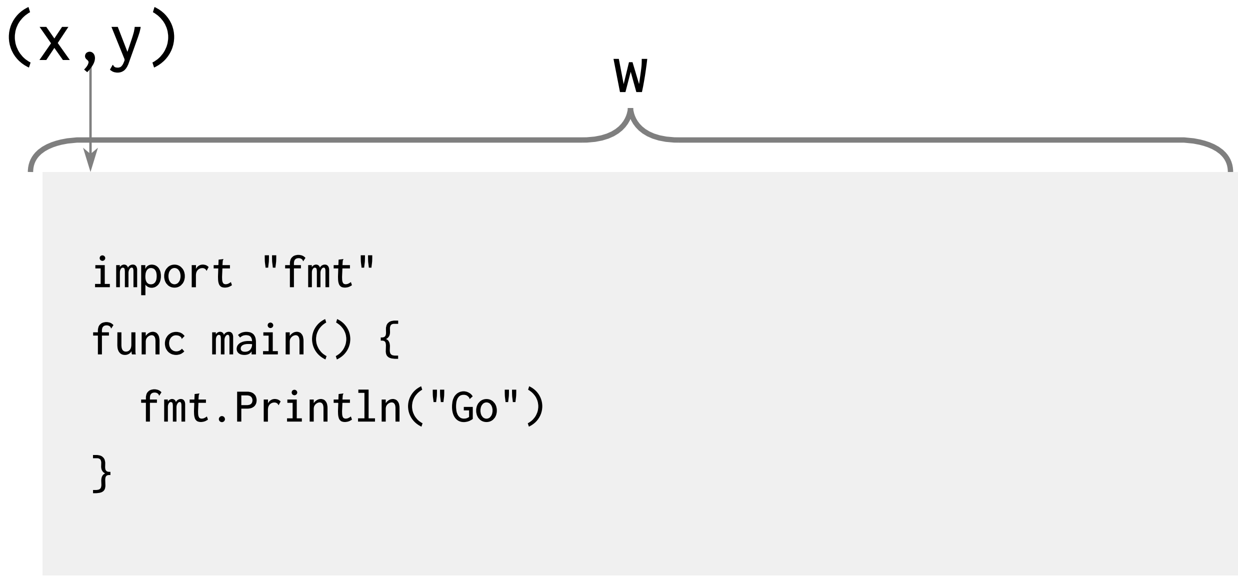
```
textfile "..." x y fontsize font color op link
```

This is the contents
of a file. it has lines of text.
Reading is fundamental.

```
import "fmt"
func main() {
  fmt.Println("Go")
}
```

```
textfile "example.txt" 10 35 2
```

```
textfile "hw-go" 55 35 1.6 "mono" "maroon"
```



textcode "..." x y w fontsize font color

```
import "fmt"
func main() {
  fmt.Println("Go")
}
```

textcode "hw-go" 10 35 25 1.0

```
import "fmt"
func main() {
  fmt.Println("Go")
}
```

textcode "hw-go" 55 35 40 1.6 "maroon"

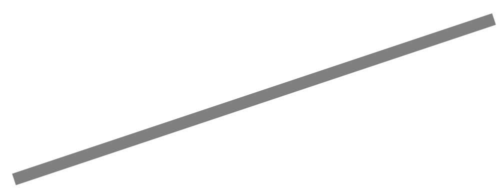
Graphical Elements

description	keyword	mandatory	optional
Line	`line`	`x1 y1 x2 y2 lw`	`color op`
Dotted Line	`dline`	`x1 y1 x2 y2`	`lw gap color op`
Horizontal line	`hline`	`x y w`	`lw color op`
Vertical line	`vline`	`x y h`	`lw color op`
Elliptical arc	`arc`	`x y w h a1 a2`	`lw color op`
Quadratic Bezier	`curve`	`bx by cx cy ex ey`	`lw color op`
Circle	`circle`	`x y w`	`color op`
Area circle	`acircle`	`x y area`	`color op`
Ellipse	`ellipse`	`x y w h`	`color op`
Square	`square`	`x y w`	`color op`
Rectangle	`rect`	`x y w h`	`color op`
Rounded rectangle	`rrect`	`x y w h radius`	`color`
Pill shape	`pill`	`x y w h`	`color`
Polygon	`polygon`	`"x1 x2...xn" "y1 y2...yn"`	`color op`
Polyline	`polyline`	`"x1 x2...xn" "y1 y2...yn"`	`lw color op`
N-sided star	`star`	`x y sides inner outer`	`color op`

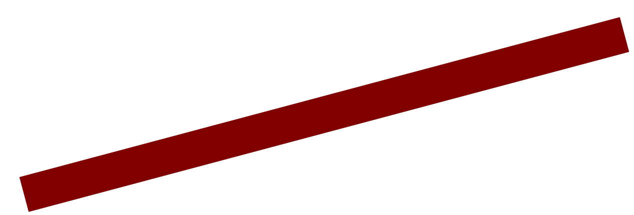

```
line x1 y1 x2 y2 lw color op
```



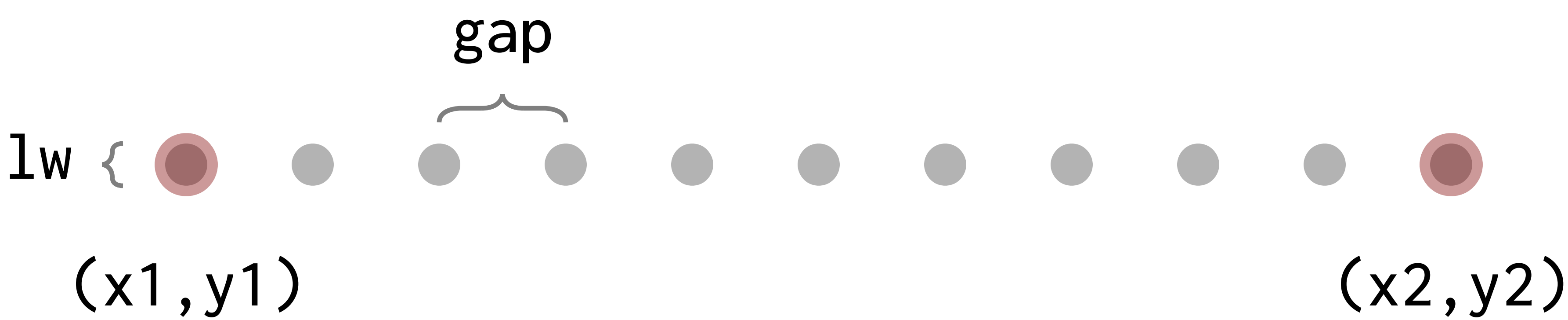
```
line 10 20 30 20
```



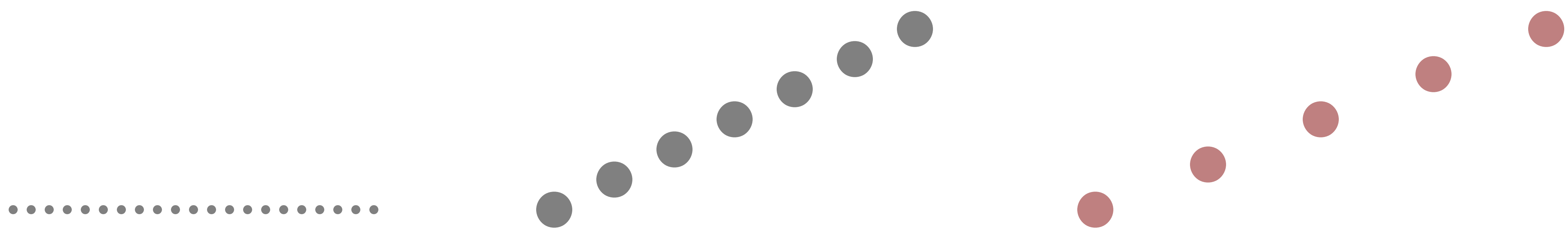
```
line 40 20 60 30 0.5
```



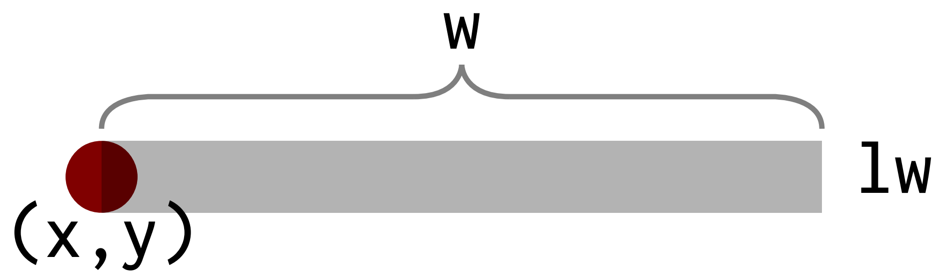
```
line 70 20 95 30 1.5 "maroon"
```




```
dline x1 y1 x2 y2 lw gap color op
```



`dline 10 20 30 20`   `dline 40 20 60 35 2 4`   `dline 70 20 95 35 2 6 "maroon" 50`

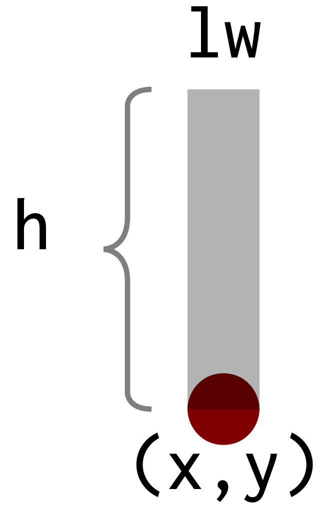


hline x y w lw color op

hline 15 20 10

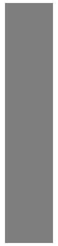
hline 40 20 20 1

hline 70 20 20 5 "maroon" 20



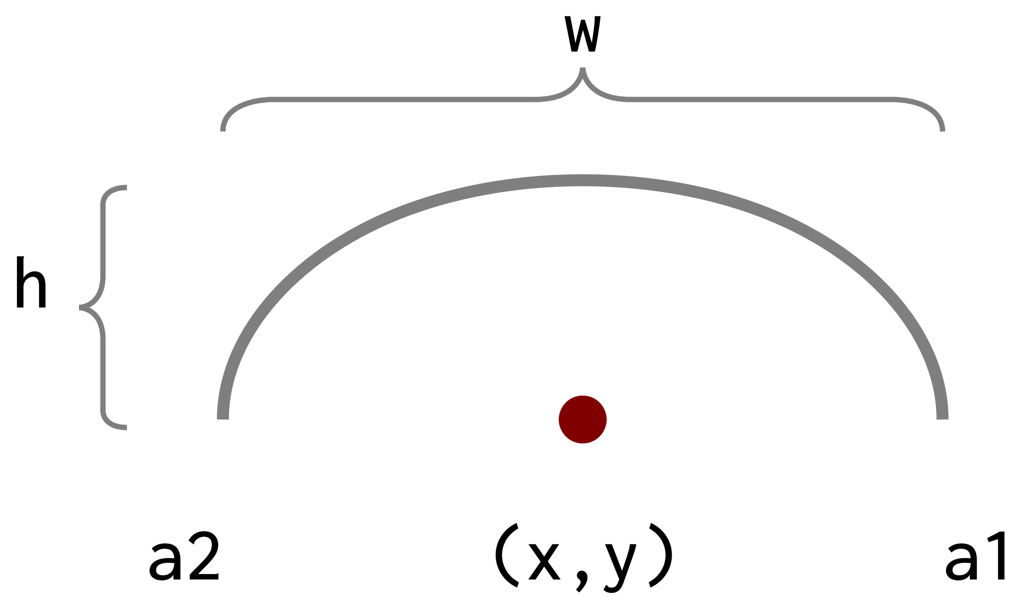
vline x y w lw color op

vline 20 20 15

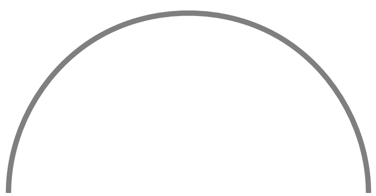


vline 50 20 15 2

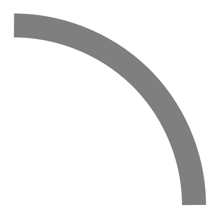
vline 80 20 15 10 "maroon" 20



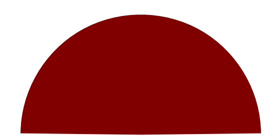
`arc x y w h a1 a2 lw color op`



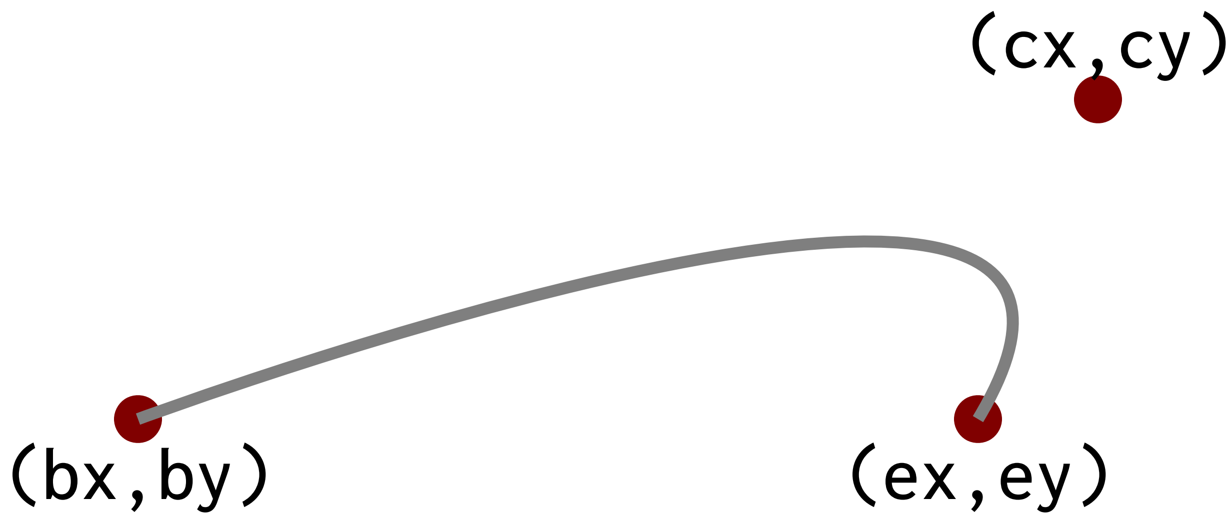
`arc 20 20 15 15 0 180`



`arc 50 20 15 15 0 90 1`



`arc 80 20 5 5 0 180 5 "maroon"`

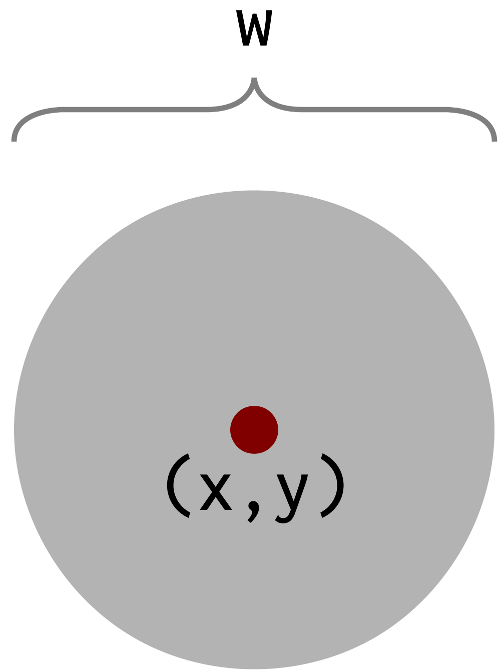



```
curve bx by cx cy ex ey lw color op
```

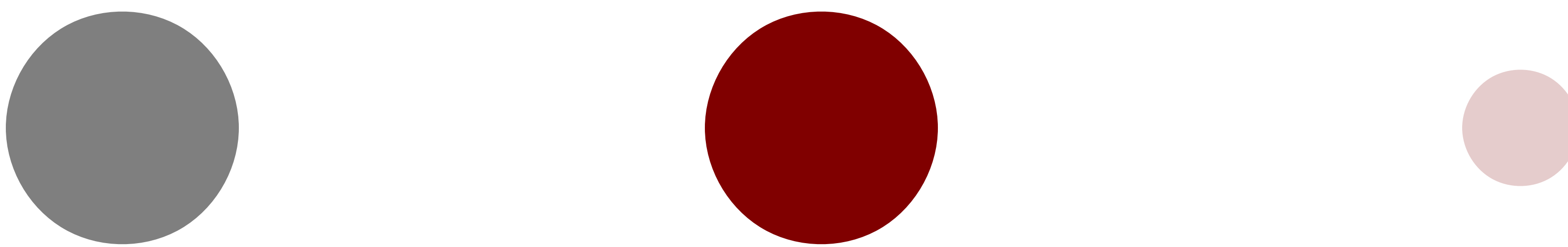
```
curve 15 20 25 30 30 25
```

```
curve 15 20 25 30 30 25
```

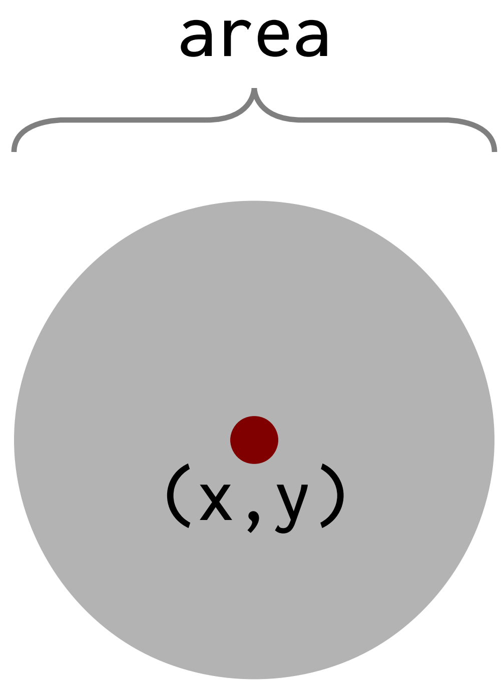
```
curve 70 20 70 30 90 25 0.5 "maroon"
```



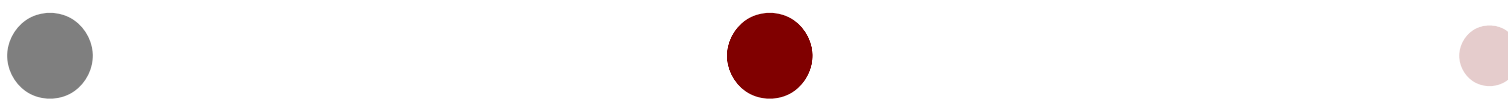
`circle x y w color op`



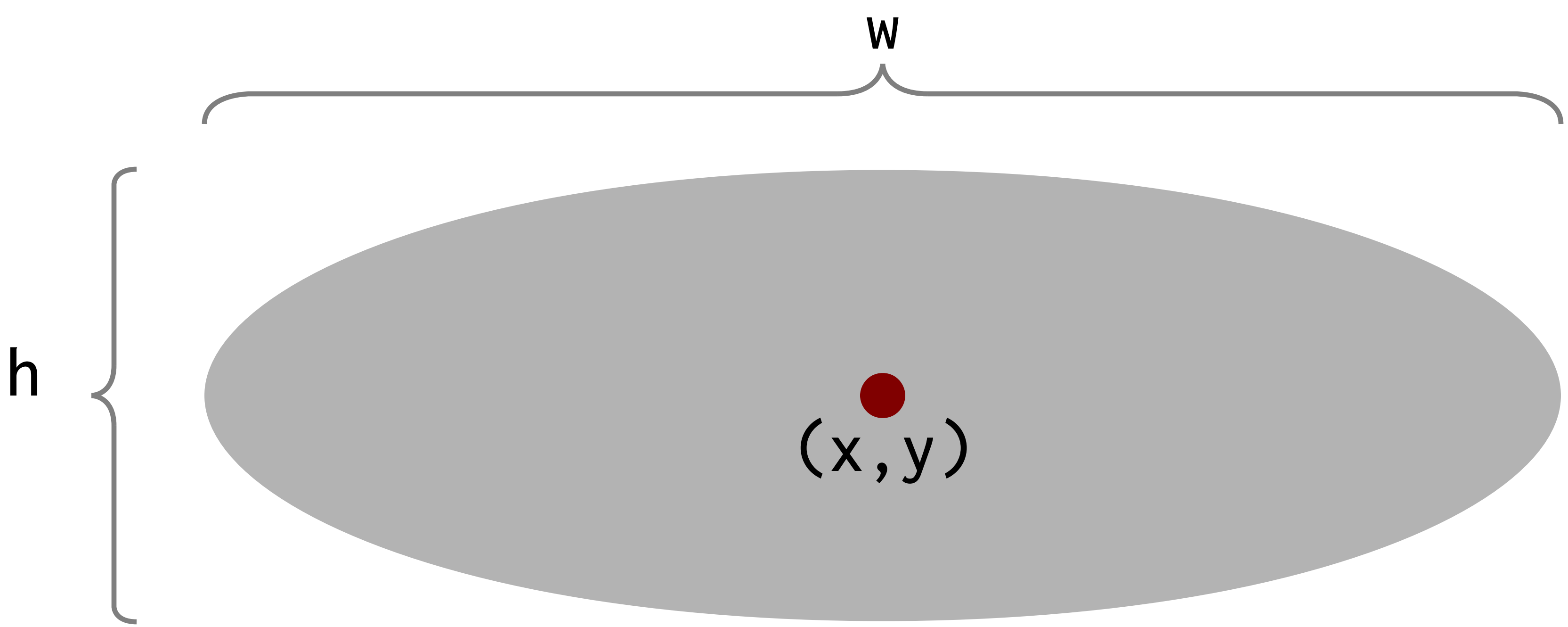
`circle 20 20 10` `circle 50 20 10 "maroon"` `circle 80 20 5 "maroon" 20`



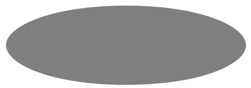
`circle x y area color op`



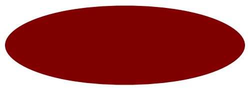
`acircle 20 20 10`   `acircle 50 20 10 "maroon"`   `acircle 80 20 5 "maroon" 20`



`ellipse x y w h color op`

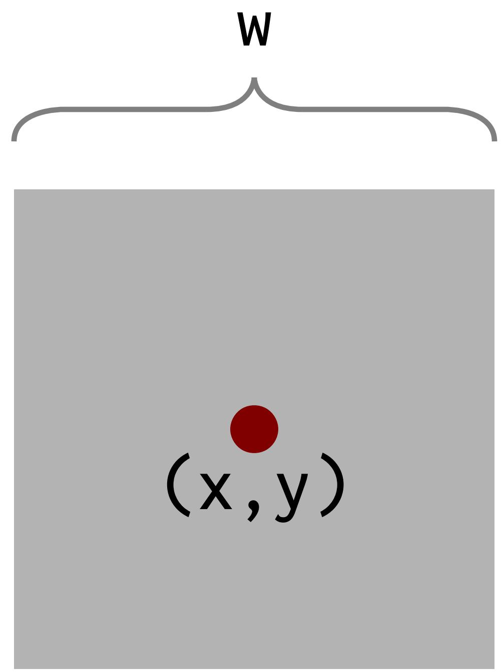


`ellipse 20 20 10 5`



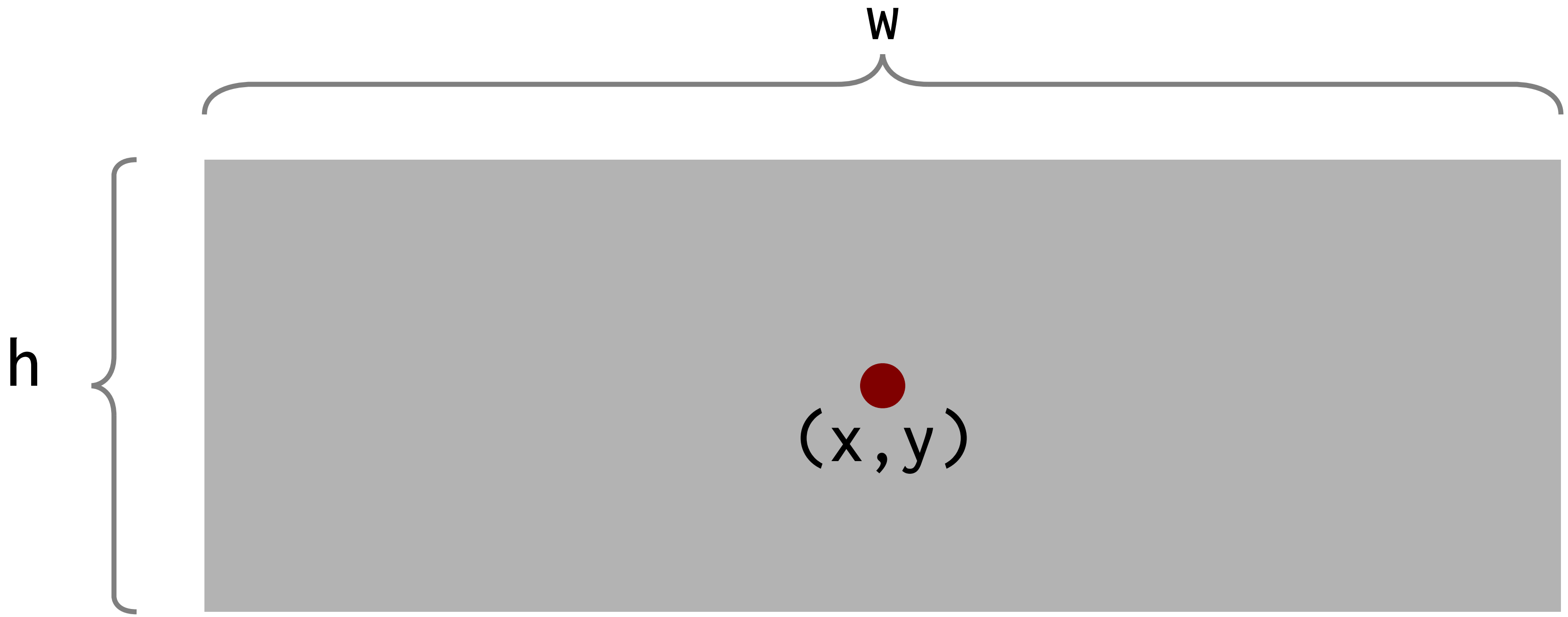
`ellipse 50 20 10 5 "maroon"`

`ellipse 80 20 5 10 "maroon" 20`




```
square x y w color op
```

```
square 20 20 10
square 50 20 10 "maroon"
square 80 20 5 "maroon" 20
```

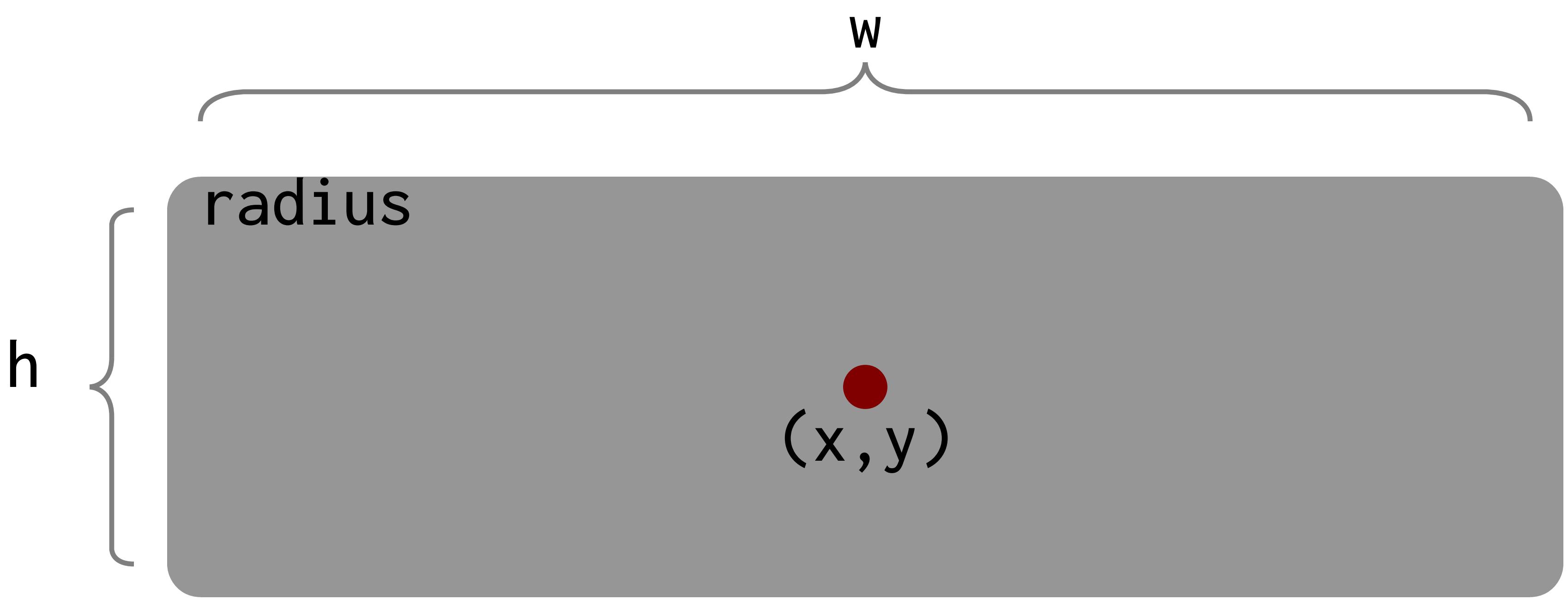



```
rect x y w h color op
```

`rect 20 20 10 5`

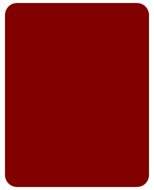
`rect 50 20 10 5 "maroon"`

`rect 80 20 5 10 "maroon" 20`

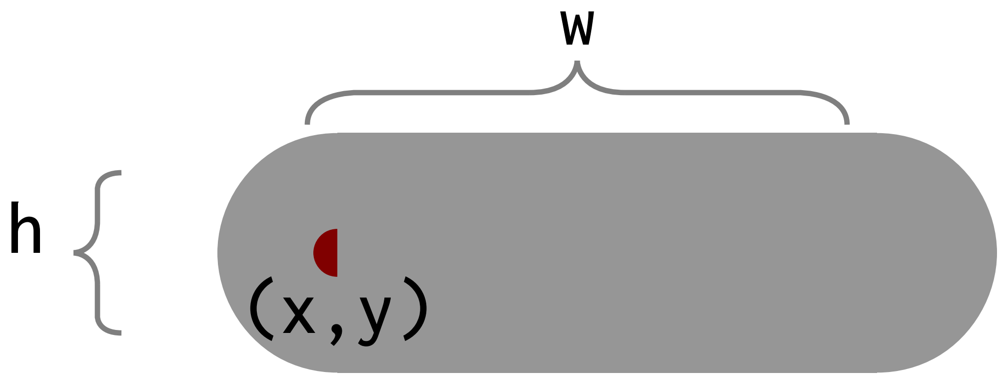



```
rrect x y w h radius color op
```

```
rrect 20 20 10 5 1
```

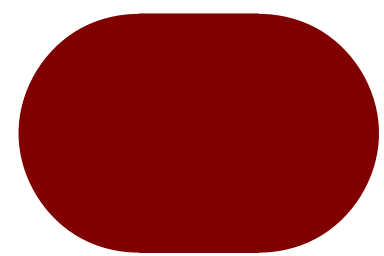


```
rrect 80 20 5 10 1 "maroon"
```

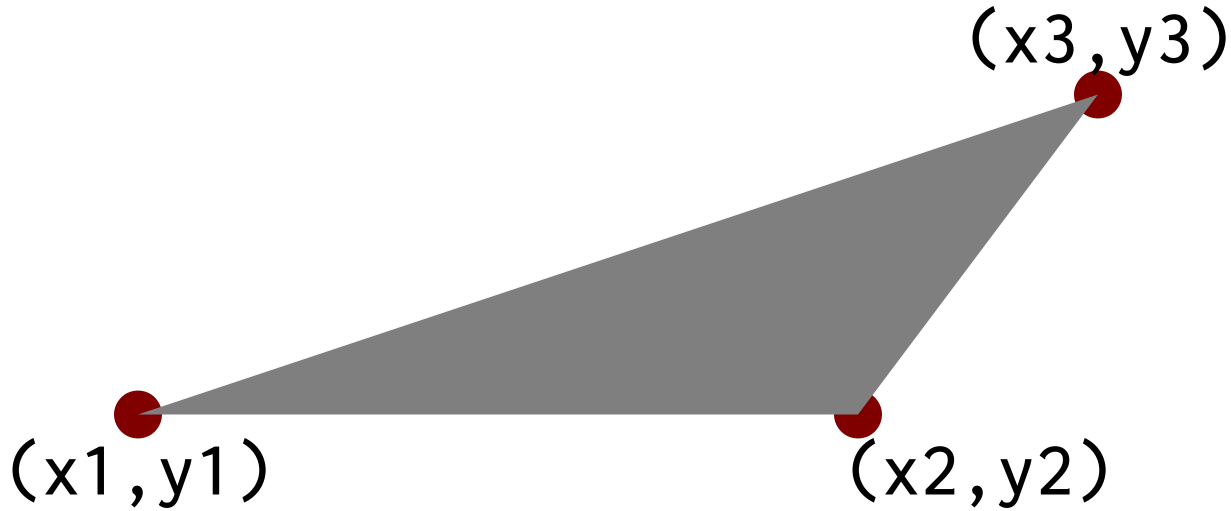


pill x y w h color

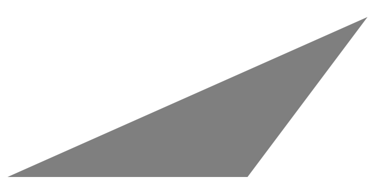
pill 20 20 10 5



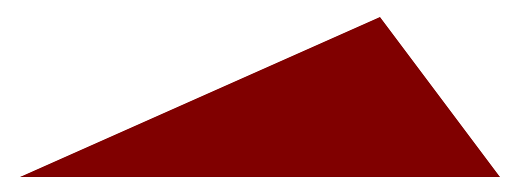
pill 80 20 5 10 "maroon"



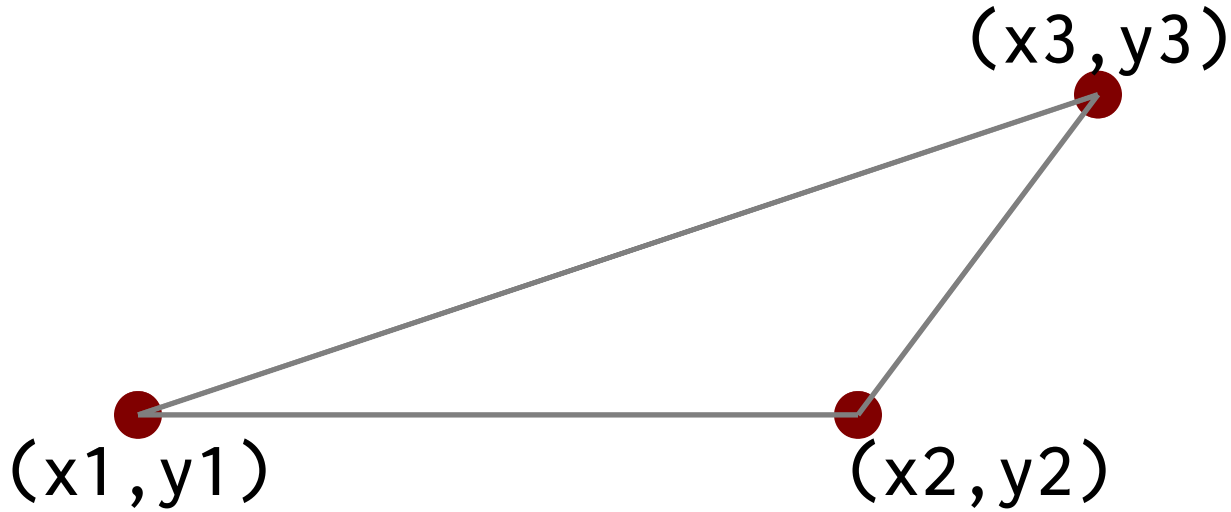
`polygon "x1 x2...xn" "y1 y2...yn" color op`



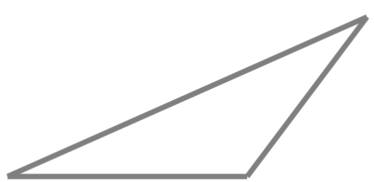
`polygon "10 25 20" "20 30 20"`



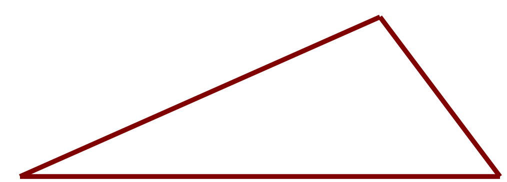
`polygon "70 85 90" "20 30 20" "maroon"`



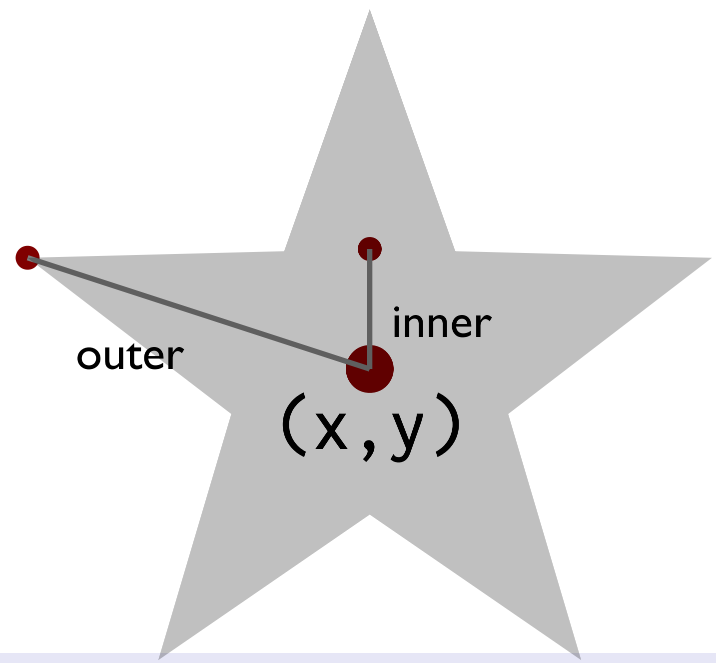

```
polyline "x1 x2...xn" "y1 y2...yn" lw color op
```



```
polyline "10 25 20" "20 30 20"
```



```
polyline "70 85 90" "20 30 20" 0.2 "maroon"
```



`star x y sides inner outer color op`

`star 20 20 5 2 6`

`star 50 20 12 2 5 "maroon"`

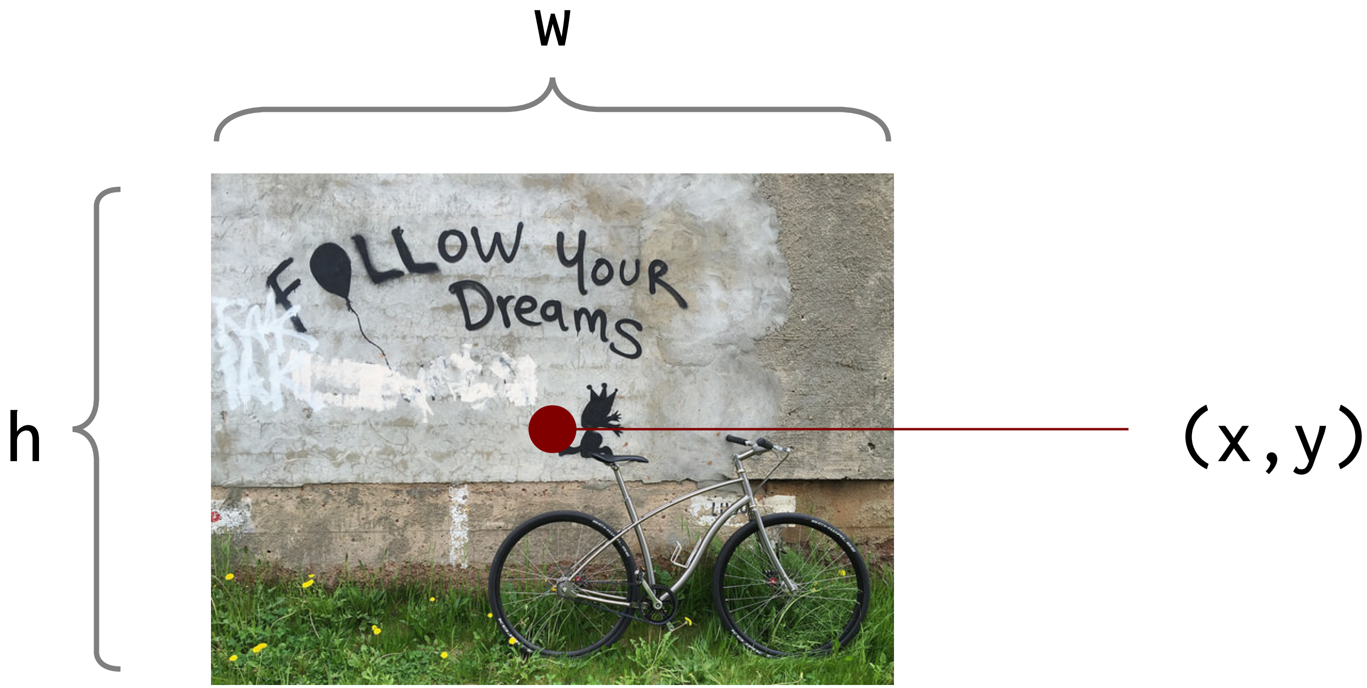
`star 80 ey 24 2 8 "maroon" 20`

Images

description	keyword	mandatory	optional
Image	`image`	`"file" x y w h`	`scale "link"`
Captioned image	`cimage`	`"file" "caption" x y w h`	`scale "link" capsize`

If h = 0, w specifies the image width in terms of canvas width.

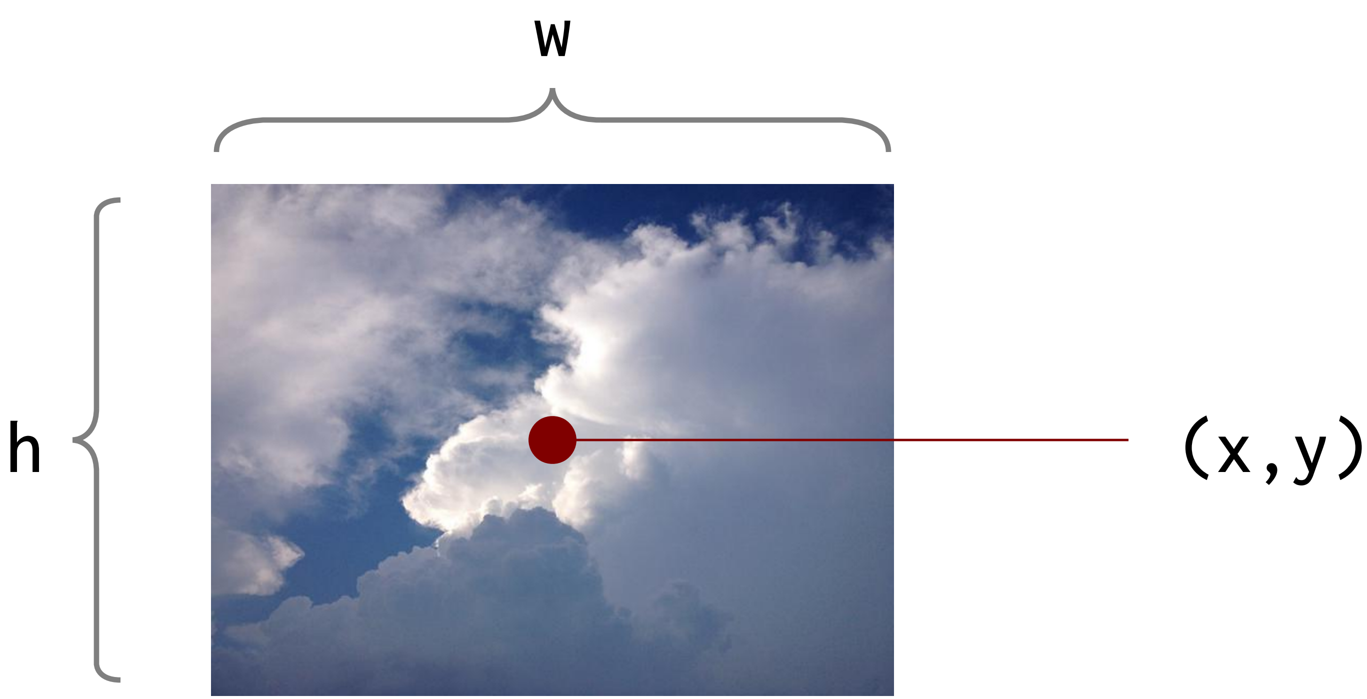
The scale value is a percentage from 1-100, and link is a URL. capsize is the text size of the caption




```
image "file" x y w h scale link
```

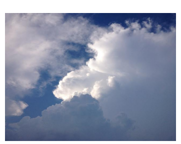
`image "follow.jpg" 20 25 640 480 10`

`image "follow.jpg" 75 25 640 480 30`



sky

```
cimage "file" x y w h scale link
```



sky

sky

```
cimage "cloudy.jpg" "sky" 20 25 640 480 10
```

```
cimage "cloudy.jpg" "sky" 75 25 640 480 30 "" 1.5
```

Lists

description	keyword	mandatory	optional
Plain list	`list`	`x y fontsize`	`font color op spacing`
Bullet list	`blist`	`x y fontsize`	`font color op spacing`
Numbered list	`nlist`	`x y fontsize`	`font color op spacing`
Centered list	`clist`	`x y fontsize`	`font color op spacing`

(x,y)

```
list x y fs
    li "first"
    li "second"
    li "third"
elist
```

list x y fontsize font color op spacing

```
list 20 30 2.5
    li "one"
    li "two"
    li "three"
elist
```

one

two

three

```
list 85 30 2.5 "serif" "maroon" 100 1.0
    li "one"
    li "two"
    li "three"
elist
```

one
two
three

(x,y)

```
blist x y fs
    li "first"
    li "second"
    li "third"
elist
```

blist x y fontsize font color op spacing

```
blist 20 30 2.5
    li "one"
    li "two"
    li "three"
elist
```

- one
- two
- three

```
blist 85 30 2.5 "serif" "maroon" 100 1.0
    li "one"
    li "two"
    li "three"
elist
```

- one
- two
- three

```
  nlist x y fs
(x,y) li "first"
      li "second"
      li "third"
  elist
```

nlist x y fontsize font color op spacing

```
nlist 20 30 2.5
    li "one"
    li "two"
    li "three"
elist
```

1. one
2. two
3. three

```
nlist 85 30 2.5 "serif" "maroon" 100 1.0
    li "one"
    li "two"
    li "three"
elist
```

1. one
2. two
3. three

```
clist x y fs
  li "first"
  li "second"
  li "third"
elist
```

(x,y)

`clist x y fontsize font color op spacing`

```
clist 30 30 2.5
    li "first one"
    li "next"
    li "and last"
elist
```

first one

next

and last

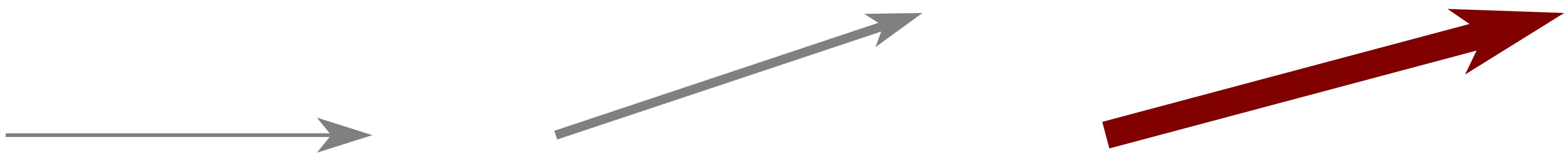
```
clist 90 30 2.5 "serif" "maroon" 100 1.0
    li "first"
    li "next"
    li "and last"
elist
```

first
next
and last

Arrows

description	keyword	mandatory	optional
Straight	`arrow`	`x1 y1 x2 y2`	`lw aw ah color op`
Left curved	`lcarrow`	`bx by cx cy ex ey`	`lw aw ah color op`
Right curved	`rcarrow`	`bx by cx cy ex ey`	`lw aw ah color op`
Up curved	`ucarrow`	`bx by cx cy ex ey`	`lw aw ah color op`
Down curved	`dcarrow`	`bx by cx cy ex ey`	`lw aw ah color op`

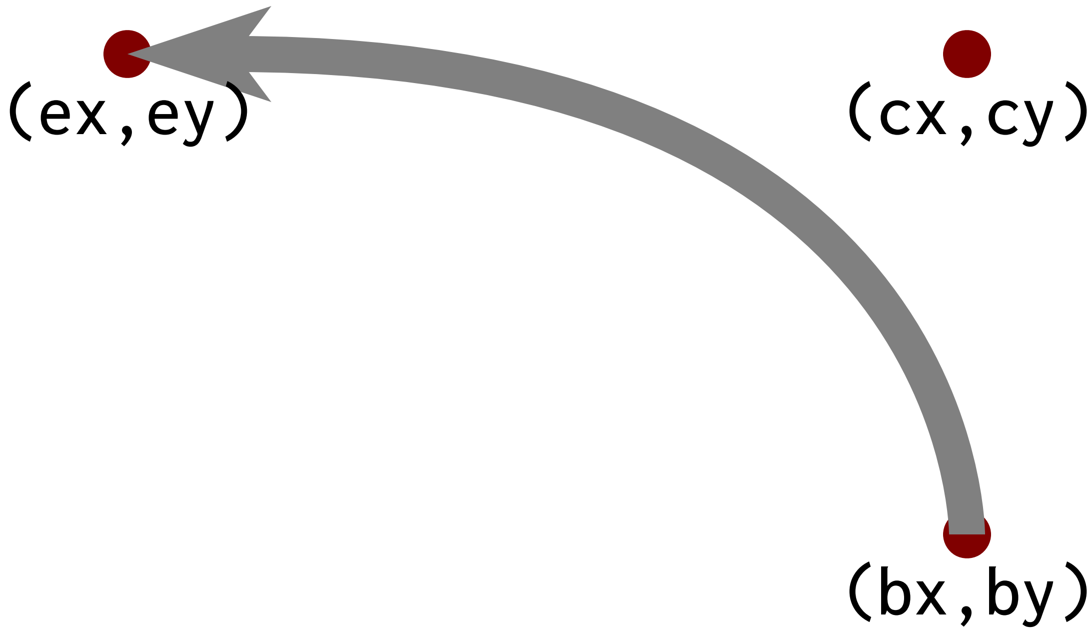
`arrow x1 y1 x2 y2 lw aw ah color op`



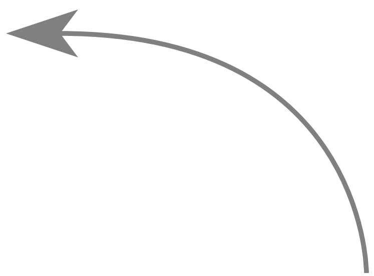
`arrow 10 20 30 20`

`arrow 40 20 60 30 0.5`

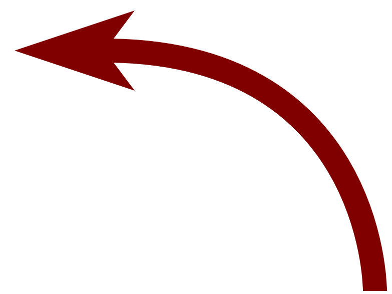
`arrow 70 20 95 30 1.5 6 6 "maroon"`



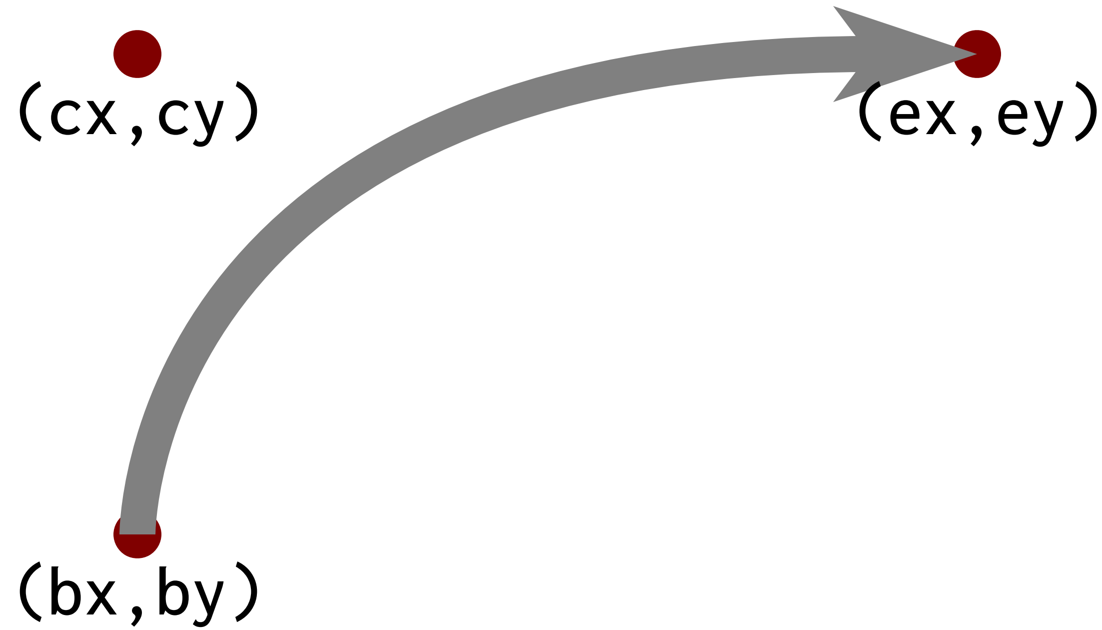

```
lcarrow bx by cx cy ex ey lw aw ah color op
```



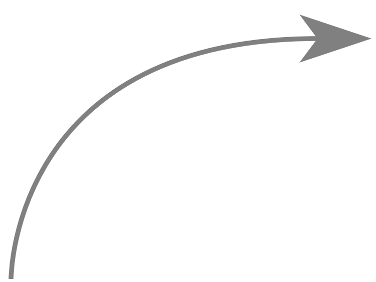
```
lcarrow 30 20 30 35 15 35
```

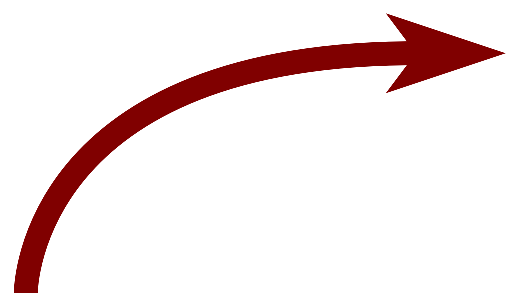


```
lcarrow 70 20 70 35 55 35 1 5 5 "maroon"
```



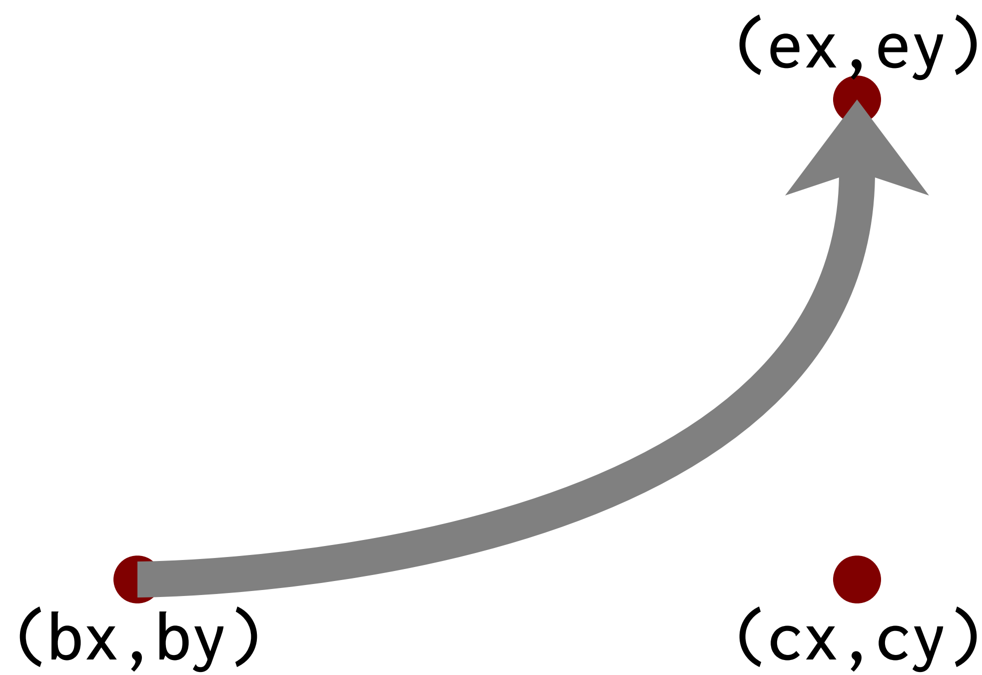

```
rcarrow bx by cx cy ex ey lw aw ah color op
```



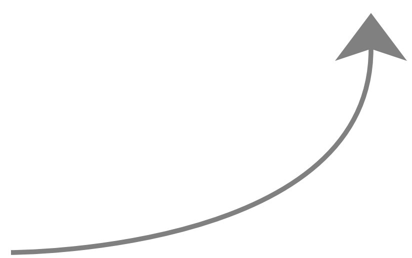


```
rcarrow 15 20 15 35 30 35
```

```
rcarrow 50 20 50 35 70 35 1 5 5 "maroon"
```

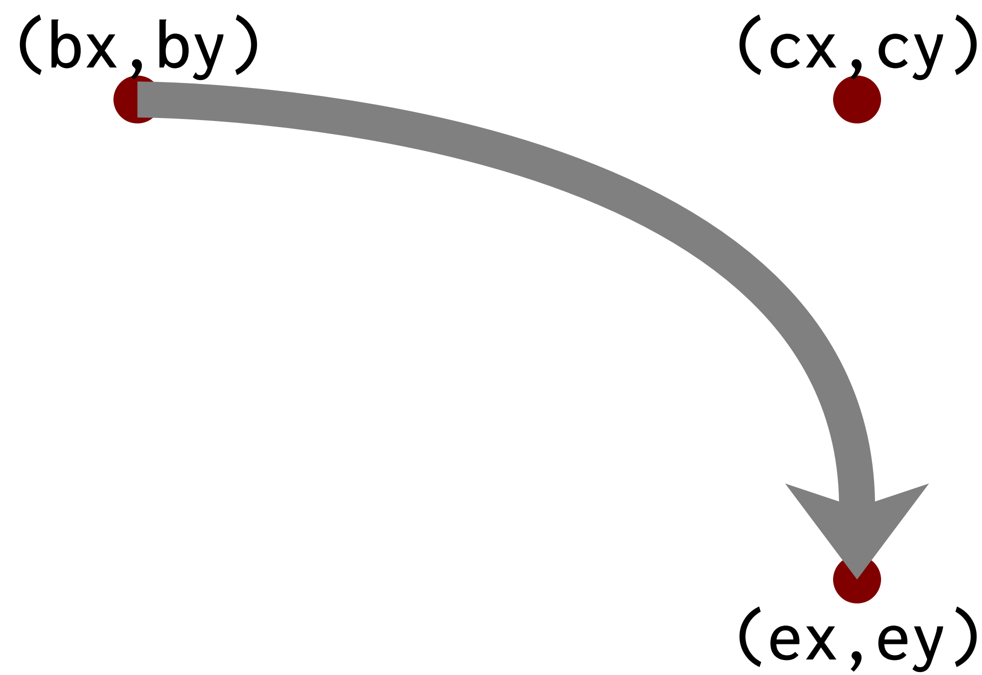



```
ucarrow  bx by cx cy ex ey lw aw ah color op
```



```
ucarrow 15 20 30 20 30 35
```

```
rcarrow 50 20 70 20 70 35 1 5 5 "maroon"
```



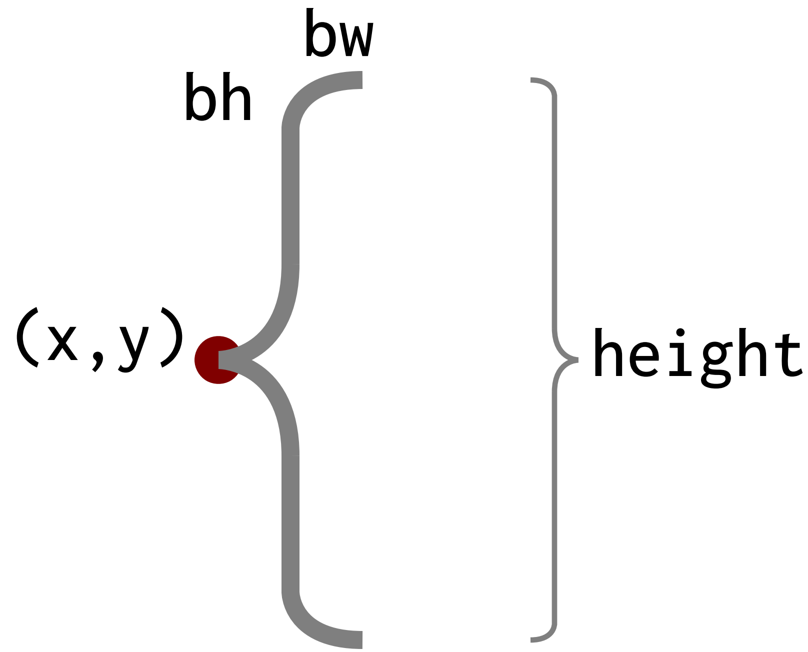
dcarrow bx by cx cy ex ey lw aw ah color op

`dcarrow 15 35 30 30 20`

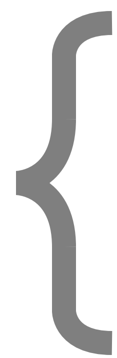
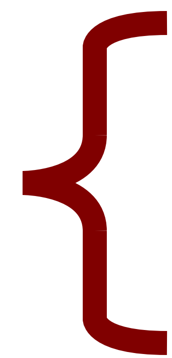
`dcarrow 50 35 70 35 70 20 1 5 5 "maroon"`

Braces and Brackets

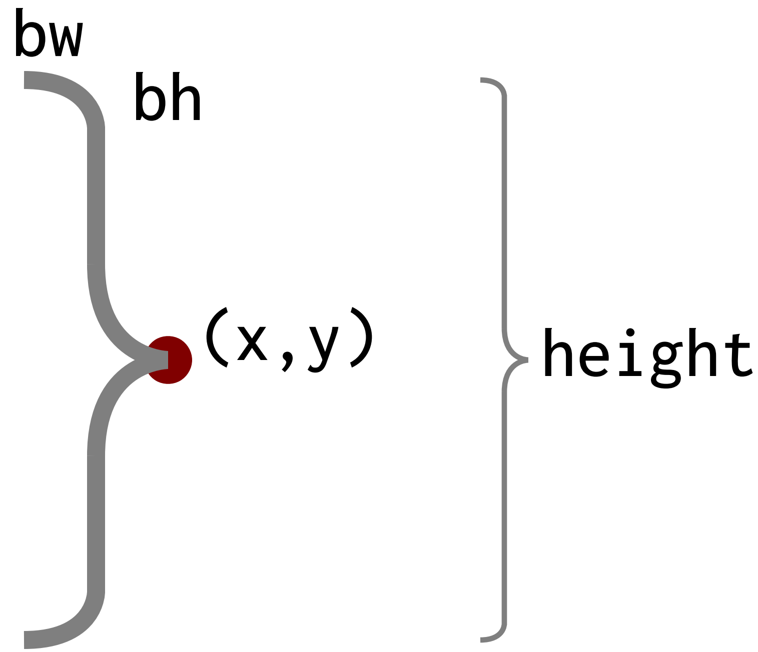
description	keyword	mandatory	optional
Left brace	`lbrace`	`x y height bw bh`	`lw color op`
Right brace	`rbrace`	`x y height bw bh`	`lw color op`
Up brace	`ubrace`	`x y width bw bh`	`lw color op`
Down brace	`dbrace`	`x y width bw bh`	`lw color op`
Left bracket	`lbracket`	`x y width height`	`lw color op`
Right bracket	`rbracket`	`x y width height`	`lw color op`
Up bracket	`ubracket`	`x y width height`	`lw color op`
Down bracket	`dbracket`	`x y width height`	`lw color op`




```
lbrace  x y height bw bh  lw color op
```

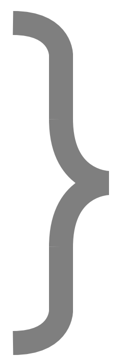




```
lbrace 20 25 20 2 2        lbrace 50 25 20 4 4 1     lbrace 80 25 20 6 3 1 "maroon"
```

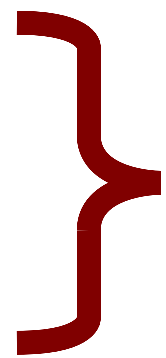



```
rbrace   x y height bw bh   lw color op
```

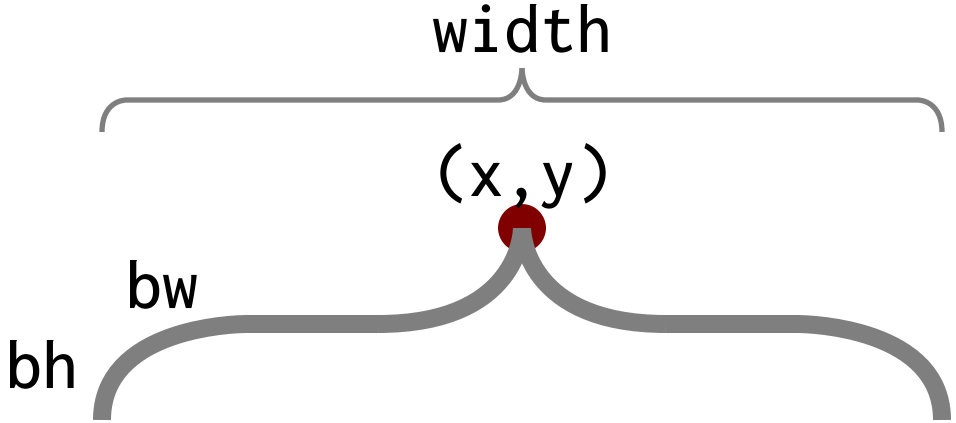
`rbrace 20 25 20 2 2`



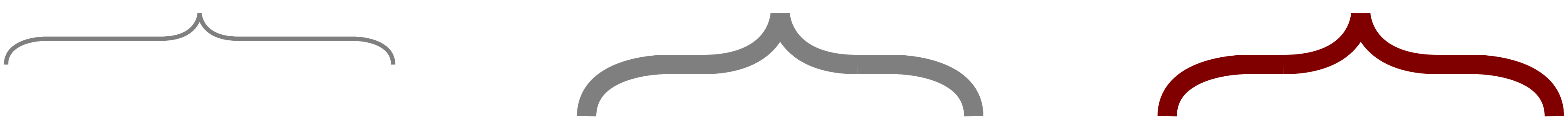
`rbrace 50 25 20 4 4 1`



`rbrace 80 25 20 6 3 1 "maroon"`



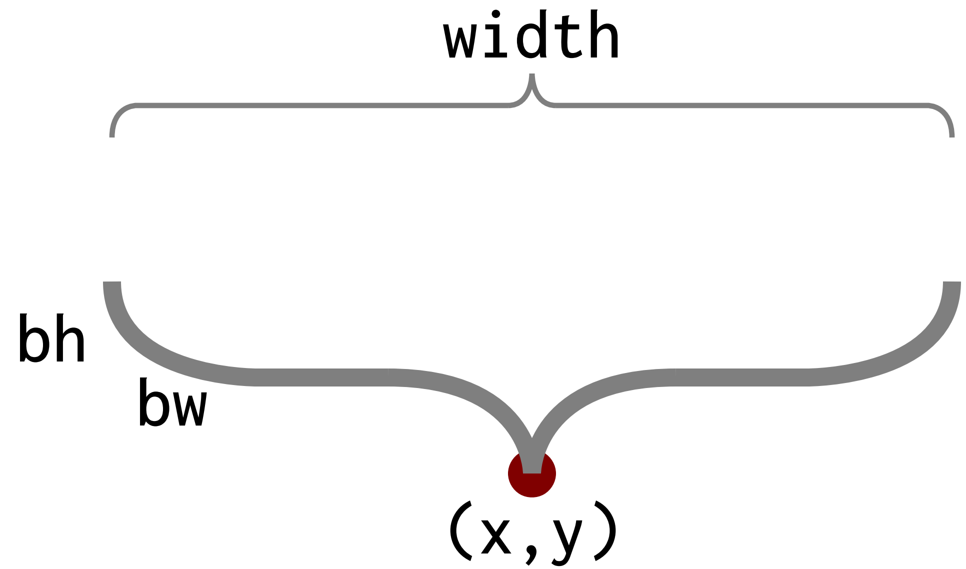

```
ubrace   x y width bw bh   lw color op
```



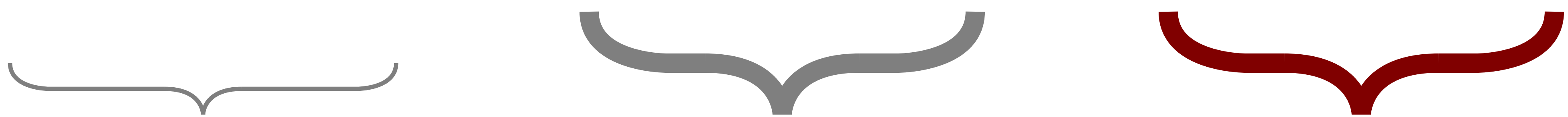
```
ubrace 20 25 20 2 4
```

```
ubrace 50 25 20 4 8 1
```

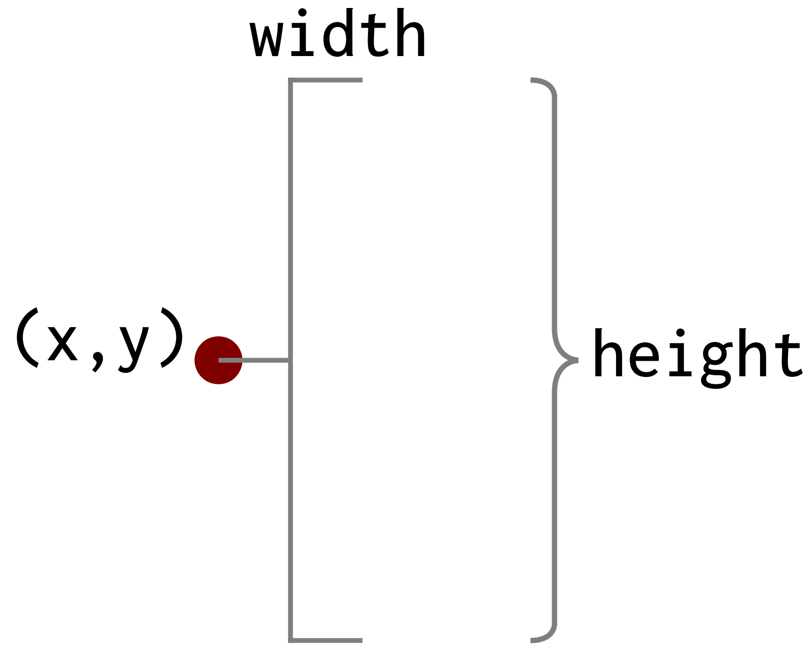
```
ubrace 80 25 20 4 8 1 "maroon"
```



`dbrace x y width bw bh lw color op`

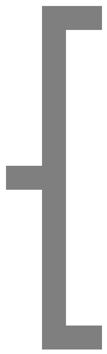


`dbrace 20 25 20 2 4`   `dbrace 50 25 20 4 8 1`   `dbrace 80 25 20 4 8 1 "maroon"`

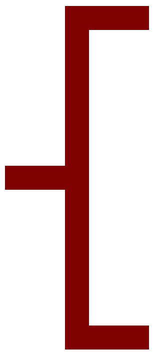


lbracket x y width height lw color op

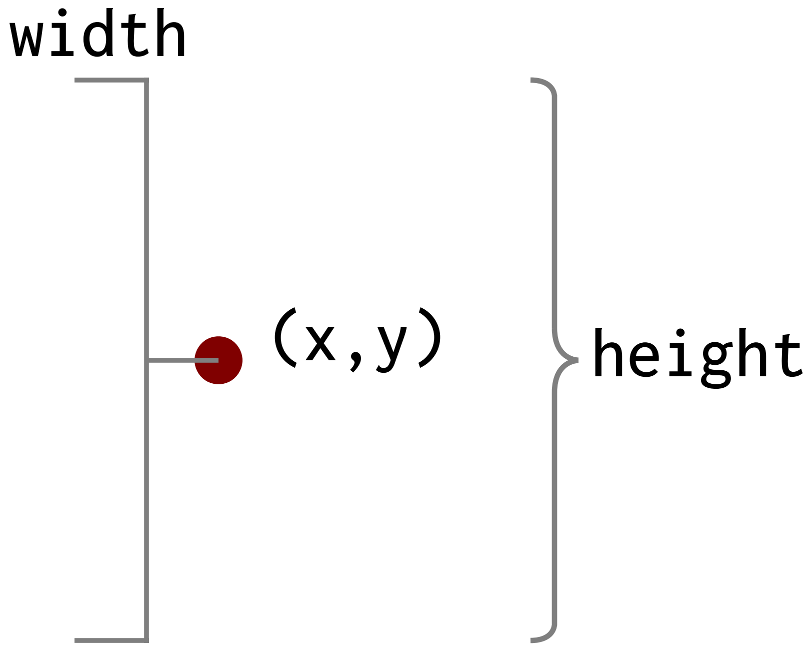
lbrace 20 25 2 20



lbracket 50 25 4 20 1

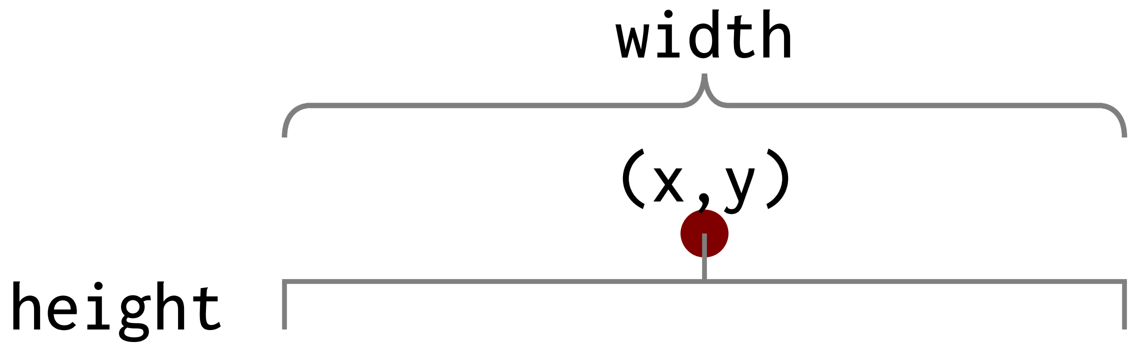


lbracket 80 25 6 20 1 "maroon"



rbracket x y width height lw color op

lbrace 20 25 2 20   rbracket 50 25 4 20 1   rbracket 80 25 6 20 1 "maroon"

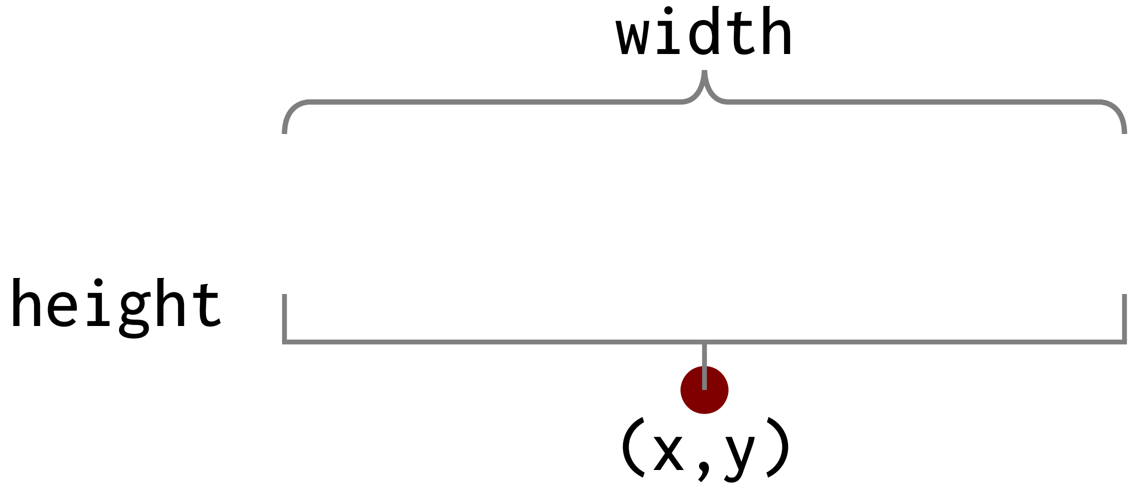


ubracket x y width height lw color op

ubracket 20 25 20 4

ubracket 50 25 20 4 0.3

ubracket 80 25 20 4 1 "maroon"




```
dbracket x y width height lw color op
```

```
dbracket 20 25 20 4
dbracket 50 25 20 4 0.3
dbracket 80 25 20 4 1 "maroon"
```

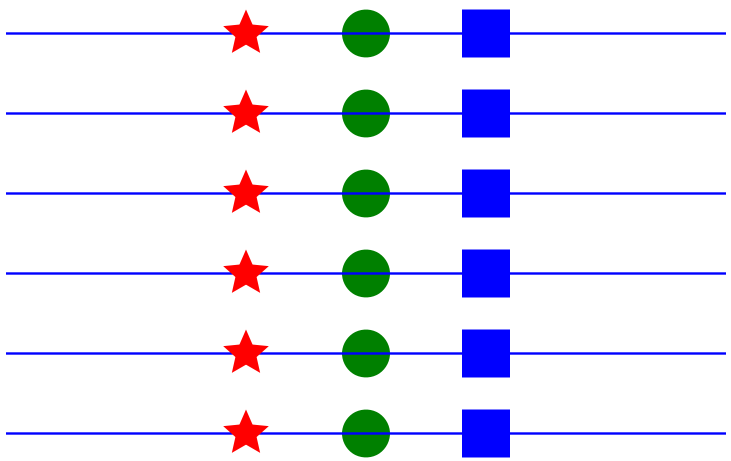
Loop, If, Built-ins

description	keyword	arguments
Loop	`for v=`	`begin end [increment] ... efor`
Conditional	`if`	`condition ... [else] ... eif`
Polar coordinate (x)	`x=polarx`	`x y radius angle`
Polar coordinate (y)	`y=polary`	`x y radius angle`
Polar coordinates	`value=polar`	`x y radius angle`
Geo coordinates	`value=geocoord`	`geo: URL or lat long pair`
Area	`value=area`	`expression`
Formatted text	`value=format`	`fmt expression or up to 5 args`
Substring	`value=substr`	`string begin end`
Random number	`value=random`	`min max`
Value mapping	`value=vmap`	`data min1 max1 min2 max2`
Define function	`def`	`name arg1 ... argn ... edef`
Import function	`import`	`"file"`
In-line data	`data`	`"file" ["plain"] ... edata`
Objects on a grid	`grid`	`"file" x y hspace vspace edge`
Rulers	`ruler/georuler`	`increment [color]`

```
for v=begin end [increment]
...items to repeat using v
efor
```

for v=begin end increment ...efor

```
for v=10 35 5
    hline  50 v 30 0.1  "blue"
    star   60 v 5 1 0.4 "red"
    circle 65 v 2 "green"
    square 70 v 2 "blue"
efor
```



```
if condition
...statements when true
else
...statements when false
eif
```

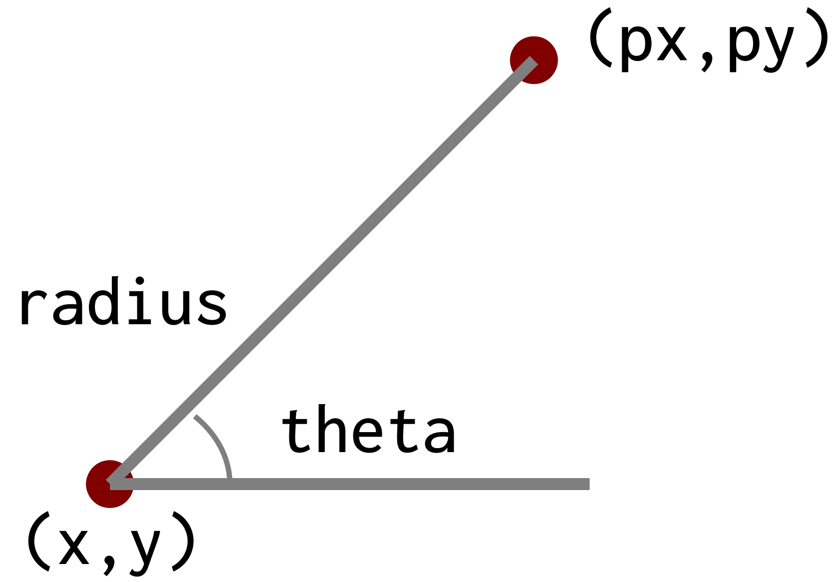
else block
may be omitted

```
if condition ... else ... eif
```

Condition	Description
`if a == b`	if a equals b
`if a != b`	if a not equal to b
`if a > b`	if a less than b
`if a < b`	if a greater than b
`if a >= b`	if a greater than or equal to b
`if a <= b`	if a less than or equal to b
`if a >< b c`	if a is between b and c

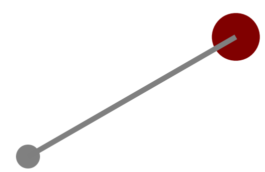
```
a=60
b=3
if a > b
    text "hello" a b 2.5 "sans" "red"
else
    text "bye"   a b 2.5 "sans" "blue"
eif
```

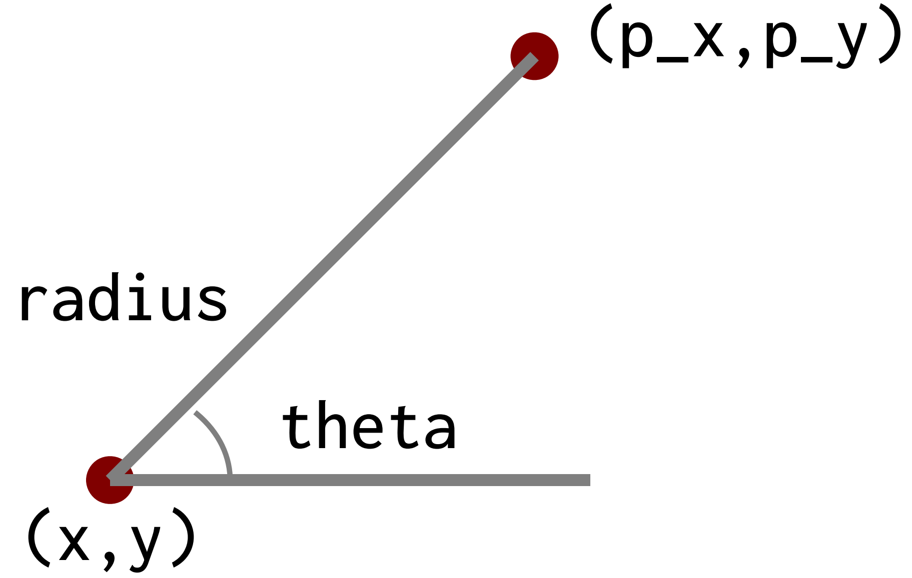
hello



px=polarx x y radius theta
py=polary x y radius theta

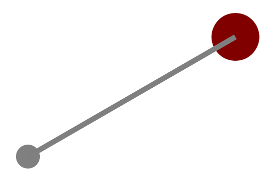
```
cpx=60
cpy=20
px1=polarx cpx cpy 10 30
py1=polary cpx cpy 10 30
line cpx cpy px1 py1
circle cpx cpy 1  "gray"
circle px1 py1 2 "maroon"
```





p=polar x y radius theta

```
cpx=60
cpy=20
point=polar cpx cpy 10 30
line cpx cpy point_x point_y
circle cpx cpy 1  "gray"
circle point_x point_y 2 "maroon"
```

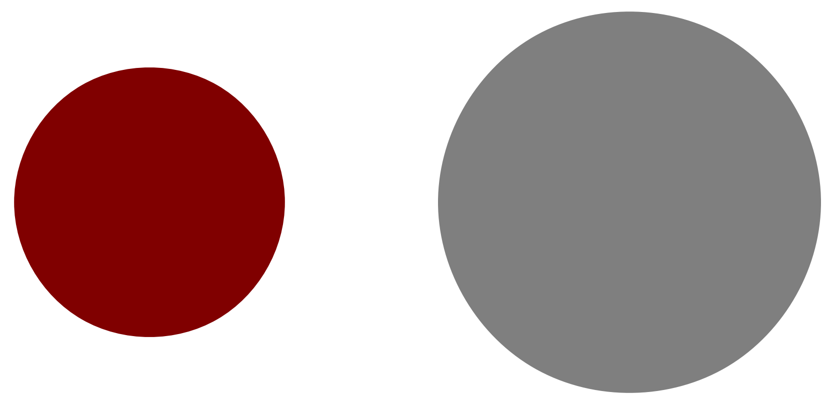


```
v=123.45
a=area v
```

area original value

```
value=area expression
```

```
m1=100
m2=200
a1=area m1
a2=area m2
circle 60 20 a1 "maroon"
circle 80 20 a2
```



```
x=3.14159
y=2.0
title=format "Value=%.2f" x*y
```

$\underbrace{\texttt{title}}_{\texttt{Value=6.28}}\texttt{=format }\underbrace{\texttt{"Value=\%.2f"}}_{\texttt{format string}}\ \underbrace{\texttt{x*y}}_{\texttt{expression}}$

```
value=format fmt expression or up to 5 args
```

```
v1=100.3
v2=200.234
pi=3.1415926
title=format "%.2f Million (USD)" v1
subtitle=format "Total value: %.2f" v1+v2
args=format "Multiple args (%v,%v,%.3f)" 80 10 pi
ctext title    80 30 4 "sans" "maroon"
ctext subtitle 80 20 3 "sans" "gray"
ctext args     80 10 2 "mono"
```

100.30 Million (USD)

Total value: 300.53

Multiple args (80,10,3.142)

```
s="hello, world"
h=substr s 3 8
```

"lo, wo" begin end

```
value=substr string begin end
```

```
now="Now is the time for all good men"
s1=substr now 0 14
s2=substr now 16 18
s3=substr now 24 end
ctext s1 70 34 3
ctext s2 70 24 3
ctext s3 70 15 3
```

Now is the time

for

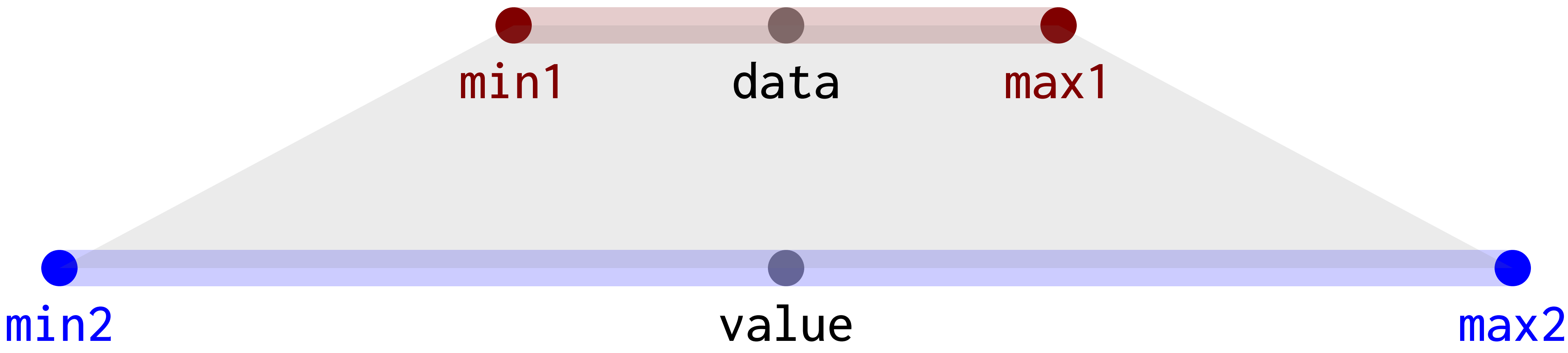
good men

value=random min max

```
rx1=random 5 30
ry1=random 15 35
circle rx1 ry1 3 "maroon"
```

```
rx2=random 40 60
ry2=random 15 35
circle rx2 ry2 3 "green"
```

```
rx1=random 75 95
ry1=random 15 35
circle rx3 ry3 3 "blue"
```



value=vmap data min1 max1 min2 max2

```
yrmin=1776
yrmax=2021
smin=60
smax=90
vp=vmap 1945 yrmin yrmax smin smax
line   smin 20 smax 20 0.5 "gray" 20
circle smin 20 1
circle smax 20 1
circle vp 20 2 "maroon"
```

import "doit.dsh"

contents of "doit.dsh"

```
def doit fx fy fs ft
    ctext ft fx fy fs "serif" "purple"
edef
```

doit 50 20 2.5 "hello"

```
fx=50
fy=20
fs=2.5
ft="hello"
ctext ft fx fy fs "serif" "purple"
```

```
import "file"
```

calling the function

call again

```
doit 50 30 5 "calling the function"
doit 50 20 4 "call again"
```

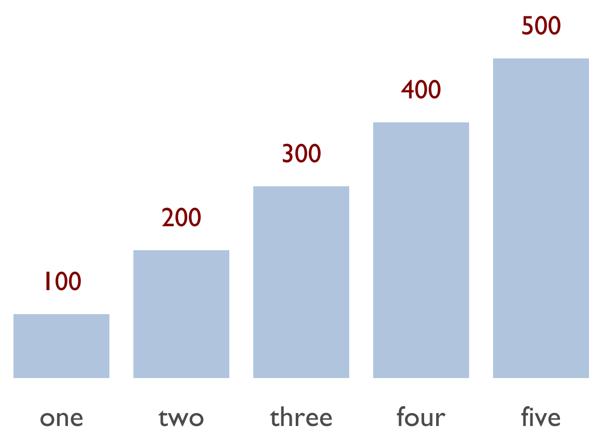
```
data "file.d"
first 20
second 100
third 200
edata
```

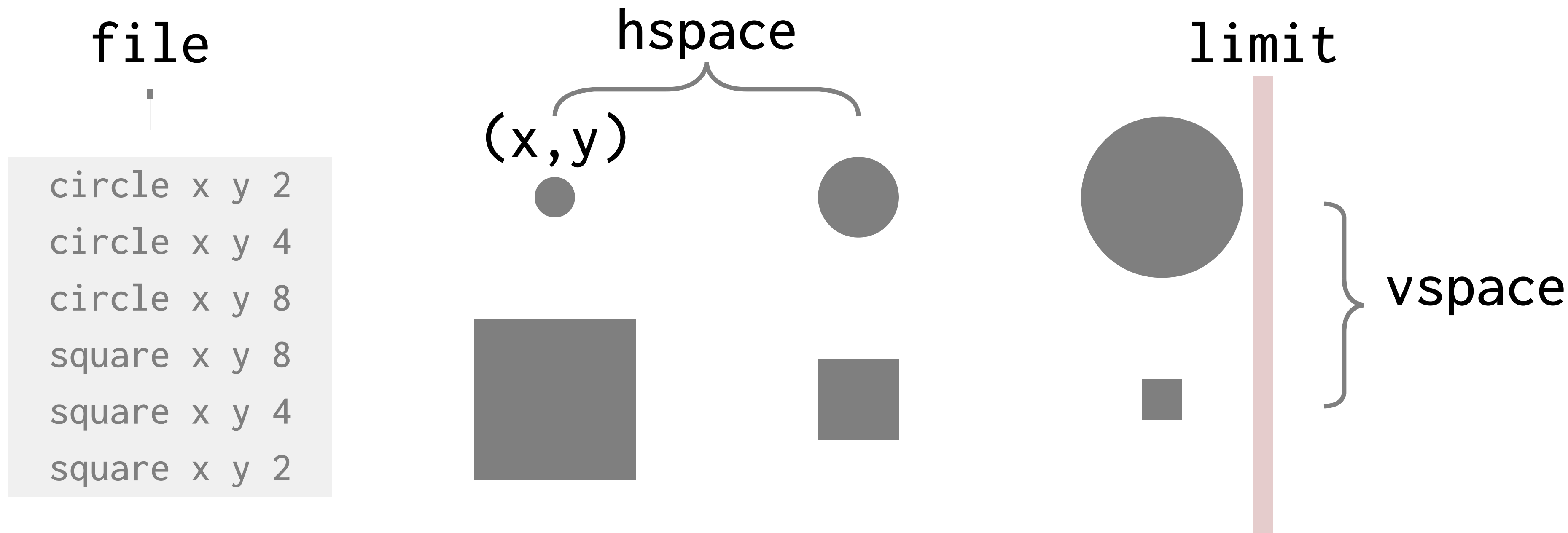
data file

data values

```
data "file" ... edata
```

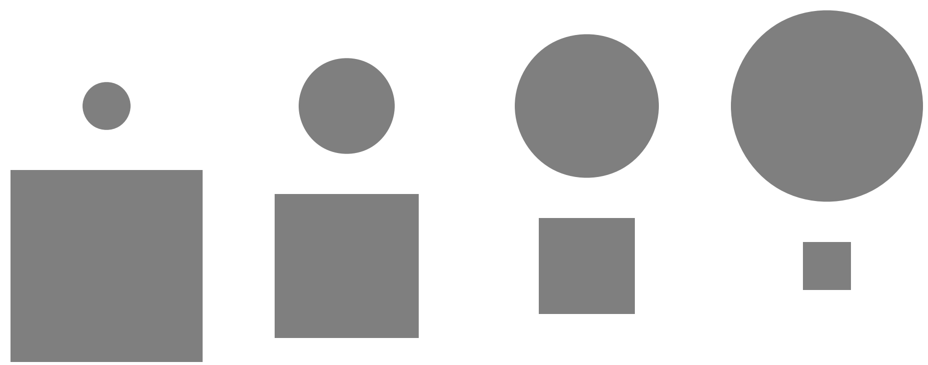
```
data "test.d"
    one   100
    two   200
    three 300
    four  400
    five  500
edata
dchart -bar -left 50 -bottom 15 -right 70 -top 35 "test.d"
```



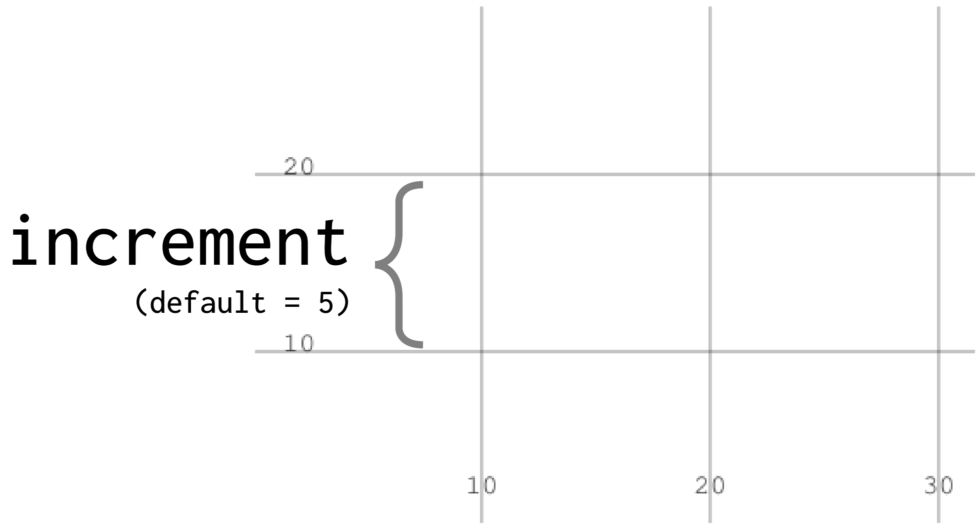



```
grid "file" x y hspace vspace limit
```

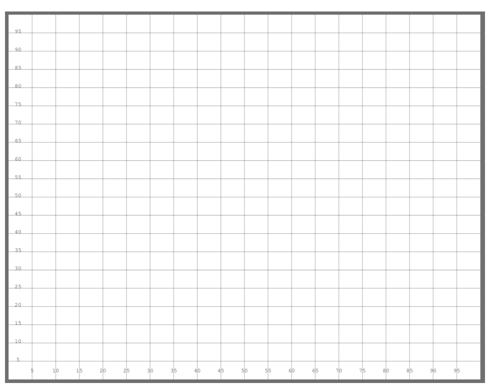
```
circle x y 2
circle x y 4
circle x y 6
circle x y 8
square x y 8
square x y 6
square x y 4
square x y 2
```



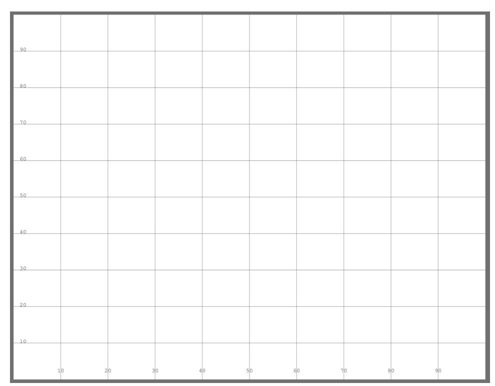
```
grid "code/grid-ex.dsh" 35 33 10 10 65
```



ruler/georuler increment color



ruler



ruler 10

ruler 2 "red"

Math Functions

description	keyword	mandatory
Cosine	`cosine`	`number or expression`
Sine	`sine`	`number or expression`
Square Root	`sqrt`	`number or expression`
Tangent	`tangent`	`number or expression`

```
a=3.14159265359
b=2*a
x=cosine 4       <----  x = -1
y=cosine a       <----  y = -0.65364
n=cosine a+b     <----  n = -1
```

```
value=cosine  number or expression
```

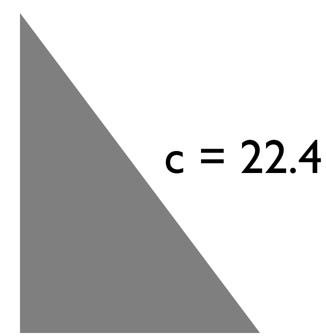
```
a=10
b=71
x=sin 4        x = -0.75680
y=sin a        y = -0.5440
n=sin a+b      n = -0.62989
```

```
value=sine number or expression
```

```
a=10
b=71
x=sqrt 4          ← x = 2
y=sqrt a          ← y = 3.1622776
n=sqrt a+b        ← n = 9
```

```
value=sqrt number or expression
```

```
a_squared=10*10
b_squared=20*20
c=sqrt a_squared + b_squared
w=70+10
h=15+20
polygon "70 w 70" "15 15 h"
clabel=format "c = %.1f" c
text clabel 76 25 2
```



```
a=10
b=71
x=tangent 4      <-- x = 1.1578213
y=tangent a      <-- y = 0.6483608
n=tangent a+b    <-- n = -0.8109944
```

```
value=tangent number or expression
```

Charts

description	keyword	mandatory	optional
General Charts	dchart	options "file"	
Chart Legends	legend	"text" x y size font color	
Chart Bounding Box	chartbbox	left	right top bottom
Area Chart	areachart	"file"	color lcolor vcolor
Bar Chart	barchart	"file"	color lcolor vcolor
Dot Chart	dotchart	"file"	color lcolor vcolor
Horizontal Bar Chart	hbarchart	"file"	color lcolor vcolor
Line Chart	linechart	"file"	color lcolor vcolor
Word Bar Chart	wbarchart	"file"	color lcolor vcolor
Slope Chart	slopechart	"file"	color lcolor vcolor
Proportional Map	pmap	"file"	size
Pie Chart	pie	"file"	size
Donut Chart	donut	"file"	width size
Proportional Grid	pgrid	"file"	size
Waffle Chart	waffle	"file"	size
Fan Chart	fanchart	"file"	size
Bowtie Chart	bowtie	"file"	size

Chart Variables

description	variable	default value
Chart Top	chartTop	80
Chart Bottom	chartBottom	30
Chart Left	chartLeft	10
Chart Right	chartRight	90
Chart text size	chartTextSize	1.5
Chart line width	chartLineWidth	0.2
Chart bar width	chartBarWidth	0 (autoscale)
Chart line spacing	chartLineSpacing	2.4
Chart Volume Opacity	chartVolOp	50
X-axis label interval	chartXLabel	1 (1: all, 0: none, every nth label)
Y-axis range	chartYRange	"" (min,max,interval)
CSV Columns	chartCSVCols	""
Data Format	chartDataFmt	%.1f
Show % Value	chartPercent	0 (0: off, 1: on)
Show data values	chartVal	1 (0: off, 1: on)
Show notes	chartNote	0 (0: off, 1: on)
Show Y-axis grid	chartGrid	0 (0: off, 1: on)
Show title	chartTitle	1 (0: off, 1: on)
Read CSV	chartReadCSV	0 (0: off, 1: on)
Show last X-axis label	chartXLast	0 (0: off, 1: on)

dchart options: chart types

option	default	description
`-bar`	true	bar chart
`-wbar`	false	word bar chart
`-hbar`	false	horizontal bar chart
`-donut`	false	donut chart
`-dot`	false	dot chart
`-lego`	false	lego chart
`-line`	false	line chart
`-pgrid`	false	proportional grid
`-pmap`	false	proportional map
`-bowtie`	false	bowtie chart
`-fan`	false	fan chart
`-radial`	false	radial chart
`-scatter`	false	scatter chart
`-slope`	false	slope chart
`-vol`	false	volume (area) chart

dchart options: elements

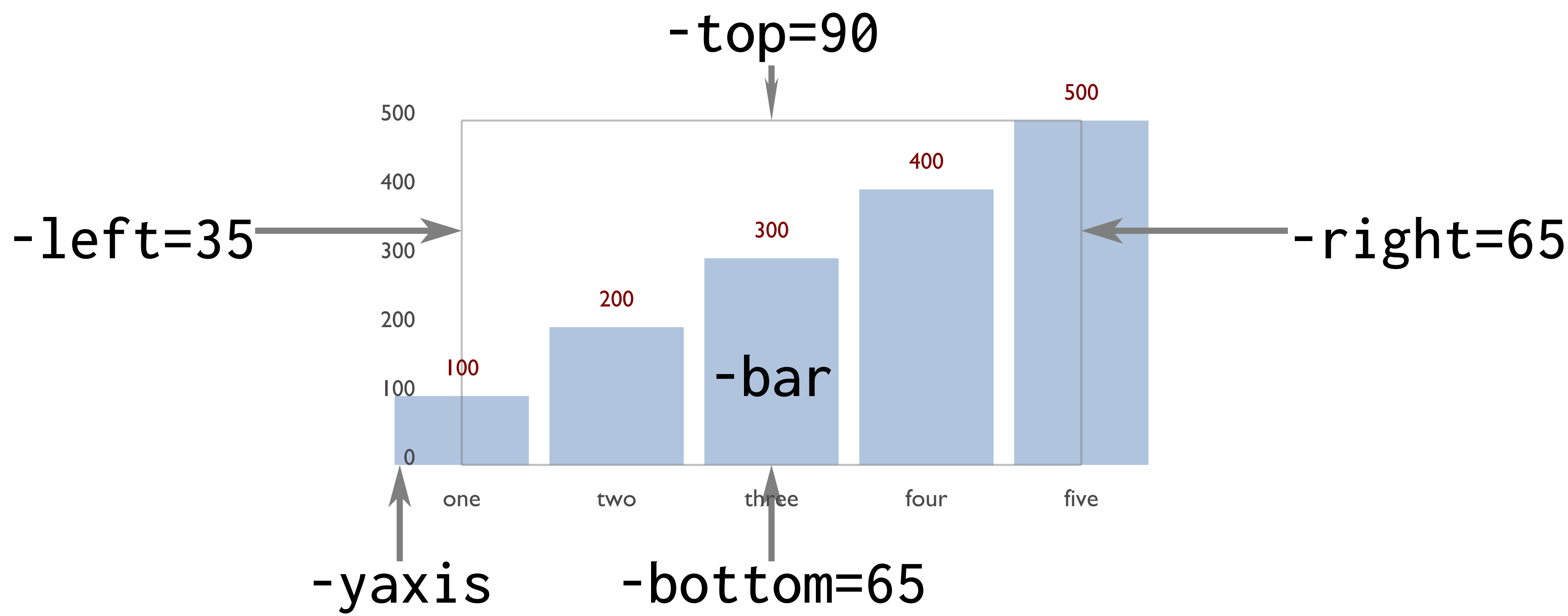
option	default	description
-csv	false	read CSV files
-frame	false	show a colored frame
-fulldeck	true	generate full deck markup
-grid	false	show gridlines on the y axis
-note	true	show annotations
-pct	false	show computed percentage
-rline	false	show a regression line
-solidpmap	false	show solid pmap colors
-spokes	false	show spokes in radial chart
-title	true	show the title
-val	true	show values
-xlast	false	show the last x label
-xstagger	false	stagger x axis labels
-yaxis	false	show a y axis
-chartitle	override title in data	specify the title
-datacond	low,high,color	conditional data colors
-hline	value,label	label horizontal line at value
-valpos	t=top, b=bottom, m=middle	value position
-xlabel	default=1, 0 to suppress	x axis label interval
-yrange	min,max.step	specify the y axis label range

dchart options: measures and attributes

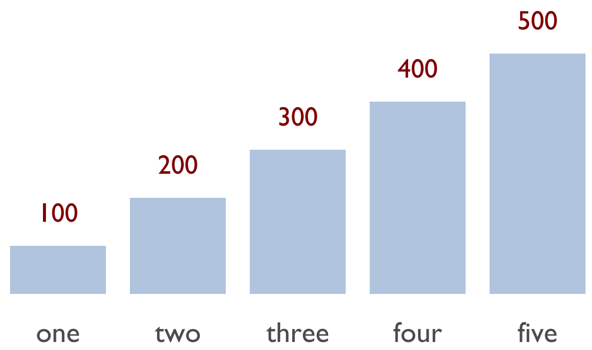
option	default	description
-bgcolor	white	background color
-barwidth	computed from data size	barwidth
-color	lightsteelblue	data color
-csvcol	label1,label2	specify csv columns
-datafmt	%.1f	data format for values
-dmin	false	use data minimum, not zero
-framecolor	rgb(127,127,127)	frame color
-lcolor	rgb(75,75,75)	label color
-linewidth	0.2	linewidth
-ls	2.4	linespacing
-noteloc	c=center, r=right, l=left	annotation location
-pmlen	20	pmap label length
-psize	30	diameter of the donut
-pwidth	3	width of the donut or pmap
-rlcolor	rgb(127,0,0)	regression line color
-textsize	1.5	text size
-xlabrot	0	xlabel rotation (deg.)
-vcolor	rgb(127,0,0)	value color
-volop	50	volume opacity %

dchart options: position and scaling

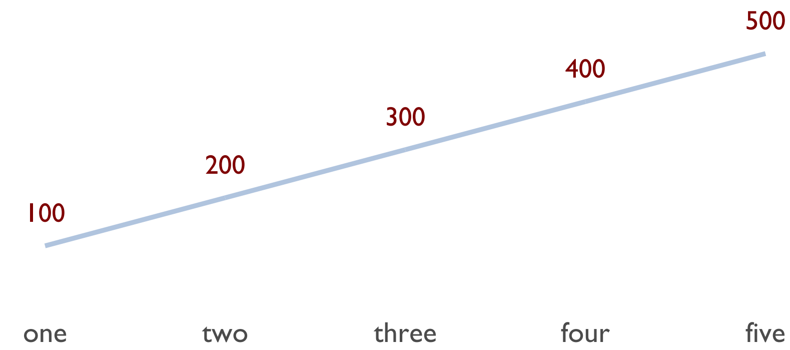
option	default	description
`-top`	80	top of the chart
`-bottom`	30	bottom of the chart
`-left`	20	left margin
`-right`	80	right margin
`-min`	data min	set the minimum data value
`-max`	data max	set the maximum data value
`-bounds`	""	set left,right,top,bottom



dchart options "file"



`dchart -left=10 -right=30 -top=35 -bottom=20 "test.d"`



`dchart -left=55 -right=85 -top=35 -bottom=20 -bar=f -line "test.d"`

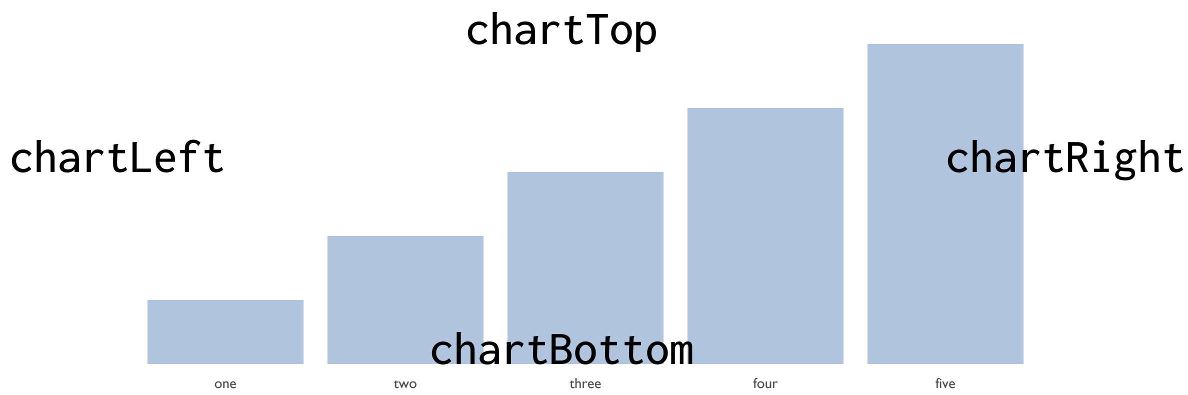

```
legend "text" x y fontsize font color
```

Item on the chart

Thing

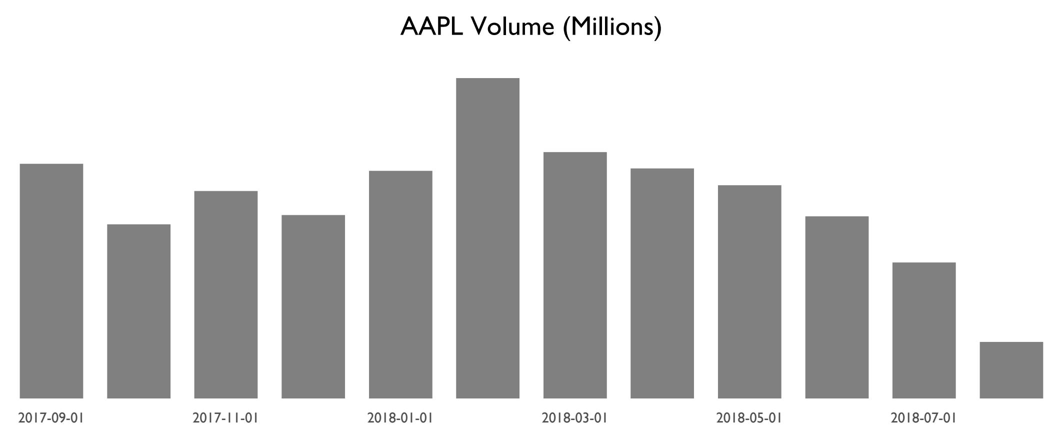
```
legend "Item on the chart" 20 30 3 "sans" "red"
```

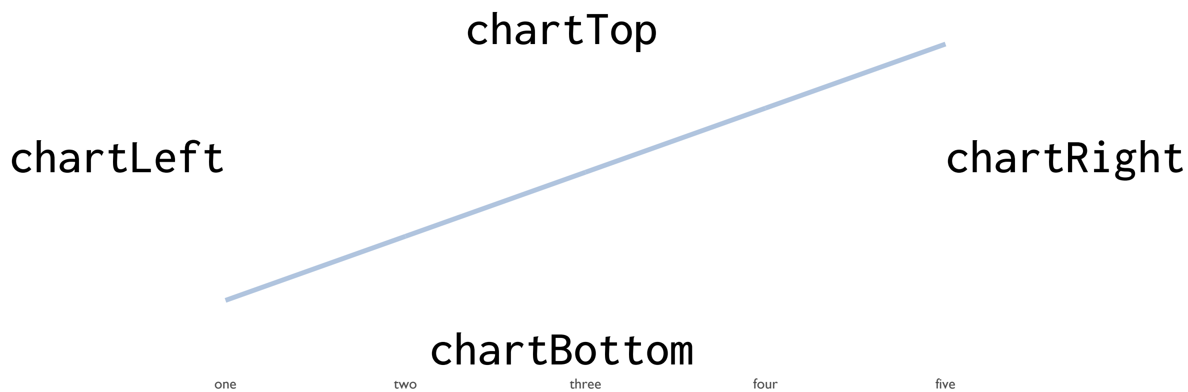
```
legend "Thing" 70 30 2 "serif" "blue"
```




```
barchart "file" color lcolor vcolor
```

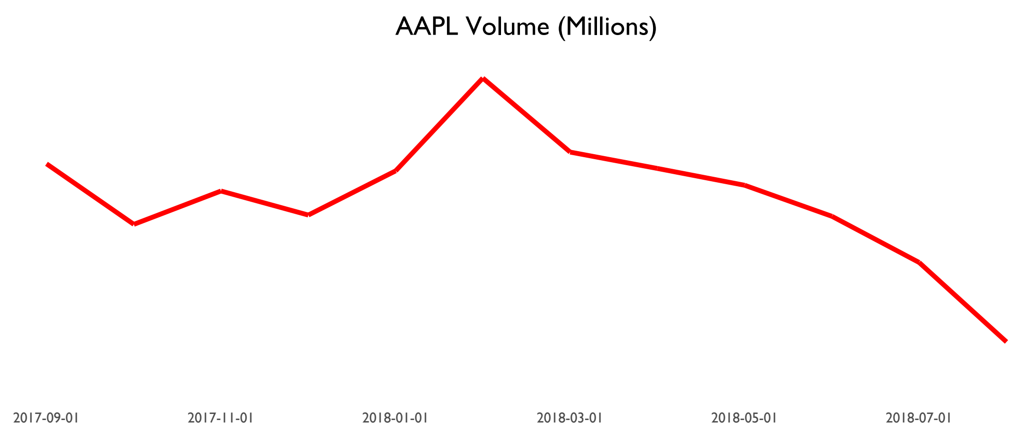
```
chartLeft=50
chartRight=90
chartTop=30
chartBottom=10
chartXLabel=2
barchart "AAPL.d" "gray"
```

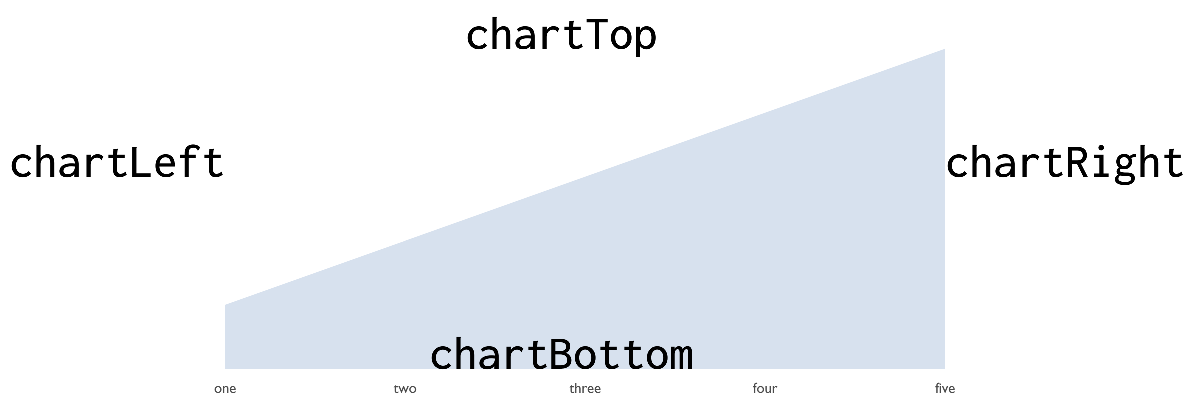





```
linechart "file" color lcolor vcolor
```

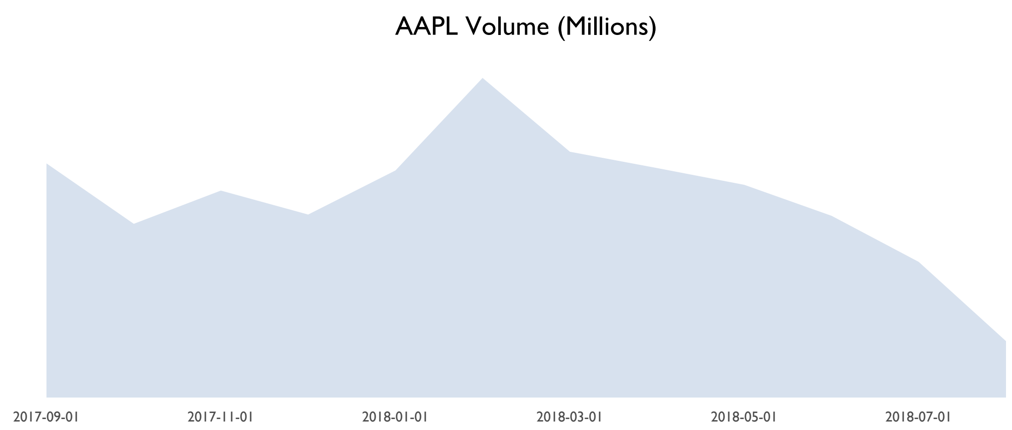
```
chartLeft=50
chartRight=90
chartTop=30
chartBottom=10
chartXLabel=2
linechart "AAPL.d" "red"
```

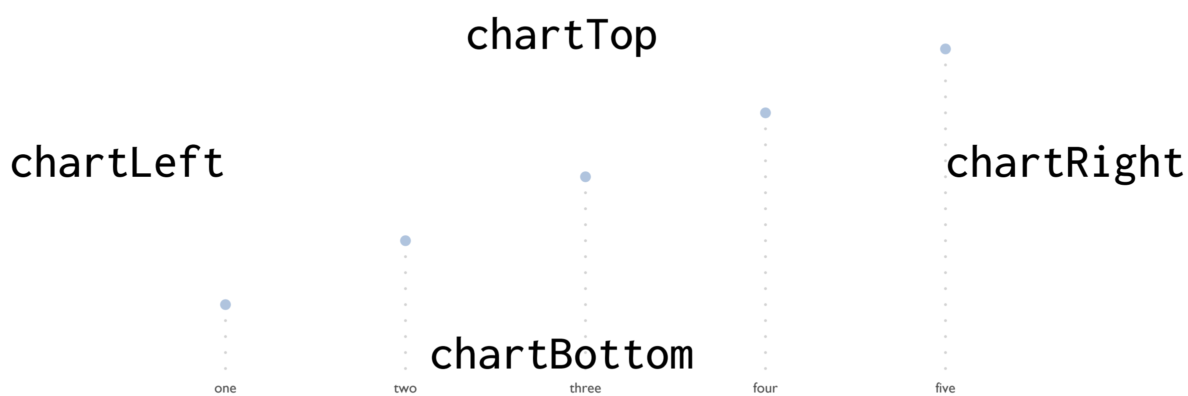




areachart "file" color lcolor vcolor

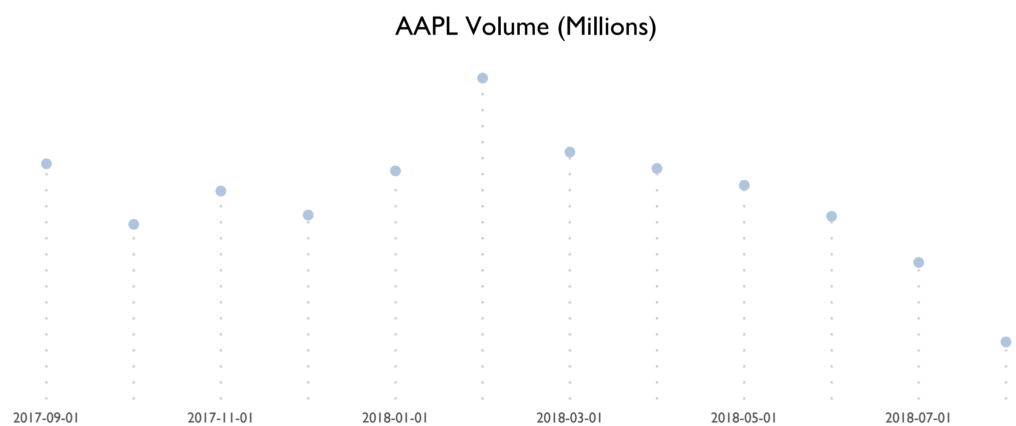
```
chartLeft=50
chartRight=90
chartTop=30
chartBottom=10
chartXLabel=2
areachart "AAPL.d"
```

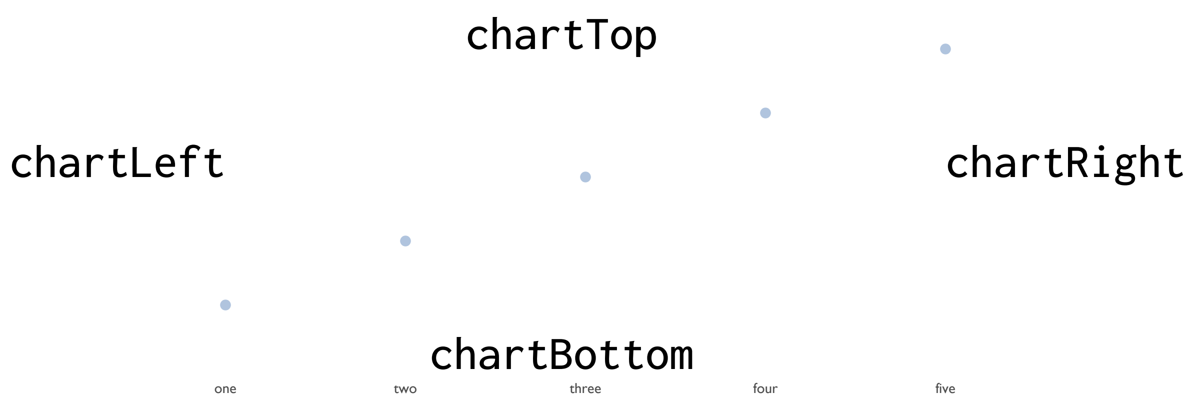





```
dotchart "file" color lcolor vcolor
```

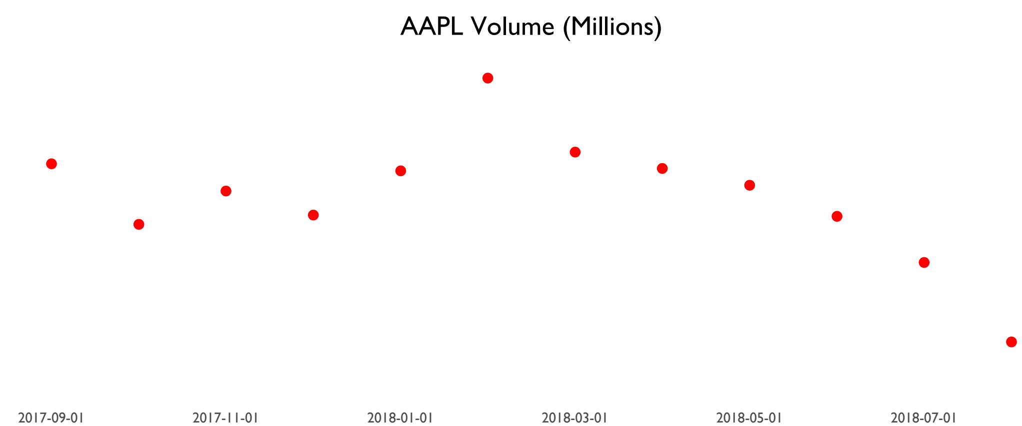
```
chartLeft=50
chartRight=90
chartTop=30
chartBottom=10
chartXLabel=2
dotchart "AAPL.d"
```

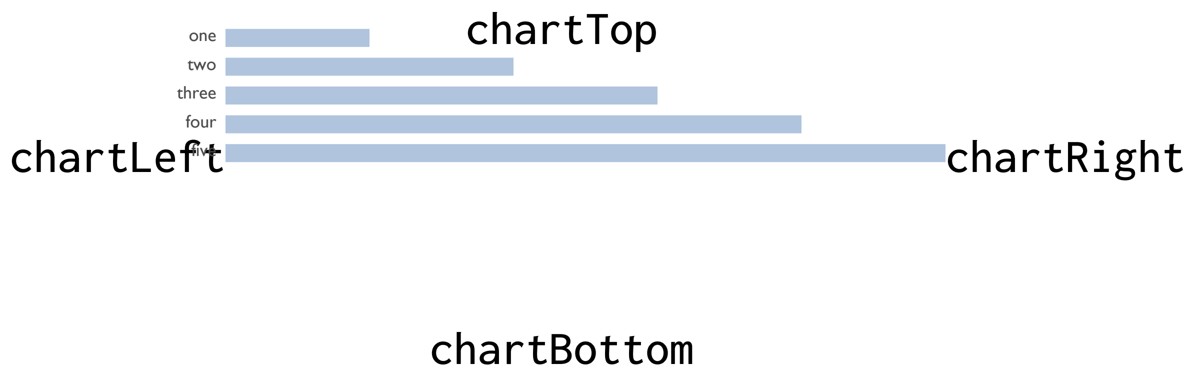




scatterchart "file" color lcolor vcolor

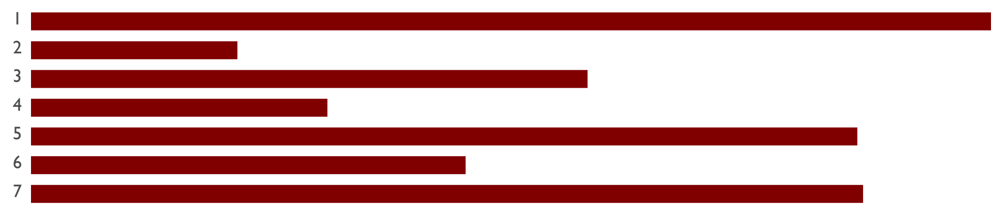
```
chartLeft=50
chartRight=90
chartTop=30
chartBottom=10
chartXLabel=2
scatterchart "AAPL.d" "red"
```

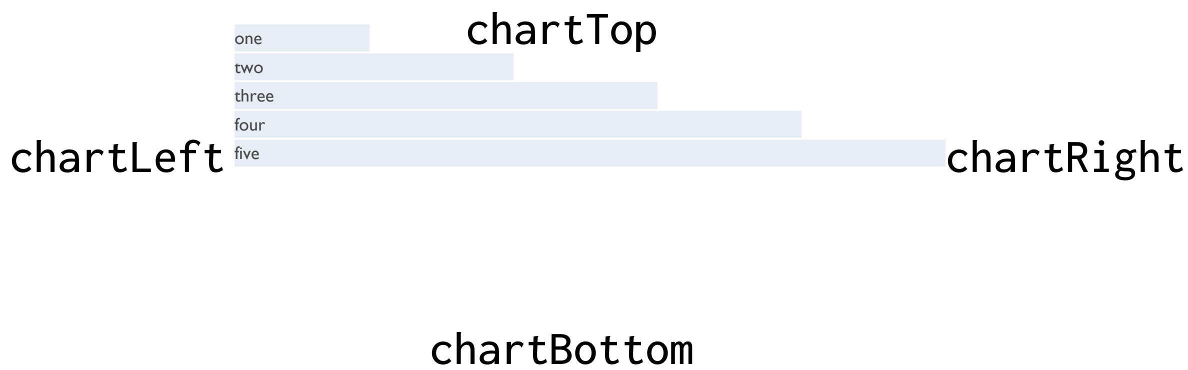




hbarchart "file" color lcolor vcolor

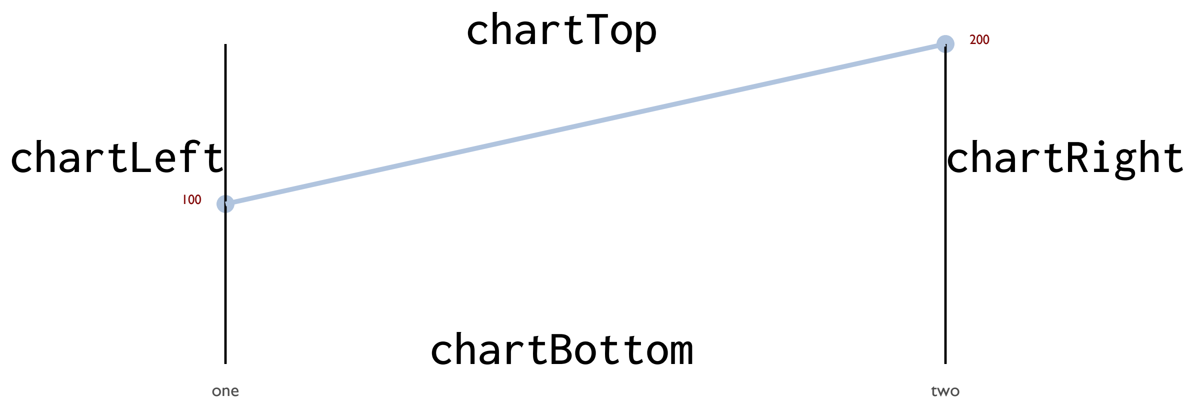
```
chartLeft=50
chartRight=90
chartTop=30
chartBottom=10
chartXLabel=2
hbarchart "rand.d" "maroon"
```





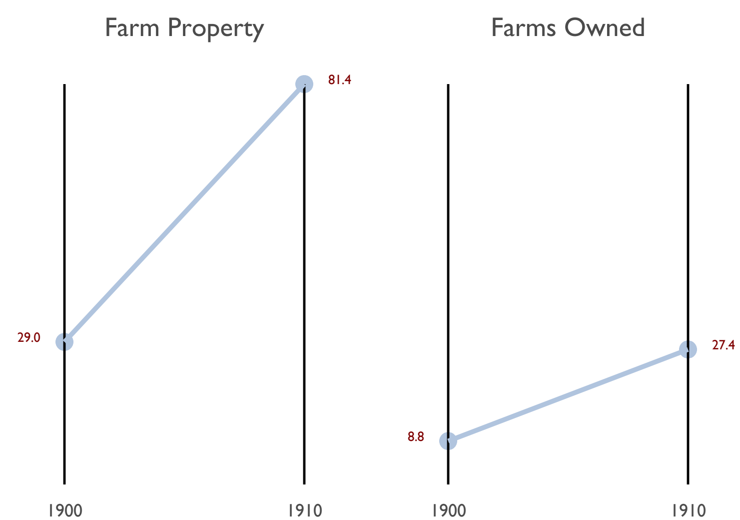
`wbarchart "file" color lcolor vcolor`

```
chartLeft=50
chartRight=90
chartTop=30
chartBottom=10
chartXLabel=2
wbarchart "rand.d"
```

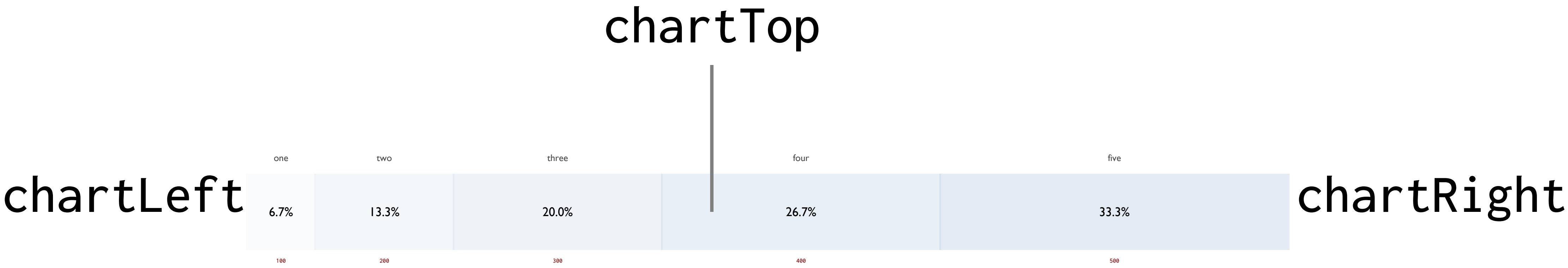


slopechart "file" color lcolor vcolor

```
chartLeft=40
chartRight=50
chartTop=35
chartBottom=10
slopechart "slopex.d"
```

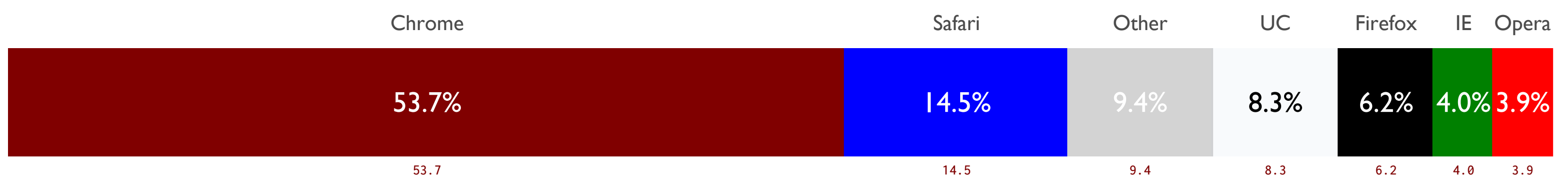


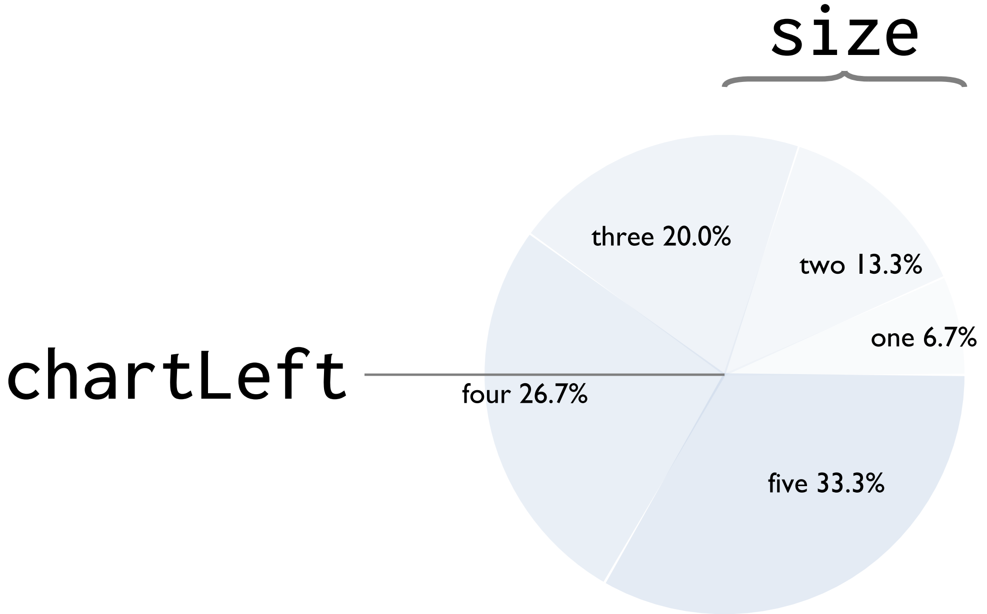

```
1900    28.99    Farm Property
1910    81.42

1900    8.83    Farms Owned
1910    27.45
```




```
pmap "file" size
```

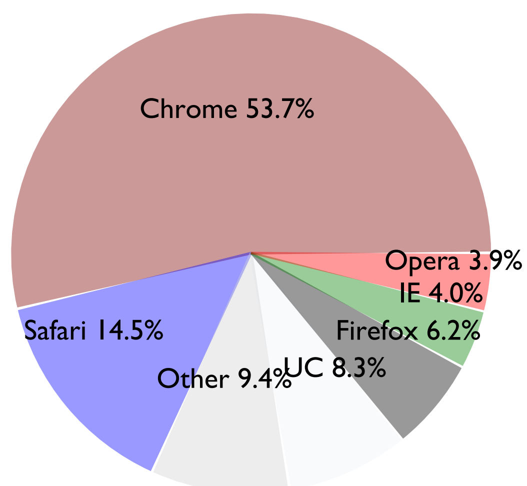
```
chartLeft=25
chartTop=25
chartRight=90
chartTitle=0
chartTextSize=1.2
pmap "browser.d"
```

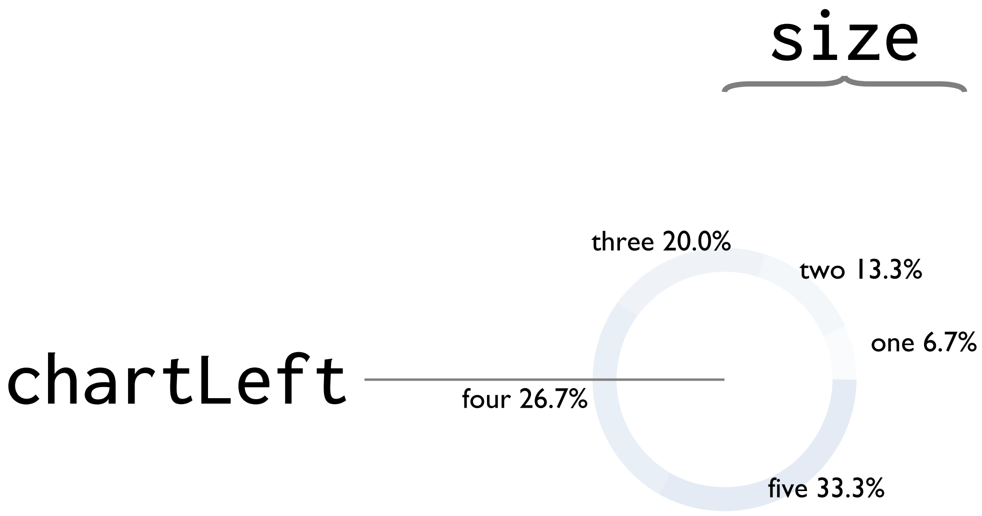





```
pie "file" size
```

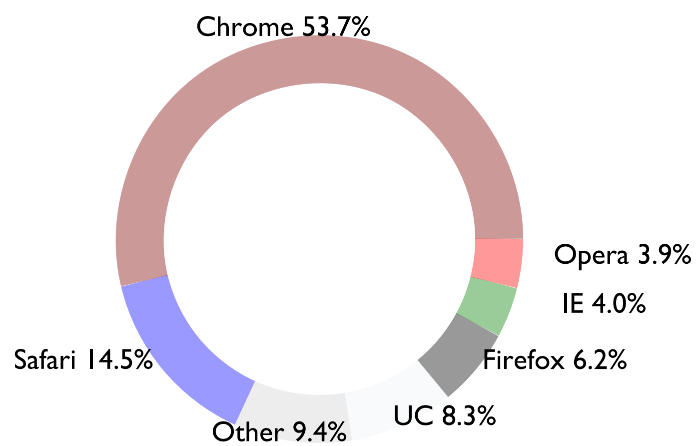
```
chartLeft=50
chartTop=25
chartRight=95
chartTextSize=1.2
chartTitle=0
pie "browser.d" 10
```

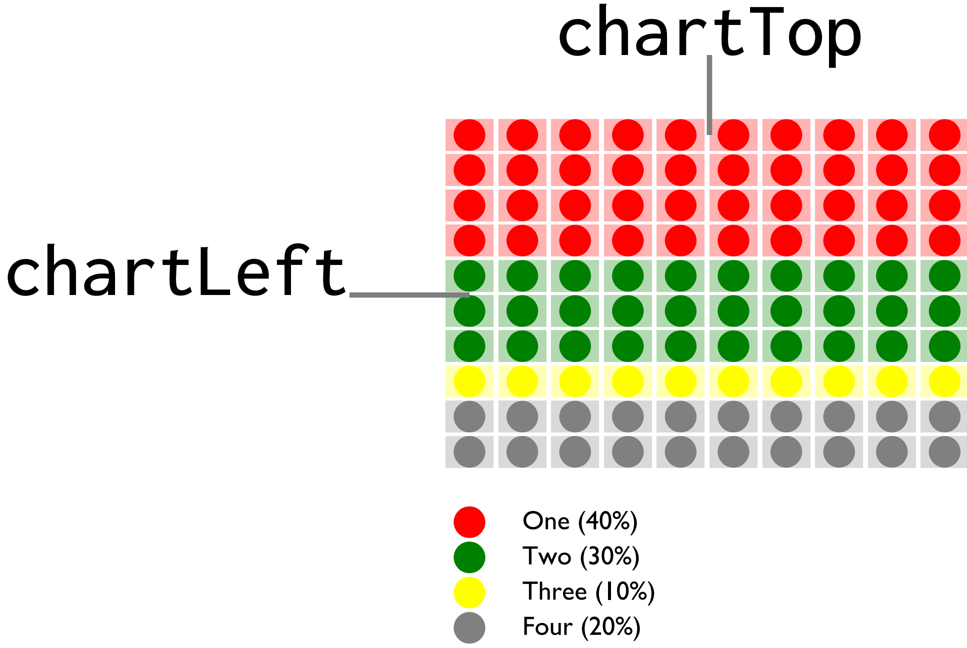





```
donut "file" size width
```

```
chartLeft=50
chartTop=30
chartRight=60
chartTextSize=1.2
chartTitle=0
donut "browser.d" 2 15
```

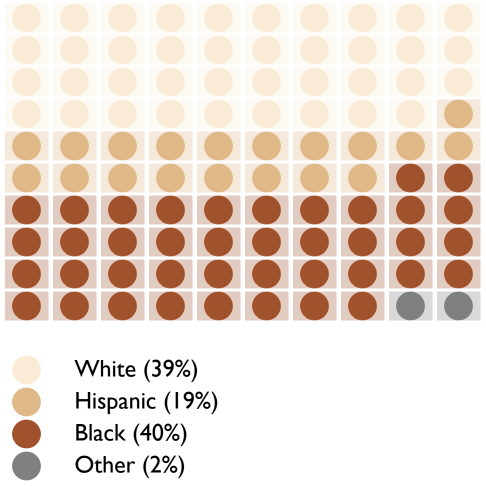


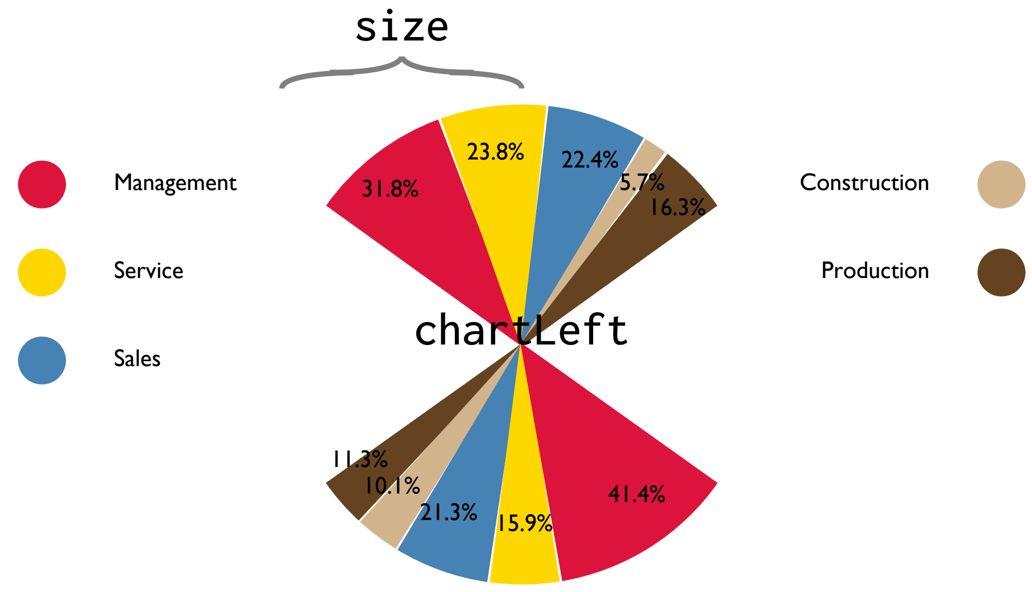


lego "file"

```
chartLeft=65
chartTop=35
chartTextSize=2
lego "incar.d"
```

```
# US Incarceration Rate
White      39     antiquewhite
Hispanic      19     burlywood
Black      40     sienna
Other      2     gray
```

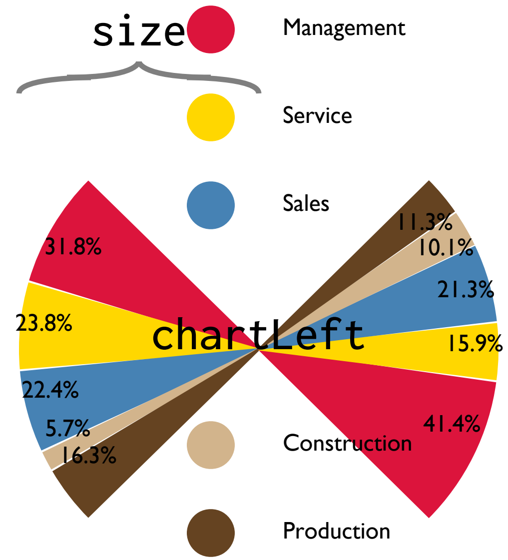





```
fanchart "file" size
```

```
chartLeft=50
chartTop=80
chartVal=1
chartTextSize=1
fanchart "occupation.d" 10
```

```
# Occupations of African-Americans and Whites (2019)
Management    31.8    crimson
Service    23.8    gold
Sales    22.4    steelblue
Construction    5.7    tan
Production    16.3    rgb(101,67,33)
Management    41.4    crimson
Service    15.9    gold
Sales    21.3    steelblue
Construction    10.1    tan
Production    11.3    rgb(101,67,33)
```




```
bowtie "file" size
```

```
chartLeft=50
chartTop=80
chartVal=1
chartTextSize=1
bowtie "occupation.d" 10
```

```
# Occupations of African-Americans and Whites (2019)
Management    31.8    crimson
Service    23.8    gold
Sales    22.4    steelblue
Construction    5.7    tan
Production    16.3    rgb(101,67,33)
Management    41.4    crimson
Service    15.9    gold
Sales    21.3    steelblue
Construction    10.1    tan
Production    11.3    rgb(101,67,33)
```

Geographic Functions

description	keyword	mandatory	optional
Geographic Regions	georegion	"file"	color op
Geographic Borders	geoborder|geoline	"file"	lw color op
Text labels	geolabel	"loc"	size font color op
Dot markers	geomark	"loc"	size color op
Text with markers	geoloc	"loc"	align size font color op
Place images	geoimage	"loc" w h	
Lines between points	geopath	"p1" "p2"	lw color op
Arcs between points	geoarc	"p1" "p2"	lw color op
Lines between points	geopathfile	"file"	lw color op gap
Geographic boundaries	geobbox|geobound	latmin or "p1" "p2"	latmax longmin longmax
Geographic canvas	geocanvas	xmin	xmax ymin ymax

"file" refer to Shapefile (.shp), GeoJSON (.json or .geojson) or Keyhole Markup Language (.kml) files from opendatasoft

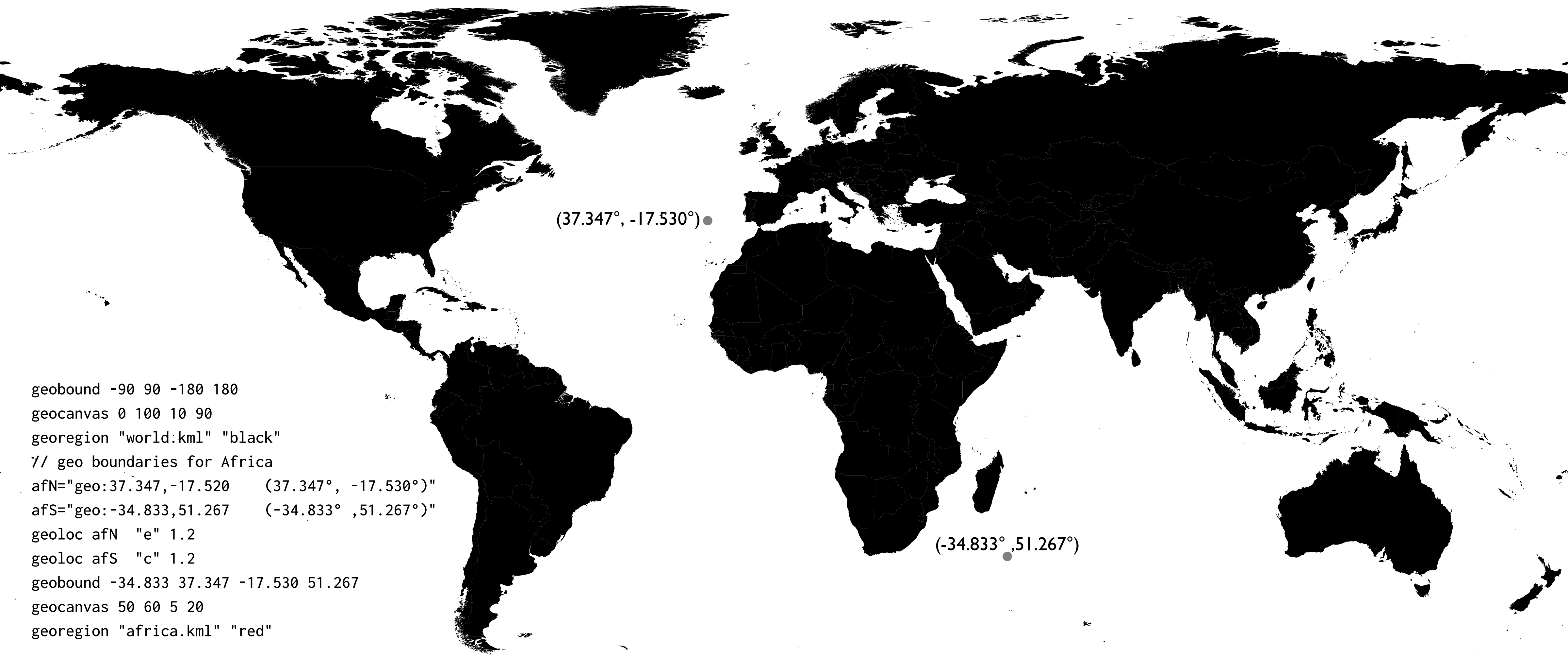
"loc" refers to a geo URI string ("geo:lat,long"), or a filename containing multiple locations

"p1" and "p2" are geo URIs

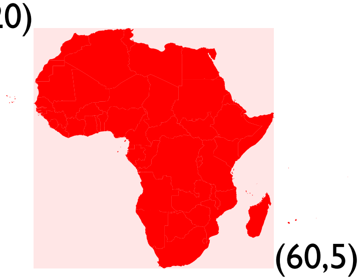
Geo Variables

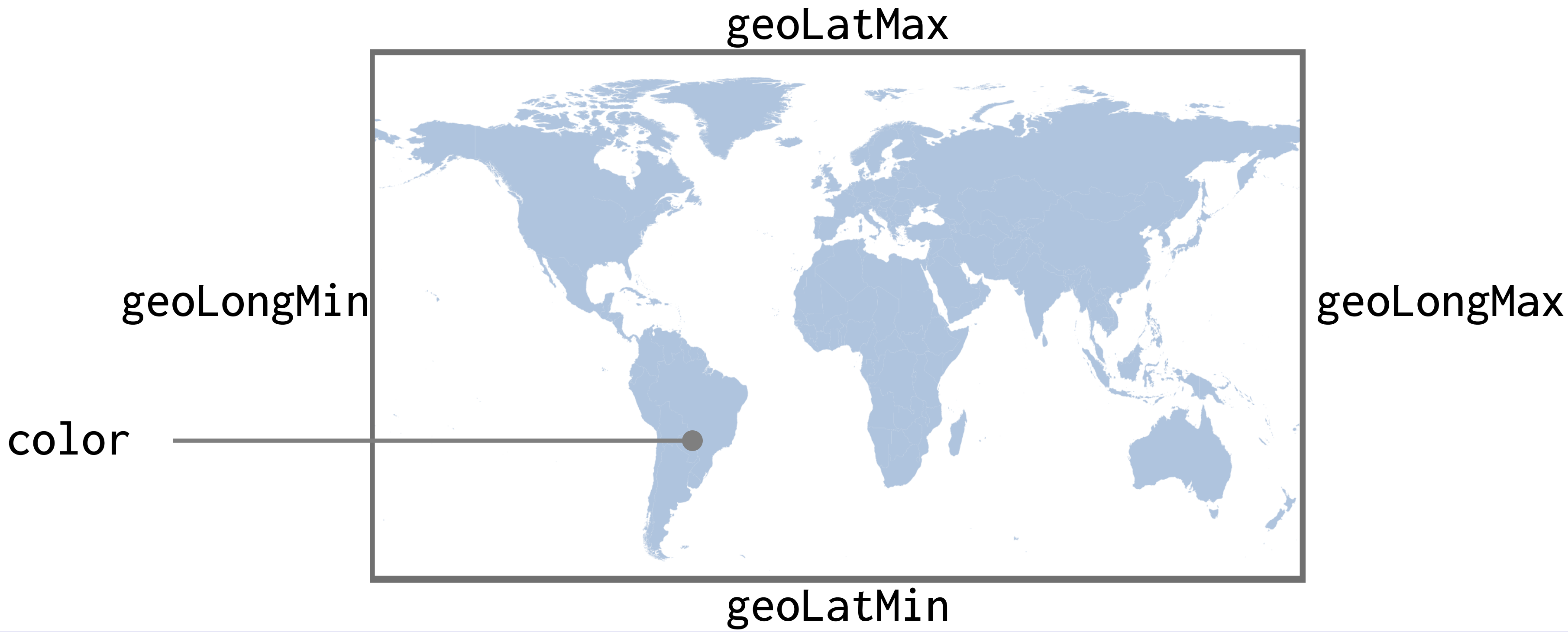
description	variable	default value
Minimum latitude	`geoLatMin`	-90
Maximum latitude	`geoLatMax`	90
Minimum longitude	`geoLongMin`	-180
Maximum longitude	`geoLongMax`	180
Geo canvas x minimum	`geoXmin`	0
Geo canvas x maximum	`geoXmax`	100
Geo canvas y minimum	`geoYmin`	0
Geo canvas y maximum	`geoYmax`	100

Geographic Scaling



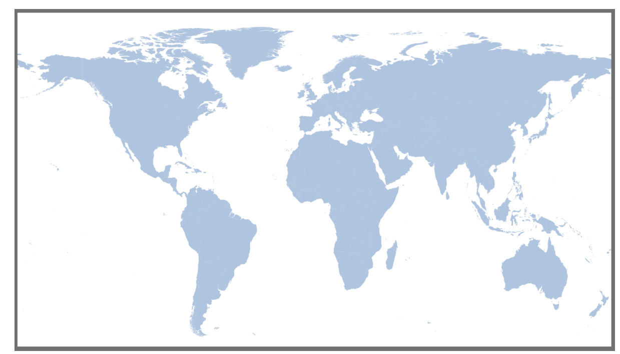

```
geobound -90 90 -180 180
geocanvas 0 100 10 90
georegion "world.kml" "black"
// geo boundaries for Africa
afN="geo:37.347,-17.520    (37.347°, -17.530°)"
afS="geo:-34.833,51.267    (-34.833° ,51.267°)"
geoloc afN  "e" 1.2
geoloc afS  "c" 1.2
geobound -34.833 37.347 -17.530 51.267
geocanvas 50 60 5 20
georegion "africa.kml" "red"
```

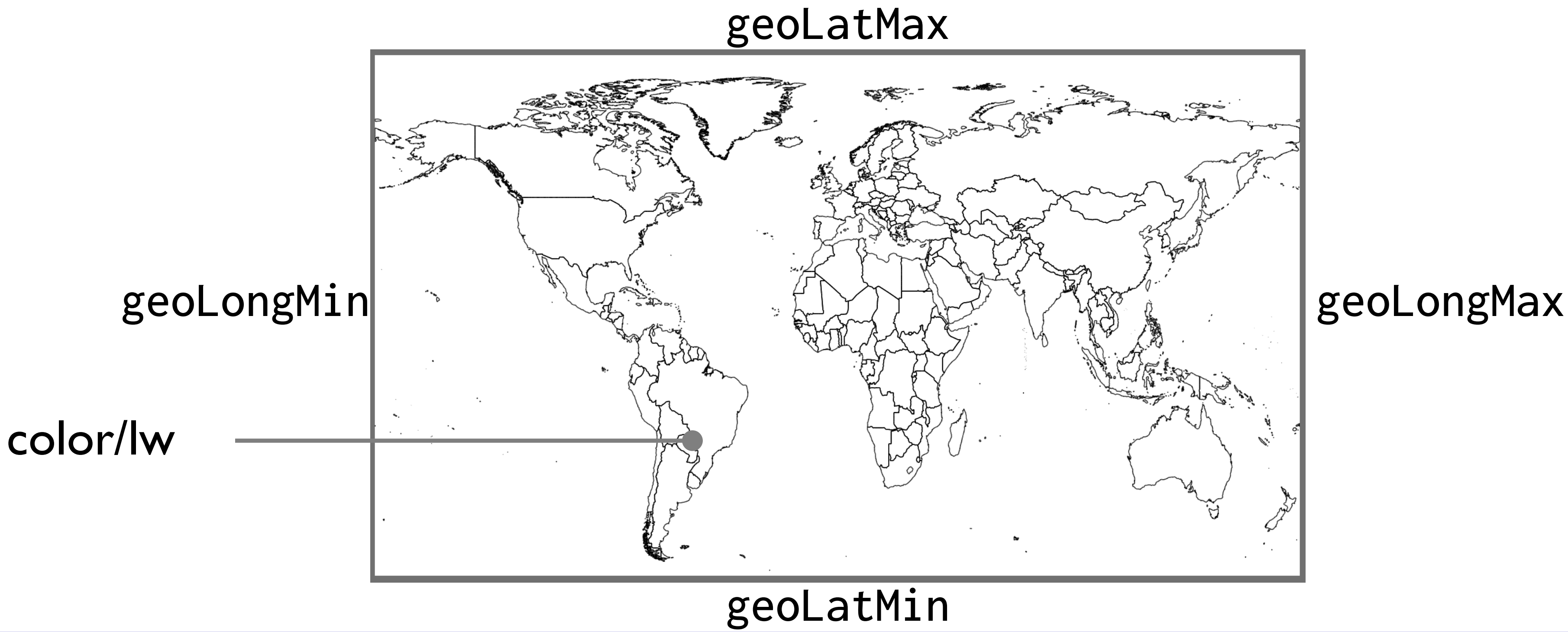




`georegion "file"     color op`

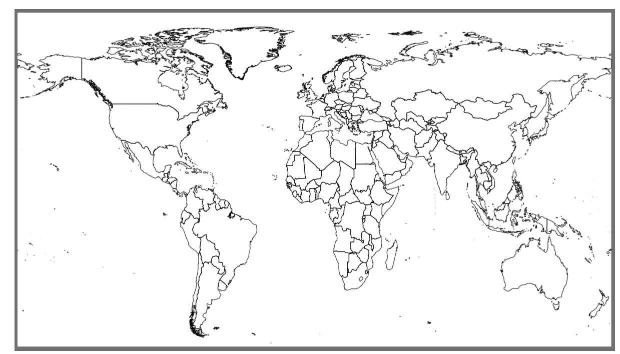
```
geoLatMin=0-60
geoLatMax=90
geoLongMin=0-180
geoLongMax=180

georegion "world.kml"  "white"
```






```
geoborder "file"      lw color
```

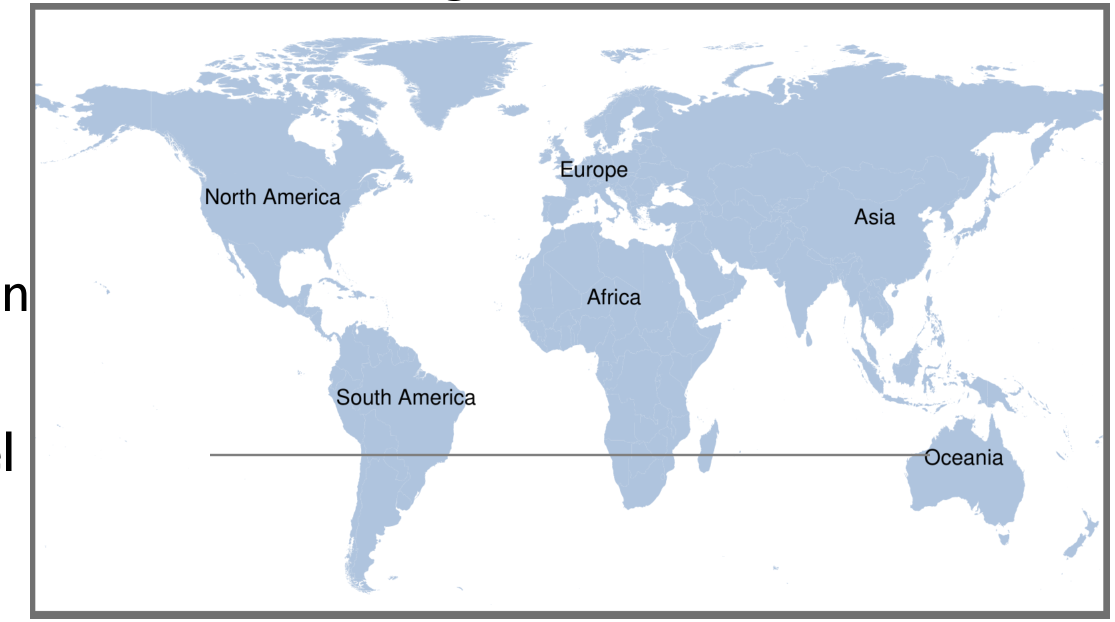
```
geoLatMin=0-60
geoLatMax=90
geoLongMin=0-180
geoLongMax=180

geoborder "world.kml"  0.1 "black"
```



geoLatMax

geoLongMin

geo:lat,long<tab>Label



geoLongMax

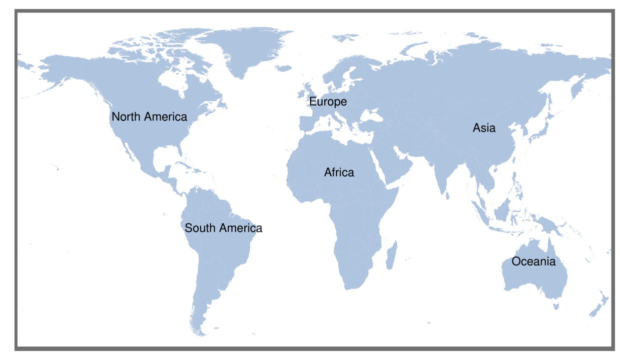
geoLatMin

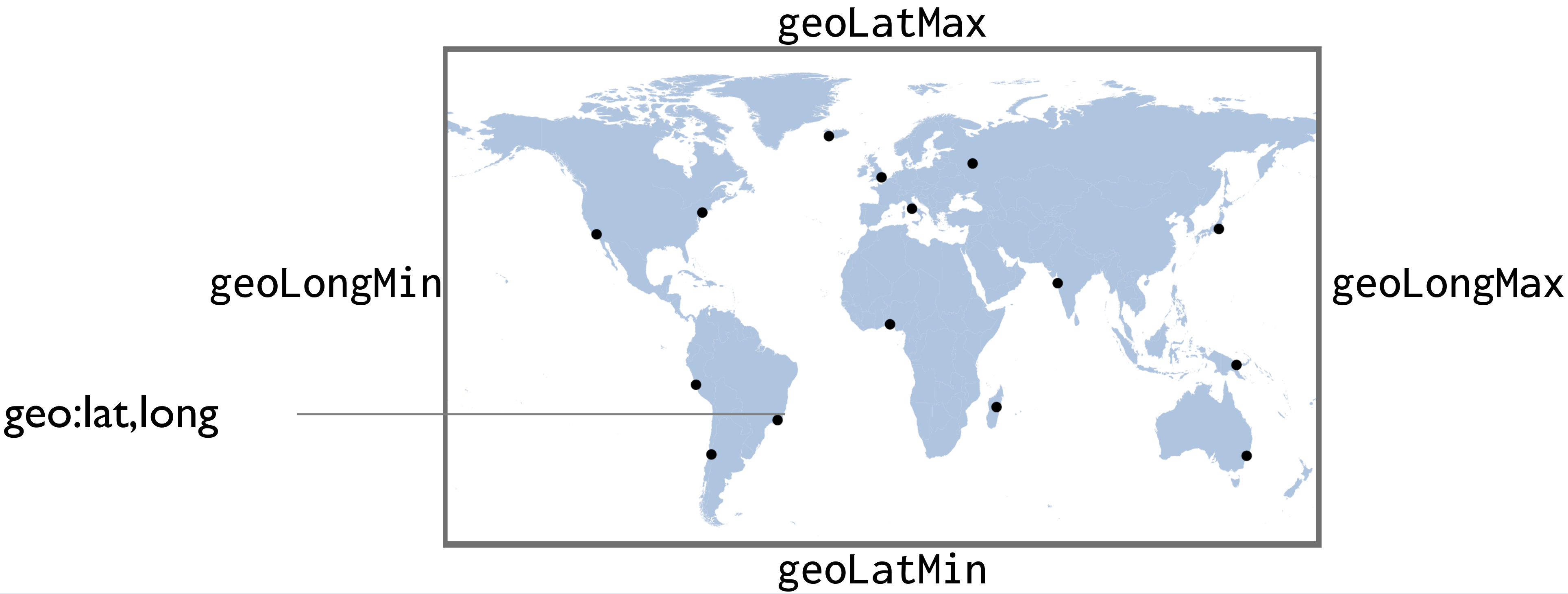
```
geolabel  "loc"  size color op
```

```
geoLatMin=0-60
geoLatMax=90
geoLongMin=0-180
geoLongMax=180
```

```
geo:-10,-55	South America
geo:40,-100	North America
geo:46.8,8.3	Europe
geo:15,15	Africa
geo:35,103	Asia
geo:-25,133	Oceania
```

```
georegion "world.kml"  "white"
geolabel "continents.d"  2 "sans" "black"
```

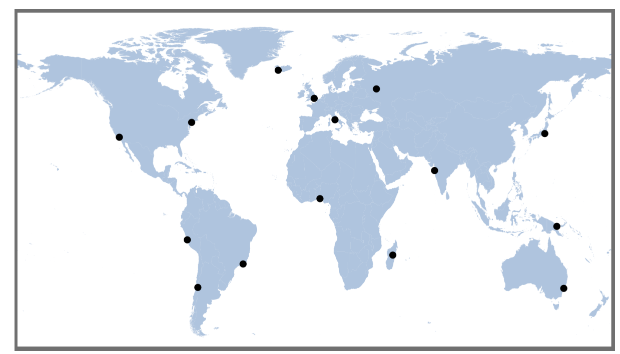




geomark "loc" size color op

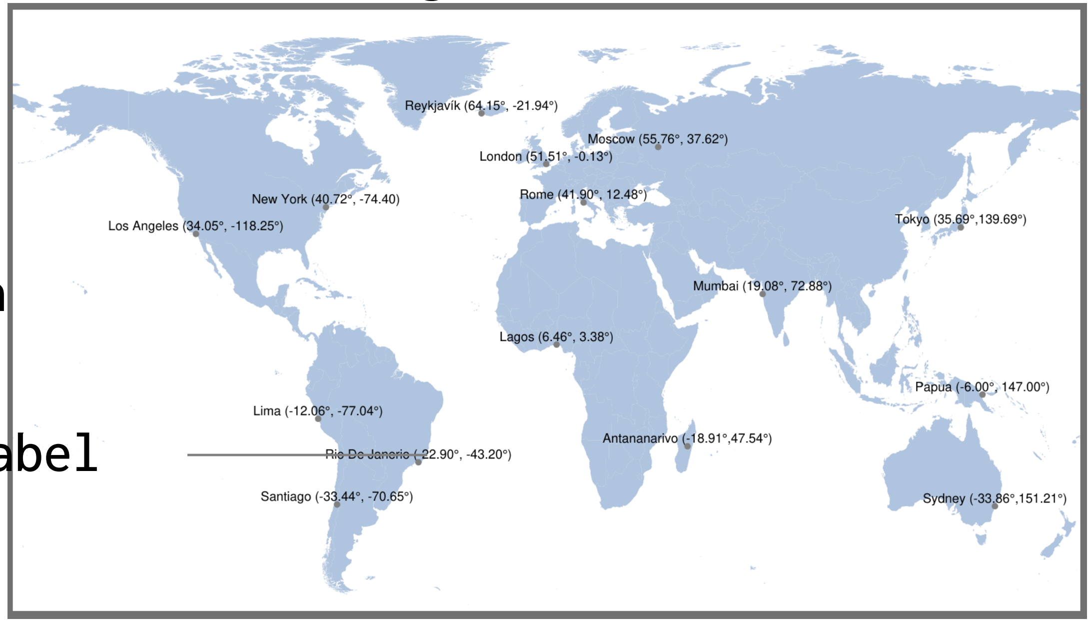
geoLatMin=0-60
geoLatMax=90
geoLongMin=0-180
geoLongMax=180

```
geo:41.8967,12.4822000     Rome (41.90°, 12.48°)
geo:-18.91368,47.53610     Antananarivo (-18.91°,47.54°)
geo:-33.8559,151.20670     Sydney (-33.86°,151.21°)
geo:40.7167,-74.400000     New York (40.72°, -74.40)
...
geo:34.0500,-118.25000     Los Angeles (34.05°, -118.25°)
```



georegion "world.kml" "white"
geolabel "cities.d" 2 "sans" "black"

geoLatMax



geoLongMin

geoLongMax

geo:lat,long<tab>Label

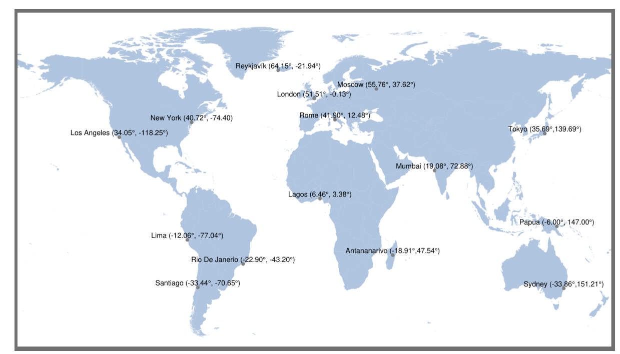
Label Position (pos):
"c" Center above
"b" Begin Aligned
"e" End Aligned
"u" Center below

geoLatMin

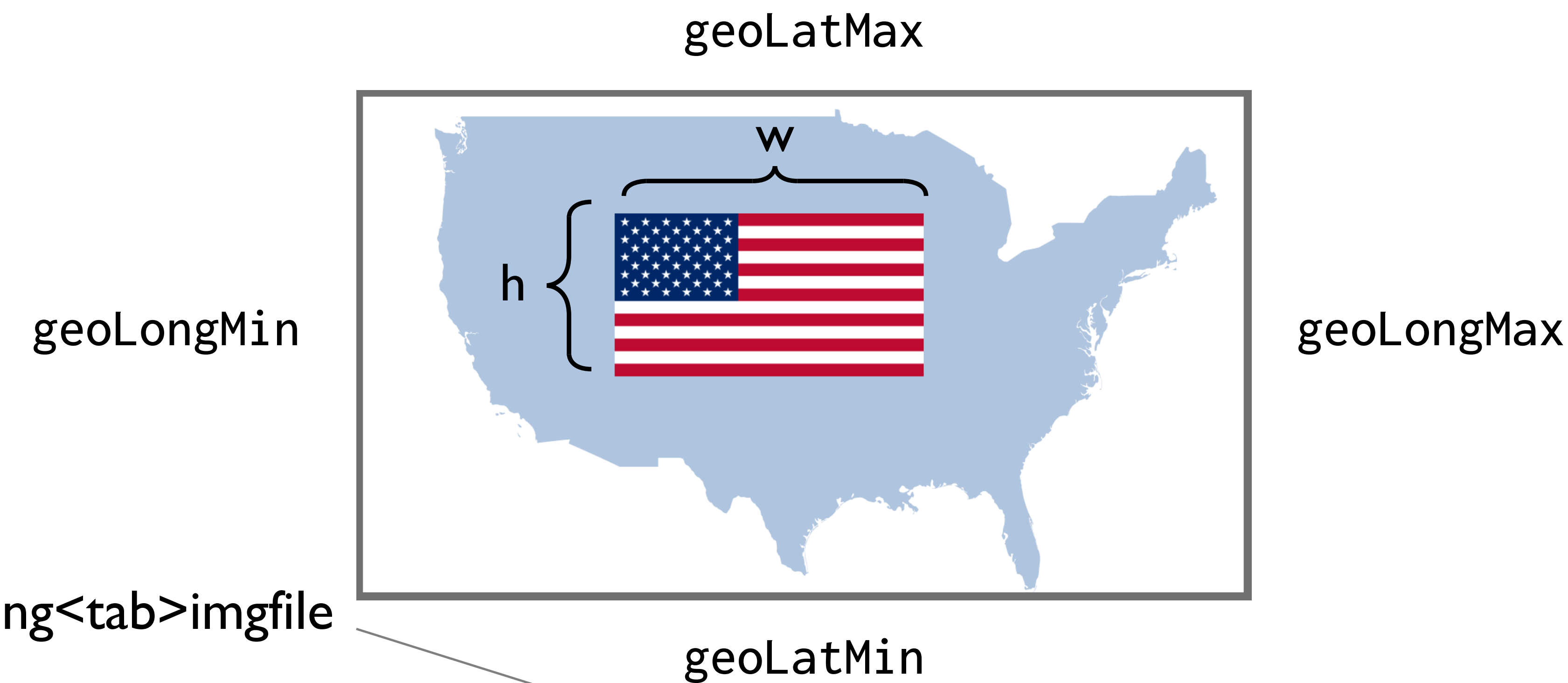
```
geoloc "loc"   pos size font color op
```

```
geoLatMin=0-60
geoLatMax=90
geoLongMin=0-180
geoLongMax=180
```

```
geo:41.8967,12.4822000    Rome (41.90°, 12.48°)
geo:-18.91368,47.53610    Antananarivo (-18.91°,47.54°)
geo:-33.8559,151.20670    Sydney (-33.86°,151.21°)
geo:40.7167,-74.400000    New York (40.72°, -74.40)
...
geo:34.0500,-118.25000    Los Angeles (34.05°, -118.25°)
```



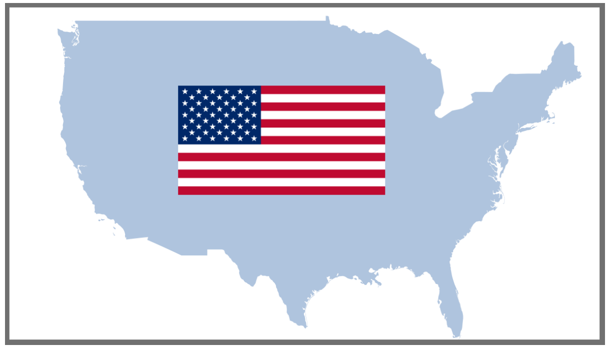
```
georegion "world.kml"   "white"
geoloc "cities.d"       "c" ts
```

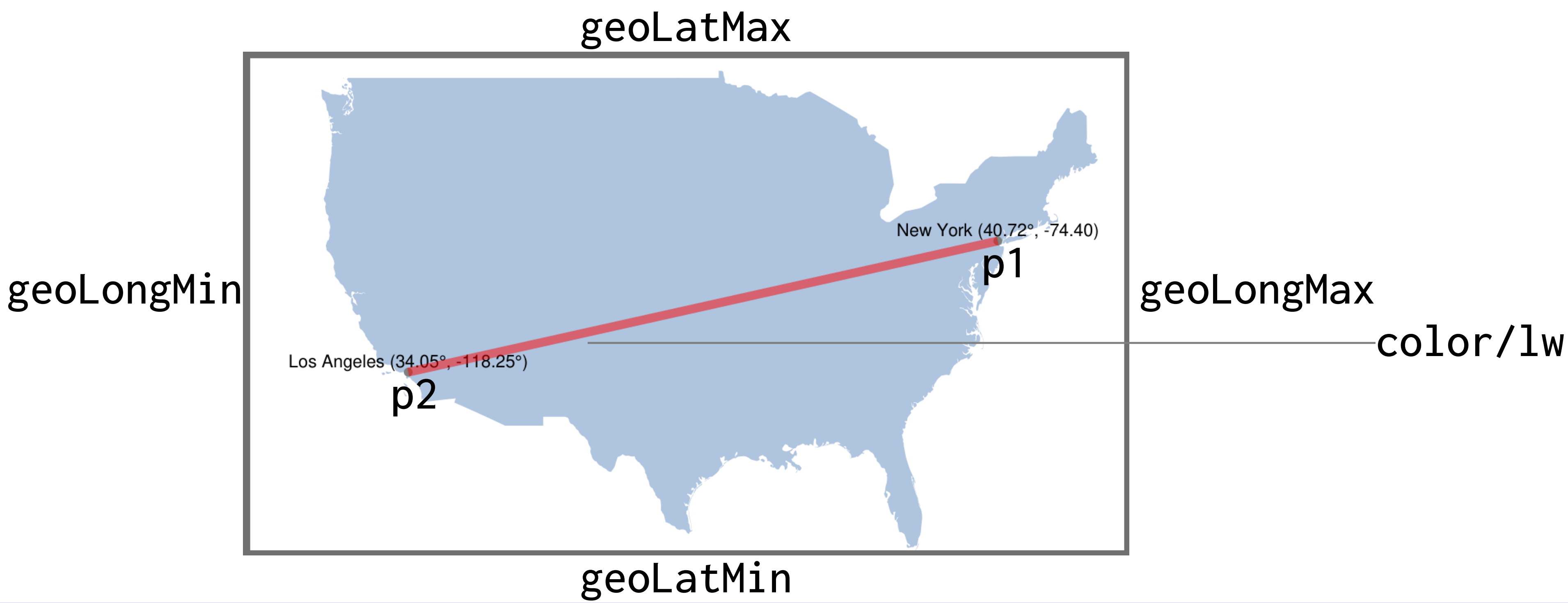


geo:lat,long<tab>imgfile

```
geoimage  "loc" w h
```

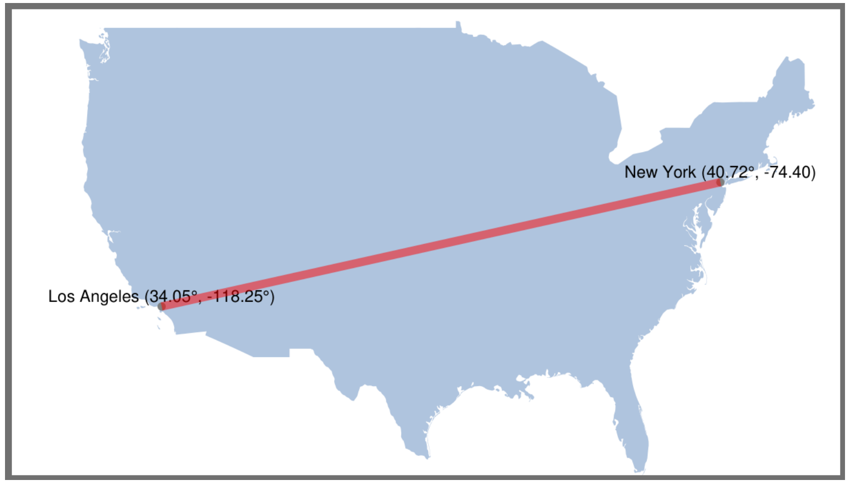
```
geoLatMin=25
geoLatMax=50
geoLongMin=0-130
geoLongMax=0-65

georegion  "usa.kml"  "lightsteelblue"
geoimage   "geo:40,-100   usa.png"  35 0
```

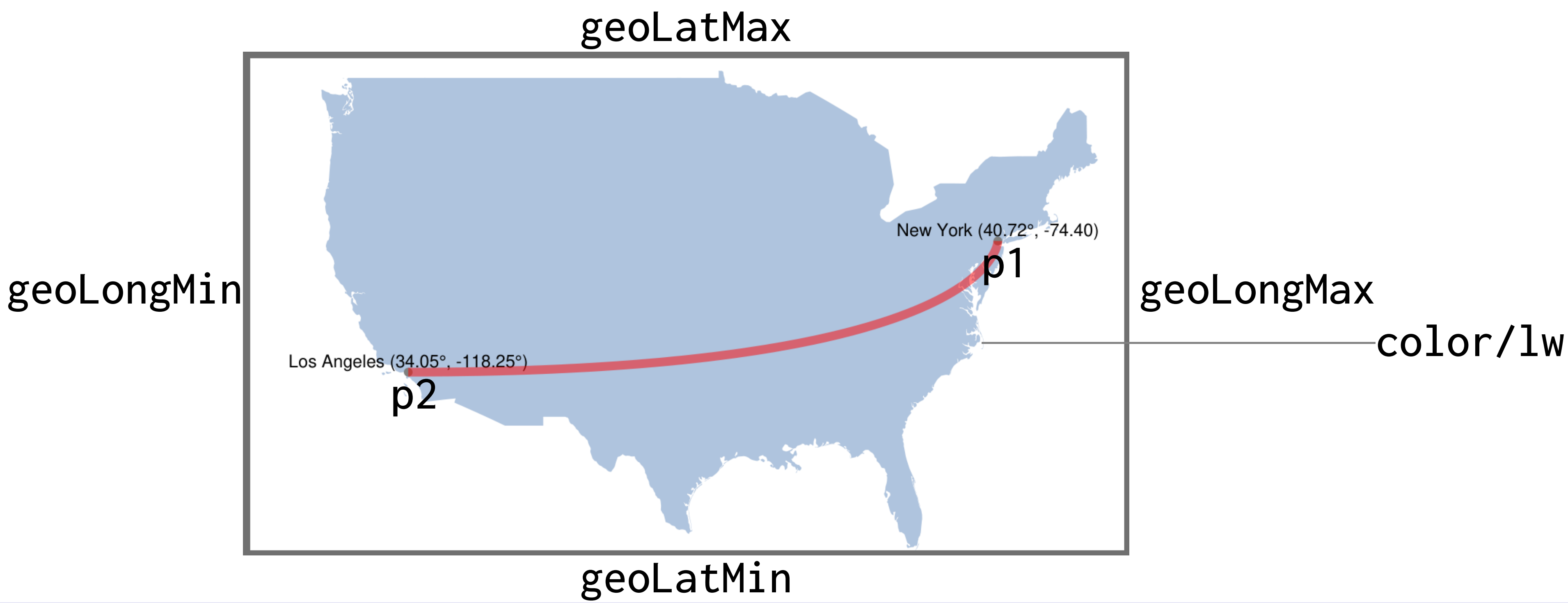




geopath "p1" "p2" lw lw color op

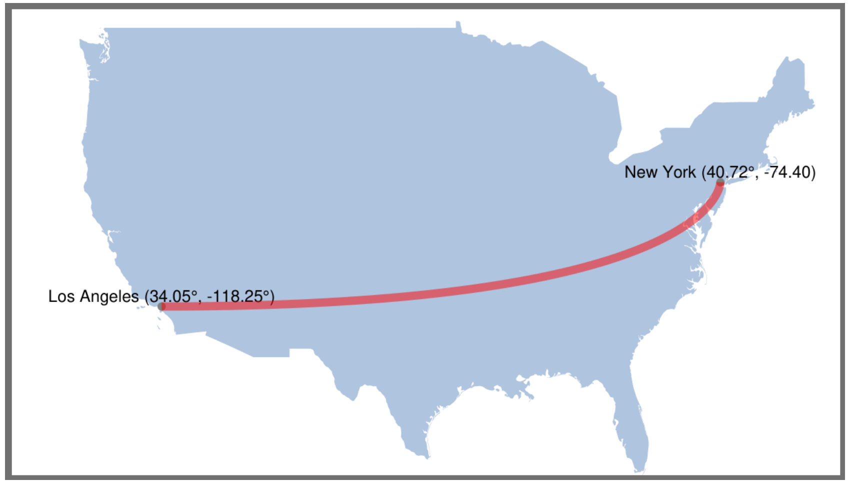
```
geoLatMin=25
geoLatMax=50
geoLongMin=0-130
geoLongMax=0-65
nyc="geo:+40.7167,-74.4000    New York (40.72°, -74.40)"
los="geo:+34.05000,-118.250    Los Angeles (34.05°, -118.25°)"

georegion "usa.kml"  "lightsteelblue"
geoloc    nyc  "c" 2
geoloc    los  "c" 2
geopath    nyc los  1 "red" 50
```



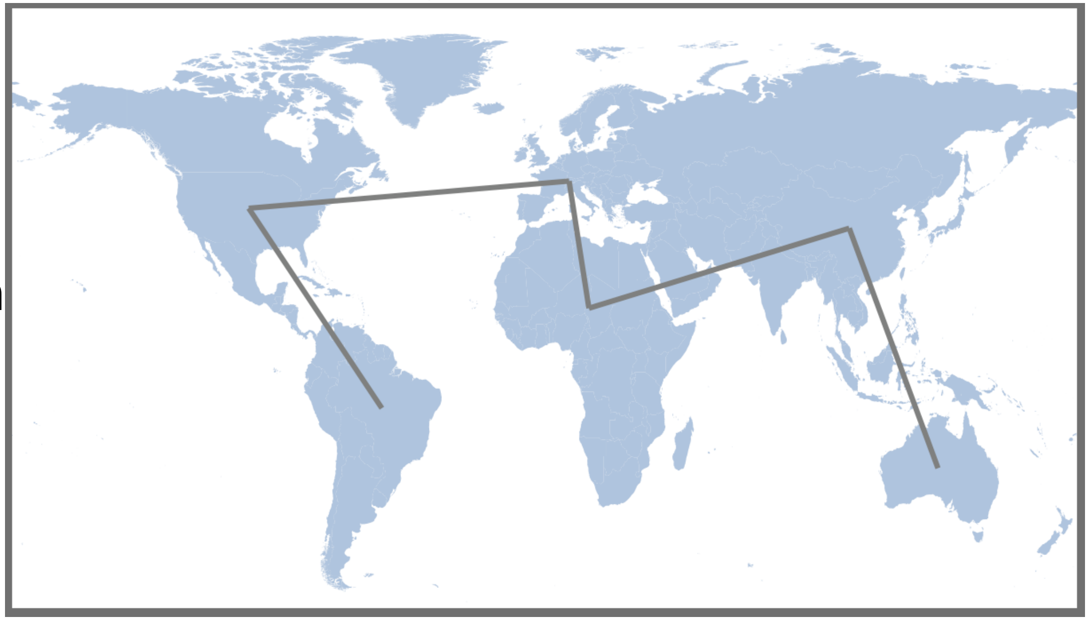


geoarc "p1" "p2" lw color op

```
geoLatMin=25
geolatmax=50
geoLongMin=0-130
geoLongMax=0-65
nyc="geo:+40.7167,-74.4000    New York (40.72°, -74.40)"
los="geo:+34.05000,-118.250    Los Angeles (34.05°, -118.25°)"

georegion "usa.kml" "lightsteelblue"
geoloc    nyc "c" 2
geoloc    los "c" 2
geoarc    nyc los 1 "red" 50
```



geoLatMax

geoLongMin  geoLongMax

geoLatMin

geopathfile "file" lw color op gap

```
geoLatMin=25
geoLatMax=50
geoLongMin=0-130
geoLongMax=0-65
```

```
geo:40.712778,-74.006111    New York
geo:39.952778,-75.163611    Philadelphia
geo:39.768611,-86.158056    Indianapolis
geo:38.627222,-90.197778    St. Louis
geo:35.468611,-97.521389    Oklahoma City
geo:35.084444,-106.650278    Albuquerque
geo:36.167222,-115.148611    Las Vegas
geo:34.050000,-118.250000    Los Angeles
```

```
georegion   "usa.kml"      "lightsteelblue"
geoloc      "roadtrip.d"   "c" 2
geopathfile "roadtrip.d"   0.5 "red" 25
```

Keyword Reference

	Keyword	Arguments	Description
Structure	deck		Begin a deck; end with "edeck"
	def	name args...	Define a function; end with "edef"
	for	x=begin end [increment]	Begin loop; end with "efor"
	if	condition...[else]...eif	Conditional; one of: a==b, a!=b, a>b, a<b, a>=b, a<=b, a<>b c
	import	"file"	import function found in a file
	include	"file"	Include the contents of a file
	slide	[bgcolor] [fgcolor]	Begin a slide; end with "eslide"
	edeck		End the deck
	edef		End the definition
	efor		End the for loop
	eif		End the conditional
	else		Begin the else clause
Utility	canvas	width height	Define with dimensions of the canvas
	content	"scheme://file"	Embed content
	dump	[name]	Dump variables
	grid	"file" x y hspace vspace edge	Define a content grid
	ruler	[increment] [color]	draw a (x,y) ruler
Graphics	acircle	x y w [color] [opacity]	Circle with size based on area
	arc	x y w h a1 a2 [lw] [color] [opacity]	Ellipical arc centered at (x,y), dimensions (w,h) between angles a1 and a2
	circle	x y w [color] [opacity]	Circle centered at (x,y), diameter w
	curve	bx by cx cy ex ey [lw] [color] [opacity]	Quadradic Bezier Curve begin (bx,by), control (cx, cy), end (ex,ey)
	ellipse	x y w h [color] [opacity]	Ellipse centered at (x,y), dimension (w,h)
	hline	x y w [lw] [color] [opacity]	Horizontal line begin at (x,y), length w
	line	x1 y1 x2 y2 [lw] [color] [opacity]	Line between (x1,y1) and (x2,y2)
	dline	x1 y1 x2 y2 [lw] [gap] [color] [opacity]	Dotted line between (x1,y1) and (x2,y2), size lw
	pill	x y w h [color]	Pill shape beginning at (x,y), dimensions (w,h)
	polygon	"x1 x2 x3...." "y1 y2 y3..." [color] [opacity]	Polygon with specified x, y coordinates
	polyline	"x1 x2 x3...." "y1 y2 y3..." [lw] [color] [opacity]	Polyline with specified x, y coordinates
	rect	x y w h [color] [opacity]	Rectangle centered at (x,y), dimensions (w,h)
	rrect	x y w h r [color] [opacity]	Rounded rectangle centered at (x,y), dimensions (w,h), corner radius r
	square	x y w [color] [opacity]	Square centered at (x,y), size w
	star	x y sides inner outer [color] [opacity]	Star centered at (x,y), with sides, innner and outer sizes
	vline	x y h [lw] [color] [opacity]	Vertical line beginning at (x,y), h high

	Keyword	Arguments	Description
Text	arctext	"string" x y radius a1 a2 fontsize [font] [color] [opacity] [link]	Text on an arc, at fontsize, center (x,y), radius r, between a1. a2
	btext	"string" x y fontsize [font] [color] [opacity] [link]	Text beginning at (x,y), at fontsize
	ctext	"string" x y fontsize [font] [color] [opacity] [link]	Centered text beginning at (x,y), at fontsize
	etext	"string" x y fontsize [font] [color] [opacity] [link]	End-aligned text at (x,y), at fontsize
	rtext	"string" x y angle fontsize [font] [color] [opacity] [link]	Rotated text centered at x,y), at angle and fontsize
	text	"string" x y fontsize [font] [color] [opacity] [link]	Text beginning at (x,y), at fontsize
	para	x y w fontsize [font] [color] [opacity] [link]...epara	Paragraphs beginning at (x,y), at fontsize with width w
	textblock	"string" x y w fontsize [font] [color] [opacity] [link]	Block of text beginning at (x,y), at fontsize, with width w
	textblockfile	"file" x y w fontsize [font] [color] [opacity] [link]	Block of text read for a file, beginning at (x,y), at fontsize, with width w
	textcode	"file" x y w fontsize [font] [color] [opacity]	Lines of code, read from a file, upper right corner at (x,y), margin at w
	textfile	"file" x y fontsize [font] [color] [opacity] [spacing]	Contents of a text file upper right corner at (x,y)
Lists	blist	x y fontsize [font] [color] [opacity] [spacing]	Bulleted list starting at (x,y), at fontsize
	clist	x y fontsize [font] [color] [opacity] [spacing]	Centered list starting at (x,y), at fontsize
	list	x y fontsize [font] [color] [opacity] [spacing]	List starting at (x,y), at fontsize
	nlist	x y fontsize [font] [color] [opacity] [spacing]	Numbered list starting at (x,y), at fontsize
	li	"item" [font] [color] [opacity]	List item
	elist		End the list
Images	cimage	"file" "caption" x y w h [scale] [link] capsize	Captioned image; center (x,y), dimensions (w,h) (h=0, w is % of canvas width)
	image	"file" x y w h [scale] [link]	Image center at (x,y), dimensions (w,h) (h=0, w is % of canvas width)
Braces/ Brackets	dbrace	x y w bw bh [lw] [color] [opacity]	Downward pointing brace
	dbracket	x y w h [lw] [color] [opacity]	Downward pointing bracket
	lbrace	x y h bw bh [lw] [color] [opacity]	Left pointing brace
	lbracket	x y w h [lw] [color] [opacity]	Left pointing bracket
	rbrace	x y h bw bh [lw] [color] [opacity]	Right pointing brace
	rbracket	x y w h [lw] [color] [opacity]	Right pointing bracket
	ubrace	x y w bw bh [lw] [color] [opacity]	Upward facing brace
	ubracket	x y w h [lw] [color] [opacity]	Upward facing bracket

	Keyword	Arguments	Description
Arrows	arrow	x1 y1 x2 y2 [lw] [aw] [ah] [color] [opacity]	Arrow starting at (x1,y1), ending at (x2,y2), aw=width, ah=height
	dcarrow	bx by bx xy ex ey [lw] [aw] [ah] [color] [opacity]	Downward curved arrow; curve specified by (bx,by), (cx,cy), (ex,ey)
	lcarrow	bx by bx xy ex ey [lw] [aw] [ah] [color] [opacity]	Left curved arrow; curve specified by (bx,by), (cx,cy), (ex,ey)
	rcarrow	bx by bx xy ex ey [lw] [aw] [ah] [color] [opacity]	Right curved arrow; curve specified by (bx,by), (cx,cy), (ex,ey)
	ucarrow	bx by bx xy ex ey [lw] [aw] [ah] [color] [opacity]	Upward curved arrow; curve specified by (bx,by), (cx,cy), (ex,ey)
Built-ins	x=area	expression	Assign an area
	x=format	"fmt" expr... (up to 5)	Assign formatting to expressions
	x=polar	x y radius angle	Assign polar coordinate centered at (x,y) at radius and angle (0-360)
	x=polarx	x y radius angle	Assign X-polar coordinate centered at (x,y) at radius and angle (0-360)
	x=polary	x y radius angle	Assign Y-polar coordinate centered at (x,y) at radius and angle (0-360)
	x=random	min max	Assign a random number between two values
	x=substr	"string" begin end	Assign a substring
	x=vmap	data min1 max1 min2 max2	Assign a value mapped to two ranges
	x=geocoord	geo URL or lat long	Map geo to canvas coordinates
Math	x=cosine	expression	Assign the cosine of expression
	x=sine	expression	Assign the sine of expression
	x=sqrt	expression	Assign the square root of expression
	x=tangent	expression	Assign the tangent of expression
Geographic	geoarc	"p1" "p2" [lw] [color] [opacity]	Draw arcs between points
	geoborder	"file" [lw] [color] [opacity]	Reads KML data from the specified file and renders the map borders
	geoimage	"loc" width height	Place an image at a geographical location
	geolabel	"loc" [size] [font] [color] [opacity]	Reads data from the specified file or location and renders the map labels
	geoloc	"loc" [align] [size] [font] [color] [opacity]	Reads data from the specified file or location and a make map point and labels
	geomark	"loc" [size] [color] [opacity]	Reads data from the specified file or location and renders map points
	geopath	"p1" "p2" [lw] [color] [opacity]	Draw line between points
	geopathfile	"file" [lw] [color] [opacity]	Reads data from the specified file and a make lines between points
	georegion	"file" [color] [opacity] [gap]	Reads KML data from the specified file and renders the map regions
	geobound	latmin [latmax] [longmin] [longmax]	Sets overall geographic bounds
	geocanvas	xmin [xmax] [ymin] [ymax]	Sets geographic canvas
	georuler	[increment] [color]	Draw a geographic ruler

	Keyword	Arguments	Description
Charts	dchart	options... file	Chart with specified options
	legend	"string" x y fontsize font color	Chart legend
	chartbbox	left [right] [top] [bottom]	Chart bounding box
	areachart	"file" [color] [lcolor] [vcolor]	Area chart with specified data
	barchart	"file" [color] [lcolor] [vcolor]	Bar chart with specified data
	dotchart	"file" [color] [lcolor] [vcolor]	Dot chart with specified data
	hbarchart	"file" [color] [lcolor] [vcolor]	Horizontal bar chart with specified data
	linechart	"file" [color] [lcolor] [vcolor]	Line chart with specified data
	wbarchart	"file" [color] [lcolor] [vcolor]	Word bar chart with specified data
	slopechart	"file" [color] [lcolor] [vcolor]	Slope chart with specified data
	pmap	"file" [size]	Proportional map chart with specified data
	pie	"file" [size]	Pie chart with specified data
	donut	"file" [width] [size]	Donut chart with specified data
	pgrid	"file" [size]	Proportional grid chart with specified data
	waffle	"file" [size]	Waffle chart with specified data
	fanchart	"file" [size]	Fan chart with specified data
	bowtie	"file" [size]	Bowtie chart with specified data